

claude-windows / 6cee04e1-f4f1-4593-86c9-2e3ca3473efd

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6cee04e1-f4f1-4593-86c9-2e3ca3473efd\subagents\workflows\wf_b7f012f8-5c3\agent-a3d433e1e8f9fa2df.jsonl`  
- SHA-256: `5fe6b0a545a7ce4fcfae5a0d0c4087ab91cd310ed99e4812b06167b89e39cf86`  
- Source modified: `2026-06-18T01:32:03+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `wf_b7f012f8-5c3`  
- session_id: `6cee04e1-f4f1-4593-86c9-2e3ca3473efd`
```

Transcript

1. user (2026-06-18T01:27:48.613Z)

You are an adversarial bug-hunter in a mature Python repo (badminton-highlight-indexer: FastAPI + SQLite + ffmpeg + GPU CV/AI rally detector). Read code READ-ONLY in this worktree (pinned to origin/master ? DO NOT modify anything):

C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hunt-base

Hunt ONLY for REAL, REACHABLE defects a maintainer would agree is a bug and want fixed:

- crash on a reachable input (unguarded dict/index/attr access, None deref, KeyError/IndexError),
- wrong logic / off-by-one / inverted condition / wrong operator,
- incorrect error handling (catching too much/little, swallowing, wrong exception type),
- resource leak (unclosed file/db/subprocess), TOCTOU, mutable-default-arg,
- contract mismatch between a function and its callers/tests,
- wrong default / unit / sign that produces incorrect output.

HARD RULES:

- PRECISION over recall. Report a finding ONLY if you traced the actual code and a concrete trigger reaches it. No style, no "could be cleaner", no speculative "might", no perf-nits, no missing-test observations, no naming. If you are not sure it is a real bug, DROP it.
- For each finding give an EXACT file path + line, the bug_class, a CONCRETE trigger (the input or call path that hits it), why it is wrong, severity, and a minimal proposed fix.
- This repo has 848 passing tests and several hardening sweeps ? easy bugs are already gone. Expect FEW or ZERO real bugs in your area. Returning an empty candidates list is a VALID, respected answer. Do NOT manufacture findings to look productive.
- EXCLUDE these files entirely (just fixed / fix already in flight): backend/eval/ablation.py, backend/eval/fusion_compare.py, backend/eval/fusion_audio.py.
- Verify your trigger by reading the surrounding code and any related test, not from memory.

YOUR AREA: backend/eval/ core scoring/stats ?

rally_seg_eval.py, tolerance_metrics.py, experiment.py, eval_stats.py, regression.py, calibration.py, per_user_gate.py (whichever exist). Focus on numeric edge cases (empty inputs, div-by-zero, n=0/1, NaN), Hungarian/greedy matching boundaries, bootstrap/sign-test correctness, and loader/IO key access.

2. assistant (2026-06-18T01:27:51.366Z)

I'll hunt for bugs in the eval scoring/stats area. Let me start by exploring the target directory.

3. user (2026-06-18T01:27:52.825Z)

```
total 461  
drwxr-xr-x 1 avidu 197609    0 Jun 18 06:56 .  
drwxr-xr-x 1 avidu 197609    0 Jun 18 06:56 ..  
-rw-r--r-- 1 avidu 197609 465 Jun 18 06:56 __init__.py
```

```

-rw-r--r-- 1 avidu 197609 31899 Jun 18 06:56 ablation.py
-rw-r--r-- 1 avidu 197609 2356 Jun 18 06:56 annotations.py
drwxr-xr-x 1 avidu 197609 0 Jun 18 06:56 audio
-rw-r--r-- 1 avidu 197609 4887 Jun 18 06:56 batch.py
-rw-r--r-- 1 avidu 197609 6135 Jun 18 06:56 calibrate_local.py
-rw-r--r-- 1 avidu 197609 5050 Jun 18 06:56 calibrate_wasb.py
-rw-r--r-- 1 avidu 197609 7992 Jun 18 06:56 calibration.py
-rw-r--r-- 1 avidu 197609 4670 Jun 18 06:56 classifier.py
-rw-r--r-- 1 avidu 197609 8602 Jun 18 06:56 distill_local.py
-rw-r--r-- 1 avidu 197609 11164 Jun 18 06:56 eval_partial_labels.py
-rw-r--r-- 1 avidu 197609 6836 Jun 18 06:56 eval_stats.py
-rw-r--r-- 1 avidu 197609 28380 Jun 18 06:56 experiment.py
-rw-r--r-- 1 avidu 197609 10959 Jun 18 06:56 export_wasb_segments.py
-rw-r--r-- 1 avidu 197609 6799 Jun 18 06:56 fusion_audio.py
-rw-r--r-- 1 avidu 197609 5265 Jun 18 06:56 fusion_compare.py
-rw-r--r-- 1 avidu 197609 10353 Jun 18 06:56 fusion_golden.py
-rw-r--r-- 1 avidu 197609 11843 Jun 18 06:56 gemini_refine.py
-rw-r--r-- 1 avidu 197609 57601 Jun 18 06:56 golden_fixtures.py
-rw-r--r-- 1 avidu 197609 4642 Jun 18 06:56 gt_loader.py
-rw-r--r-- 1 avidu 197609 5583 Jun 18 06:56 harness.py
-rw-r--r-- 1 avidu 197609 5803 Jun 18 06:56 metrics.py
-rw-r--r-- 1 avidu 197609 8094 Jun 18 06:56 per_user_eval.py
-rw-r--r-- 1 avidu 197609 9247 Jun 18 06:56 per_user_gate.py
-rw-r--r-- 1 avidu 197609 13822 Jun 18 06:56 per_user_regression.py
-rw-r--r-- 1 avidu 197609 10251 Jun 18 06:56 promotion.py
-rw-r--r-- 1 avidu 197609 22490 Jun 18 06:56 rally_seg_eval.py
-rw-r--r-- 1 avidu 197609 7508 Jun 18 06:56 rally_seq_proto.py
-rw-r--r-- 1 avidu 197609 9072 Jun 18 06:56 regression.py
-rw-r--r-- 1 avidu 197609 6839 Jun 18 06:56 score_calibration.py
-rw-r--r-- 1 avidu 197609 10619 Jun 18 06:56 segmentation_metrics.py
-rw-r--r-- 1 avidu 197609 8414 Jun 18 06:56 serve_contrast.py
-rw-r--r-- 1 avidu 197609 9205 Jun 18 06:56 served_gate_a.py
-rw-r--r-- 1 avidu 197609 3647 Jun 18 06:56 splits.py
-rw-r--r-- 1 avidu 197609 14713 Jun 18 06:56 tolerance_metrics.py
-rw-r--r-- 1 avidu 197609 3765 Jun 18 06:56 training.py
-rw-r--r-- 1 avidu 197609 11642 Jun 18 06:56 windowing.py

```

4. assistant (2026-06-18T01:27:55.236Z)

Let me read the core scoring/stats files in my assigned area.

5. user (2026-06-18T01:27:56.232Z)

```

1      """Honest small-N evaluation statistics (course-corrections C1 + C3).
2
3      The ~6-match golden corpus + a regularised logistic arbiter is exactly the regime where
4      a
5      single leave-one-video-out *mean* lies: leave-one-out anti-correlates each fold's
6      training
7      and held-out label means at small N, and one lucky/unlucky fold swings the headline. The
8      ``compare``/LOVO machinery already produces **per-video held-out scores** ? this module
9      turns those into an honest summary:
10
11      - **C1 ? never a single mean.** ``mean_with_ci`` / ``bootstrap_ci`` report a confidence
12      interval; ``paired_sign_test`` and ``paired_delta_ci`` grade a B?A improvement by how
13      many *videos* it wins on (the distribution-free test the repo already leans on:
14      "sign test needs n ? ~n=10/9-wins for p<0.05"), so the promotion gate can't promote
15      noise.
16
17      - **C3 ? honest stacking.** ``out_of_fold_predictions`` / ``grouped_folds`` give a
18      learned
19      producer's *out-of-fold* outputs so feeding them to the arbiter doesn't leak (group by
20      match, never a held-out window from the same video).
21
22      See ``docs/ANALYZERS_AND_RECONCILERS.md`` §13/C1+C3. All **additive**, pure (stdlib
23      only),

```

```

18 deterministic (bootstrap takes an explicit ``seed``) ? nothing here changes existing
19 ``compare``/``calibration.leave_one_video_out`` behaviour; callers opt in.
20 """
21 from __future__ import annotations
22
23 import math
24 import random
25 import statistics
26 from typing import Any, Callable, Dict, List, Optional, Sequence, Tuple
27
28 Interval = Tuple[float, float]
29 FitPredict = Callable[[List[int], List[int]], List[Any]]
30
31
32 def bootstrap_ci(values: Sequence[float], confidence: float = 0.95,
33                 n_boot: int = 2000, seed: int = 0) -> Tuple[float, float]:
34     """Percentile bootstrap CI for the mean of ``values``. Deterministic given ``seed``.
35
36     ``n<=1`` returns a degenerate ``(v, v)`` / ``(0, 0)`` interval ? there is no spread
37 to
38     resample. With n=6 (our corpus) this is the honest "how wide could the mean be?"
39 band.
40     """
41     vals = [float(v) for v in values]
42     n = len(vals)
43     if n == 0:
44         return (0.0, 0.0)
45     if n == 1:
46         return (vals[0], vals[0])
47     rng = random.Random(seed)
48     means: List[float] = []
49     for _ in range(max(1, n_boot)):
50         means.append(sum(vals[rng.randrange(n)] for _ in range(n)) / n)
51     means.sort()
52     alpha = (1.0 - confidence) / 2.0
53     last = len(means) - 1
54     lo = means[max(0, int(math.floor(alpha * last)))]
55     hi = means[min(last, int(math.ceil((1.0 - alpha) * last)))]
56     return (lo, hi)
57
58 def mean_with_ci(values: Sequence[float], confidence: float = 0.95,
59                 n_boot: int = 2000, seed: int = 0) -> Dict[str, float]:
60     """``{mean, std, n, ci_low, ci_high, confidence}`` ? report this, never a bare mean
61 (C1)."""
62     vals = [float(v) for v in values]
63     n = len(vals)
64     mean = statistics.mean(vals) if vals else 0.0
65     std = statistics.pstdev(vals) if n > 1 else 0.0
66     lo, hi = bootstrap_ci(vals, confidence, n_boot, seed)
67     return {"mean": mean, "std": std, "n": float(n),
68           "ci_low": lo, "ci_high": hi, "confidence": confidence}
69
70 def paired_sign_test(deltas: Sequence[float], eps: float = 0.0) -> Dict[str, float]:
71     """Two-sided exact sign test over per-fold deltas (B ? A).
72
73     ``wins`` = folds where B beat A by more than ``eps``; ``losses`` the reverse; ties
74     excluded. ``p_value`` is the exact two-sided binomial tail under p=0.5 ? the
75     distribution-free "did B win on enough *videos*" check.
76     """
77     wins = sum(1 for d in deltas if d > eps)
78     losses = sum(1 for d in deltas if d < -eps)
79     ties = len(deltas) - wins - losses
80     n_eff = wins + losses

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80     if n_eff == 0:
81         p = 1.0
82     else:
83         k = min(wins, losses)
84         tail = sum(math.comb(n_eff, j) for j in range(k + 1)) * (0.5 ** n_eff)
85         p = min(1.0, 2.0 * tail)
86     return {"wins": float(wins), "losses": float(losses), "ties": float(ties),
87           "n_effective": float(n_eff), "p_value": p}
88
89
90 def paired_delta_ci(a_values: Sequence[float], b_values: Sequence[float],
91                    confidence: float = 0.95, n_boot: int = 2000, seed: int = 0,
92                    eps: float = 0.0) -> Dict[str, Any]:
93     """Paired (by fold) B ? A summary: mean delta + bootstrap CI + the sign test (C1).
94
95     ``a_values``/``b_values`` are per-fold held-out scores for variants A and B, aligned
96     by
97     fold (same video order). A promotion is honest when the delta CI clears 0 *and* the
98     sign
99     test agrees ? not when a single mean ticked up.
100     """
101     n = min(len(a_values), len(b_values))
102     deltas = [float(b_values[i]) - float(a_values[i]) for i in range(n)]
103     summary = mean_with_ci(deltas, confidence, n_boot, seed)
104     return {
105         "mean_delta": summary["mean"],
106         "ci_low": summary["ci_low"],
107         "ci_high": summary["ci_high"],
108         "n": float(n),
109         "ci_excludes_zero": (summary["ci_low"] > 0.0) or (summary["ci_high"] < 0.0),
110         "sign_test": paired_sign_test(deltas, eps),
111     }
112
113 def grouped_folds(groups: Sequence[Any]) -> List[Tuple[List[int], List[int]]]:
114     """Leave-one-group-out splits ``(train_idx, test_idx)`` ? group by match, the honest
115     unit.
116
117     The field's name for "LOVO by match" is leave-one-group-out (LOGO). Group order is
118     first-seen (deterministic). Groups with no complement (a single group) yield no
119     fold.
120     """
121     seen: List[Any] = []
122     for g in groups:
123         if g not in seen:
124             seen.append(g)
125     n = len(groups)
126     folds: List[Tuple[List[int], List[int]]] = []
127     for g in seen:
128         test = [i for i in range(n) if groups[i] == g]
129         train = [i for i in range(n) if groups[i] != g]
130         if test and train:
131             folds.append((train, test))
132     return folds
133
134 def out_of_fold_predictions(groups: Sequence[Any], fit_predict: FitPredict) ->
135 List[Optional[Any]]:
136     """Out-of-fold predictions for honest stacking (C3).
137
138     For each leave-one-group-out fold, ``fit_predict(train_idx, test_idx)`` trains on
139     the
140     other groups and returns predictions for ``test_idx`` (in that order); results are
141     scattered back to a list aligned to the input items. So a *learned* producer's
142     outputs,

```

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138         when used as arbiter features, never include in-sample predictions ? the leakage
path
139         that inflates measured lift on a tiny corpus. Items in a singleton group stay
``None``
140         (needs >= 2 groups to predict).
141         """
142         n = len(groups)
143         oof: List[Optional[Any]] = [None] * n
144         for train_idx, test_idx in grouped_folds(groups):
145             preds = fit_predict(train_idx, test_idx)
146             for pos, ti in enumerate(test_idx):
147                 if pos < len(preds):
148                     oof[ti] = preds[pos]
149         return oof
150

```

6. user (2026-06-18T01:27:56.818Z)

```

1         """R2 ? task-faithful rally-boundary eval (tolerance-match + over-segmentation guard).
2
3         The headline rally metric (tIoU-F1 is kept as a SECONDARY diagnostic): a prediction
matches a
4         golden rally iff ``|?start| ? ?_start AND |?end| ? ?_end`` under a **strict 1:1
assignment**, so
5         over-production costs precision directly. Reported as a DECOMPOSITION ? ``P/R/F1@?`` +
start/end
6         boundary-error distributions + an over-seg guard (split/merge counts) + a ?-sweep ?
never one
7         number (a single ? misleads; see the design doc).
8
9         Locked decisions + rationale: ``docs/R2_EVAL_METRIC_DESIGN.md`` (owner-reviewed
2026-06-17):
10        asymmetric ? (``?_start=2.0`` forgives the ~1?1.5 s service window; ``?_end=1.5`` keeps
the
11        rally-end honest ? IAA-calibrated later); augment tIoU now, ablation-gate any eventual
swap.
12
13        Pure + stdlib; **numpy/scipy are OPTIONAL** ? used for the exact Hungarian assignment
when present,
14        else a deterministic greedy 1:1 fallback. Both agree on the temporally-ordered, near-
diagonal
15        eligibility graphs that rally intervals produce, so scores are environment-independent.
16        """
17        from __future__ import annotations
18
19        import statistics
20        from typing import Any, Dict, List, Optional, Tuple
21
22        Interval = Tuple[float, float]
23        #: (pred_idx, gt_idx, dstart_signed, dend_signed) ? signed = pred ? gt (positive = pred
is late).
24        Match = Tuple[int, int, float, float]
25
26        #: Provisional ? (the design's locked starting point; fixed by the IAA study later).
Always
27        #: report the sweep alongside, since the golden END convention carries >1.5 s of noise.
28        TAU_START: float = 2.0
29        TAU_END: float = 1.5
30        SWEEP_TAU: Tuple[float, ...] = (1.5, 2.0, 2.5, 3.0)
31
32
33        def _eligible(p: Interval, g: Interval, tau_start: float, tau_end: float) -> bool:
34            return abs(p[0] - g[0]) <= tau_start and abs(p[1] - g[1]) <= tau_end
35
36

```

```

37     def _greedy_match_1to1(preds: List[Interval], gts: List[Interval],
38                             tau_start: float, tau_end: float) -> List[Match]:
39         """Deterministic greedy 1:1: eligible (pred, gt) pairs in ascending combined
boundary error,
40         assigned first-come and skipping used indices. Optimal-or-near on the near-diagonal
eligibility
41         graph rally intervals produce. The numpy/scipy-free path (and the CI path)."""
42         pairs: List[Tuple[float, float, int, int]] = []
43         for i, p in enumerate(preds):
44             for j, g in enumerate(gts):
45                 if _eligible(p, g, tau_start, tau_end):
46                     # sort key: combined error, then (i, j) for a fully deterministic
tiebreak
47                     pairs.append((abs(p[0] - g[0]) + abs(p[1] - g[1]), float(i * 1e-6 + j *
1e-9), i, j))
48         pairs.sort()
49         used_p: set = set()
50         used_g: set = set()
51         matches: List[Match] = []
52         for _, _, i, j in pairs:
53             if i in used_p or j in used_g:
54                 continue
55             used_p.add(i)
56             used_g.add(j)
57             matches.append((i, j, preds[i][0] - gts[j][0], preds[i][1] - gts[j][1]))
58         matches.sort(key=lambda mm: mm[1])
59         return matches
60
61
62     def _hungarian_match_1to1(preds: List[Interval], gts: List[Interval],
63                             tau_start: float, tau_end: float) -> Optional[List[Match]]:
64         """Exact max-cardinality (then min total boundary error) via scipy. Returns ``None``
when
65         numpy/scipy are unavailable so the caller falls back to greedy."""
66         try:
67             import numpy as np
68             from scipy.optimize import linear_sum_assignment
69         except Exception:
70             return None
71         n, m = len(preds), len(gts)
72         size = max(n, m)
73         big = 1.0e6
74         cost = np.full((size, size), big, dtype=float)
75         for i, p in enumerate(preds):
76             for j, g in enumerate(gts):
77                 if _eligible(p, g, tau_start, tau_end):
78                     cost[i, j] = abs(p[0] - g[0]) + abs(p[1] - g[1])
79         rows, cols = linear_sum_assignment(cost)
80         matches: List[Match] = []
81         for i, j in zip(rows.tolist(), cols.tolist()):
82             if i < n and j < m and cost[i, j] < big: # drop forced ineligible assignments
83                 matches.append((i, j, preds[i][0] - gts[j][0], preds[i][1] - gts[j][1]))
84         matches.sort(key=lambda mm: mm[1])
85         return matches
86
87
88     def match_1to1(preds: List[Interval], gts: List[Interval],
89                   tau_start: float = TAU_START, tau_end: float = TAU_END
90                   ) -> Tuple[List[Match], float, float, float]:
91         """Strict 1:1 tolerance match ? ``(matches, precision, recall, f1)``. ``P = matched
/ n_pred``
92         (over-production costs precision), ``R = matched / n_gt``. Exact Hungarian when
scipy is
93         present; deterministic greedy otherwise."""
94         n, m = len(preds), len(gts)

```

```

95     if n == 0 or m == 0:
96         return [], 0.0, 0.0, 0.0
97     matches = _hungarian_match_1to1(preds, gts, tau_start, tau_end)
98     if matches is None:
99         matches = _greedy_match_1to1(preds, gts, tau_start, tau_end)
100    k = len(matches)
101    precision = k / n
102    recall = k / m
103    f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) else
0.0
104    return matches, precision, recall, f1
105
106
107    def _percentile(xs: List[float], p: float) -> float:
108        """Linear-interpolated percentile (pure stdlib)."""
109        if not xs:
110            return 0.0
111        s = sorted(xs)
112        k = (len(s) - 1) * p / 100.0
113        lo = int(k)
114        hi = min(lo + 1, len(s) - 1)
115        return s[lo] + (s[hi] - s[lo]) * (k - lo)
116
117
118    def _overlap(a: Interval, b: Interval) -> float:
119        return max(0.0, min(a[1], b[1]) - max(a[0], b[0]))
120
121
122    def boundary_error_report(preds: List[Interval], gts: List[Interval]) -> Dict[str,
Dict[str, float]]:
123        """Median + IQR of signed AND absolute ?start, ?end, split start/end ? the "start
solved,
124        end open" diagnostic that aims R4.
125
126        Computed over **best-overlap** matches (each GT ? its maximally-overlapping
prediction, any
127        positive overlap), NOT the within-? 1:1 matches: a ?-gated set is survivorship-
biased (loose
128        ends fail the gate and vanish from the distribution), which would HIDE the very end-
deficit
129        this diagnostic exists to surface. Unconditional best-overlap is what the day-1
validation used
130        to find |?end| ? |?start|. (The within-? deficit shows up instead as low
`tol_recall`.)"""
131        def dist(vals: List[float]) -> Dict[str, float]:
132            av = [abs(v) for v in vals]
133            return {
134                "signed_median": statistics.median(vals) if vals else 0.0,
135                "abs_median": statistics.median(av) if av else 0.0,
136                "abs_p25": _percentile(av, 25.0),
137                "abs_p75": _percentile(av, 75.0),
138                "n": float(len(vals)),
139            }
140        ds: List[float] = []
141        de: List[float] = []
142        for g in gts:
143            best = max(preds, key=lambda p: _overlap(p, g), default=None)
144            if best is not None and _overlap(best, g) > 0.0:
145                ds.append(best[0] - g[0])
146                de.append(best[1] - g[1])
147        return {"dstart": dist(ds), "dend": dist(de)}
148
149
150    def over_seg_report(preds: List[Interval], gts: List[Interval]) -> Dict[str, int]:
151        """The over-segmentation GUARD (per-video no-regression in promotion):

```

```

``split_count`` (?2
152     preds covering one rally) + ``merge_count`` (one pred ? ?2 rallies). Reuses the
bipartite
153     overlap-graph in ``segmentation_metrics.merge_split_report``. """
154     from backend.eval.segmentation_metrics import merge_split_report
155     r = merge_split_report(preds, gts)
156     return {"split_count": int(r["splits"]), "merge_count": int(r["merges"])}
157
158
159     # ----- #
160     # R1 ? net over-segmentation promotion guard
161     # ----- #
162     #: Default over-seg cost weights. A MERGE hides a rally (?2 golden rallies collapse into
one
163     #: predicted window ? the user can't reach the hidden one ? recall loss at rally
granularity); a
164     #: SPLIT only fragments one rally (it is still found, just in pieces). So a merge is the
strictly
165     #: worse fault and is weighted higher. (Equal weights also pass the milestone ? see the
R1 evidence;
166     #: the asymmetry is the principled default, not a number tuned to force a verdict.)
167     WEIGHT_MERGE: float = 2.0
168     WEIGHT_SPLIT: float = 1.0
169
170
171     def over_seg_guard(base: List[Dict[str, Any]], cand: List[Dict[str, Any]], *,
172                       weight_merge: float = WEIGHT_MERGE, weight_split: float =
WEIGHT_SPLIT,
173                       use_rates: bool = True, eps: float = 1e-9) -> Dict[str, Any]:
174         """**Net** over-segmentation promotion guard (the R1 refinement of the strict per-
video rule).
175
176         ``base``/``cand`` are aligned per-video over-seg dicts (the rows ``rally_seg_eval``
already
177         emits): each needs ``split_count``/``merge_count`` and ? when ``use_rates``
(default) ?
178         ``merge_rate``/``split_rate``. Optional ``name`` keys align the two lists by video
(else by
179         position). Returns the promote-or-block verdict for promoting ``cand`` over
``base``.
180
181         **Why the strict rule needed refining.** R2 §2.3.3's guard blocks promotion if
``cand`` raises
182         ``split_count`` OR ``merge_count`` on ANY video. That is right for an *incremental*
cue (same
183         windowing structure, an increase = a real regression) but mis-fires on a
*structural* change
184         (mega-windows ? bounded): splitting a single mega-window necessarily lifts
``split_count`` from
185         ~0, and the raw merge **count** is non-monotonic ? one mega-window swallowing 40
rallies is
186         ``merge_count=1`` yet ``merge_rate=1.0`` (every rally hidden), whereas bounding it
into pieces
187         that each glue 2 neighbours is ``merge_count=9`` yet ``merge_rate=0.5`` (FEWER
rallies hidden).
188         So the strict count rule blocks a config that strictly *reduces* the fraction of
rallies lost to
189         over-merge. (Measured: the cap6+R4 milestone trips strict on 10/11 videos yet cuts
mean
190         ``merge_rate`` 0.694 ? 0.150 with NO per-video merge_rate regression ?
RALLY_QUALITY_RESEARCH §6.)
191
192         **The net rule (default verdict ``passes``):** score over-seg as ONE weighted cost
per video and
193         require BOTH:

```



```

194         1. the corpus-mean net cost does not rise (`cand_net ? base_net + eps`), AND
195         2. NO video's **merge_rate** rises beyond `eps` ? reintroducing a mega-window
that hides MORE
196         of a video's rallies is the one over-merge regression that is never acceptable,
the honest
197         "no NEW merges" rule expressed on the *rate* (the recall-meaningful quantity),
not the count.
198         With `use_rates` the per-video cost uses `merge_rate`/`split_rate` (fraction
of GT rallies
199         affected, ? [0,1] ? comparable across a structural change); `use_rates=False`
scores raw counts.
200
201         **Revert (one line):** read `strict_passes` instead of `passes` to fall back to
the original
202         strict per-video count rule. Both verdicts are always reported, so the choice is
auditable.
203         """
204         def _aligned() -> List[Tuple[Dict[str, Any], Dict[str, Any]]]:
205             if base and cand and all("name" in b for b in base) and all("name" in c for c in
cand):
206                 cmap = {c["name"]: c for c in cand}
207                 return [(b, cmap[b["name"]]) for b in base if b["name"] in cmap]
208                 n = min(len(base), len(cand))
209                 return [(base[i], cand[i]) for i in range(n)]
210
211         def _cost(row: Dict[str, Any]) -> float:
212             if use_rates:
213                 return weight_merge * float(row["merge_rate"]) + weight_split *
float(row["split_rate"])
214             return weight_merge * float(row["merge_count"]) + weight_split *
float(row["split_count"])
215
216         pairs = _aligned()
217         n = len(pairs)
218         base_costs = [_cost(b) for b, _ in pairs]
219         cand_costs = [_cost(c) for _, c in pairs]
220         base_net = (sum(base_costs) / n) if n else 0.0
221         cand_net = (sum(cand_costs) / n) if n else 0.0
222         net_delta = cand_net - base_net
223
224         # Per-video regressions, reported for transparency.
225         new_merges: List[Dict[str, Any]] = []           # raw COUNT increases (what strict
trips on)
226         new_splits: List[Dict[str, Any]] = []           # raw COUNT increases (what strict
trips on)
227         merge_rate_regress: List[Dict[str, Any]] = []   # RATE increases (the real over-
merge regression)
228         for b, c in pairs:
229             if float(c["merge_count"]) > float(b["merge_count"]) + eps:
230                 new_merges.append({"name": b.get("name", ""),
231                                     "base": float(b["merge_count"]), "cand":
float(c["merge_count"])})
232             if float(c["split_count"]) > float(b["split_count"]) + eps:
233                 new_splits.append({"name": b.get("name", ""),
234                                    "base": float(b["split_count"]), "cand":
float(c["split_count"])})
235             if use_rates and float(c["merge_rate"]) > float(b["merge_rate"]) + eps:
236                 merge_rate_regress.append({"name": b.get("name", ""),
237                                             "base": float(b["merge_rate"]), "cand":
float(c["merge_rate"])})
238
239         strict_passes = (not new_merges) and (not new_splits)
240         net_cost_ok = net_delta <= eps
241         no_merge_rate_regress = not merge_rate_regress
242         passes = bool(net_cost_ok and no_merge_rate_regress)

```

```

243     return {
244         "passes": passes,
245         "strict_passes": strict_passes,
246         "net_cost_ok": net_cost_ok,
247         "no_merge_rate_regress": no_merge_rate_regress,
248         "base_net": base_net,
249         "cand_net": cand_net,
250         "net_delta": net_delta,
251         "n": n,
252         "use_rates": use_rates,
253         "weight_merge": weight_merge,
254         "weight_split": weight_split,
255         "new_merges": new_merges,
256         "new_splits": new_splits,
257         "merge_rate_regress": merge_rate_regress,
258     }
259
260
261     def tolerance_report(preds: List[Interval], gts: List[Interval],
262                         tau_start: float = TAU_START, tau_end: float = TAU_END) ->
Dict[str, object]:
263         """The full R2 block for one video: P/R/F1@? + boundary-error distributions + over-
seg guard."""
264         matches, precision, recall, f1 = match_1to1(preds, gts, tau_start, tau_end)
265         return {
266             "tau_start": tau_start, "tau_end": tau_end,
267             "n_pred": len(preds), "n_gt": len(gts), "matched": len(matches),
268             "precision": precision, "recall": recall, "f1": f1,
269             "boundary_error": boundary_error_report(preds, gts),
270             "over_seg": over_seg_report(preds, gts),
271         }
272
273
274     def tolerance_sweep(preds: List[Interval], gts: List[Interval],
275                       taus: Tuple[float, ...] = SWEEP_TAUS) -> Dict[float, float]:
276         """Symmetric-? F1 sweep ? the trend curve (a single ? misleads). Monotonic non-
decreasing in ?."""
277         return {t: match_1to1(preds, gts, t, t)[3] for t in taus}
278

```

7. assistant (2026-06-18T01:27:59.597Z)

Let me read the remaining core files.

8. user (2026-06-18T01:28:01.041Z)

```

1     """One-command rally-segmentation eval ? score a windowing against the golden labels
with the
2     over-segmentation-sensitive metric set, honestly.
3
4     This is the harness the rally-quality project (`docs/RALLY_QUALITY_RESEARCH.md` $4.5 /
$7)
5     measures every tier against. It closes the loop the over-merge finding exposed: plain
temporal
6     IoU-F1 is blind to merged mega-windows, so we report it alongside segmental F1@k,
the
7     segment-count ratio, and the explicit ``merge_split`` breakdown
(``segmentation_metrics``).
8
9     For each golden video it: parses the cached WASB/TrackNet trajectory CSV ? builds
candidate
10    windows under a :class:`~backend.eval.windowing.WindowingPreset` (so it tracks the
SERVED
11    operating point and any future windowing the same way) ? scores the windows-as-rallies
against

```

```

12     the ground-truth rally intervals. Aggregates per-video held-out F1 with a bootstrap CI,
and can
13     diff two windowings with a paired sign-test + CI (the per-tier ``delta``).
14
15     GROUND TRUTH: the golden manifest (``output/human_lovo_manifest.json``) gives each
video's
16     trajectory path + fps + frame_width; the GT rally intervals come from
17     ``output/<name>_golden_features.json`` (the ``gts`` key) so this needs no extra label
files.
18
19     Pure offline (CPU): trajectory parsing + windowing + interval scoring; no GPU/Gemini.
20     """
21     from __future__ import annotations
22
23     import argparse
24     import json
25     import os
26     from typing import Any, Dict, List, Optional
27
28     from backend.eval import eval_stats, metrics, segmentation_metrics, windowing
29     from backend.eval.metrics import Interval
30     from backend.eval.windowing import PRESETS, SERVED, WindowingPreset
31
32     DEFAULT_MANIFEST = os.path.join("output", "human_lovo_manifest.json")
33     DEFAULT_FEATURES_DIR = "output"
34     DEFAULT_IOU = 0.5
35
36
37     # ----- #
38     # Pure scoring core (no I/O ? unit-testable)
39     # ----- #
40     def score_intervals(preds: List[Interval], gts: List[Interval],
41                        iou: float = DEFAULT_IOU,
42                        tau_start: Optional[float] = None,
43                        tau_end: Optional[float] = None) -> Dict[str, Any]:
44         """The full per-video metric row for predicted vs ground-truth rally intervals.
45
46         Pairs temporal-IoU P/R/F1 + boundary error (``metrics.score``) with the
47         over-segmentation-sensitive set (F1@k, segment-count ratio, merge/split breakdown)
so a
48         merged mega-window is *visible* even when IoU-F1 isn't moved.
49
50         Also carries the **R2** task-faithful block (``tolerance_metrics``, the headline
going forward;
51         tIoU here is the SECONDARY trend-line per the augment-then-ablation-gate decision):
strict-1:1
52         tolerance-match ``tol_f1/tol_precision/tol_recall`` (over-production costs
precision) + the
53         start/end boundary-error medians + the over-seg guard counts. See
docs/R2_EVAL_METRIC_DESIGN.md.
54         """
55         from backend.eval import tolerance_metrics as tm
56         ts = tm.TAU_START if tau_start is None else tau_start
57         te = tm.TAU_END if tau_end is None else tau_end
58         sc = metrics.score(preds, gts, iou_threshold=iou)
59         f1k = segmentation_metrics.f1_at_overlaps(preds, gts)
60         ms = segmentation_metrics.merge_split_report(preds, gts)
61         tol_matches, tol_p, tol_r, tol_f1 = tm.match_1to1(preds, gts, ts, te)
62         be = tm.boundary_error_report(preds, gts) # unconditional best-overlap (the
R4-aiming diagnostic)
63         osr = tm.over_seg_report(preds, gts)
64         return {
65             "f1": sc.f1, "precision": sc.precision, "recall": sc.recall, "mean_iou":
sc.mean_iou,
66             "mean_start_error": sc.mean_start_error, "mean_end_error": sc.mean_end_error,

```

```

67         "f1@0.1": f1k[0.1], "f1@0.25": f1k[0.25], "f1@0.5": f1k[0.5],
68         "segment_count_ratio": segmentation_metrics.segment_count_ratio(preds, gts),
69         "merge_rate": ms["merge_rate"], "split_rate": ms["split_rate"],
70         "merges": ms["merges"], "splits": ms["splits"], "missed": ms["missed"],
71         "spurious": ms["spurious"], "num_pred": len(preds), "num_gt": len(gts),
72         # --- R2 (tolerance-match) ? the task-faithful headline block ---
73         "tau_start": ts, "tau_end": te,
74         "tol_f1": tol_f1, "tol_precision": tol_p, "tol_recall": tol_r,
75         "tol_matched": float(len(tol_matches)),
76         "dstart_abs_median": be["dstart"]["abs_median"], "dend_abs_median":
be["dend"]["abs_median"],
77         "split_count": osr["split_count"], "merge_count": osr["merge_count"],
78     }
79
80
81     _AGG_KEYS = ("f1", "f1@0.25", "f1@0.5", "merge_rate", "split_rate",
"segment_count_ratio",
82                 "precision", "recall",
83                 "tol_f1", "tol_precision", "tol_recall", "dstart_abs_median",
"dend_abs_median")
84
85
86     def aggregate(rows: List[Dict[str, Any]]) -> Dict[str, Any]:
87         """Aggregate per-video rows: mean (+ bootstrap 95% CI) of the headline metrics
across videos.
88
89         Per-video means (not pooled) keep videos the unit of evidence, matching the
harness's
90         leave-one-video-out discipline (windows within a video are correlated)."""
91         out: Dict[str, Any] = {"n": len(rows)}
92         for k in _AGG_KEYS:
93             vals = [r[k] for r in rows]
94             out[k] = eval_stats.mean_with_ci(vals) if vals else None
95         return out
96
97
98     # ----- #
99     # Golden corpus I/O
100    # ----- #
101    def load_golden(manifest_path: str = DEFAULT_MANIFEST,
102                   features_dir: str = DEFAULT_FEATURES_DIR) -> List[Dict[str, Any]]:
103        """Load each golden video's trajectory path + fps + frame_width (manifest) and GT
rally
104        intervals (`<name>_golden_features.json` `gts`). Videos with no GTs are kept but
flagged."""
105        with open(manifest_path) as fh:
106            manifest = json.load(fh)
107        out: List[Dict[str, Any]] = []
108        for v in manifest:
109            feat = os.path.join(features_dir, f"{v['name']}_golden_features.json")
110            gts: List[Interval] = []
111            if os.path.exists(feat):
112                with open(feat) as fh:
113                    gts = [(float(s), float(e)) for s, e in (json.load(fh).get("gts") or
[])]
114            out.append({"name": v["name"], "trajectory": v["trajectory"],
115                      "fps": float(v["fps"]), "frame_width": float(v["frame_width"]),
116                      "gts": gts})
117        return out
118
119    def windows_for(video: Dict[str, Any], preset: WindowingPreset,
120                   min_crossings: int = 0) -> List[Interval]:
121        """Parse a video's trajectory, window it under `preset` ? predicted rally
intervals.

```

```

122
123     ``min_crossings`` > 0 applies the **net-crossings rally-state gate** (Gate-A,
124     ``rally_gate.gate_windows``): drop windows whose shuttle crosses the net fewer than
this many
125     times (a non-rally fragment). This is the cheapest rally-state cue (W2) ? CPU-only,
derived
126     from the trajectory + an auto-estimated net axis; it cleans up the over-split that
bounded
127     windowing (W1) leaves behind.
128
129     NEVER-EMPTY safeguard (mirrors ``FusionSegmenter._select_for_handoff``): if the gate
would
130     drop EVERY window of a video that had ?1, fall back to the ungated windows. The gate
is
131     strictly *pruning* ? it never zeroes out a non-empty video ? so W2-on-W1 is "do no
harm".
132     """
133     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
134     points = TrackNetRunner.parse_trajectory_csv(video["trajectory"])
135     wins = windowing.build_windows(points, video["fps"], video["frame_width"], preset)
136     ivs = windowing.windows_to_intervals(wins)
137     if min_crossings > 0 and ivs:
138         from backend.pipeline.detectors.rally_gate import gate_windows
139         gated = [(g.start, g.end)
140                  for g in gate_windows(points, ivs, video["fps"],
min_crossings=min_crossings)
141                  if g.kept]
142         # Gating emptied a non-empty video ? keep the ungated windows.
143         ivs = gated if gated else ivs
144     return ivs
145
146
147     def evaluate(golden: List[Dict[str, Any]], preset: WindowingPreset,
148                 iou: float = DEFAULT_IOU, min_crossings: int = 0,
149                 tau_start: Optional[float] = None, tau_end: Optional[float] = None) ->
Dict[str, Any]:
150         """Score every golden video under ``preset`` (+ optional net-crossings gate); return
per-video
151         rows + the aggregate. Each row carries both the tIoU set and the R2 tolerance-match
block."""
152         rows: List[Dict[str, Any]] = []
153         for v in golden:
154             if not v["gts"] or not os.path.exists(v["trajectory"]):
155                 continue
156             preds = windows_for(v, preset, min_crossings)
157             row = score_intervals(preds, v["gts"], iou, tau_start, tau_end)
158             row["name"] = v["name"]
159             rows.append(row)
160         return {"preset": preset.to_dict(), "iou": iou, "min_crossings": min_crossings,
161                "rows": rows, "aggregate": aggregate(rows)}
162
163
164     def compare(golden: List[Dict[str, Any]], base: WindowingPreset, cand: WindowingPreset,
165                 iou: float = DEFAULT_IOU, base_min_crossings: int = 0,
166                 cand_min_crossings: int = 0,
167                 tau_start: Optional[float] = None, tau_end: Optional[float] = None,
168                 guard_weight_merge: Optional[float] = None,
169                 guard_weight_split: Optional[float] = None,
170                 guard_use_rates: bool = True) -> Dict[str, Any]:
171         """Baseline vs candidate (windowing and/or net-crossings gate): per-video paired ?
(+ CI +
172         sign-test) on both tIoU-F1 and the R2 ``tol_f1`` ? the per-tier 'number'. Also
reports ? on the
173         over-segmentation metrics AND the **R1 net over-seg promotion guard**
(``over_seg_guard``):

```

```

174         the honest promote/block verdict for cand-over-base, with the strict per-video rule
reported
175         alongside so the refinement is auditable."""
176         from backend.eval import tolerance_metrics as tm
177         b = evaluate(golden, base, iou, base_min_crossings, tau_start, tau_end)
178         c = evaluate(golden, cand, iou, cand_min_crossings, tau_start, tau_end)
179         by_name_b = {r["name"]: r for r in b["rows"]}
180         paired = [(by_name_b[r["name"]], r) for r in c["rows"] if r["name"] in by_name_b]
181         delta: Dict[str, Any] = {}
182         for k in ("f1", "f1@0.25", "tol_f1", "merge_rate", "split_rate",
"segment_count_ratio"):
183             delta[k] = eval_stats.paired_delta_ci([rb[k] for rb, _ in paired],
184                                                    [rc[k] for _, rc in paired])
185         gkw: Dict[str, Any] = {"use_rates": guard_use_rates}
186         if guard_weight_merge is not None:
187             gkw["weight_merge"] = guard_weight_merge
188         if guard_weight_split is not None:
189             gkw["weight_split"] = guard_weight_split
190         guard = tm.over_seg_guard([rb for rb, _ in paired], [rc for _, rc in paired], **gkw)
191         return {"base": b, "cand": c, "n_paired": len(paired), "delta": delta,
"over_seg_guard": guard}
192
193
194         # ----- #
195         # Reporting
196         # ----- #
197         def _ci(m: Optional[Dict[str, Any]]) -> str:
198             return "?" if not m else f"{m['mean']:.3f} [{m['ci_low']:.3f},{m['ci_high']:.3f}]"
199
200
201         def _format_guard(g: Dict[str, Any]) -> str:
202             """Render the R1 net over-seg promotion-guard verdict (with the strict rule for
contrast)."""
203             basis = "rate" if g["use_rates"] else "count"
204             lines = [f"\n## R1 over-seg promotion guard ({basis}-based, "
205                     f"w_merge={g['weight_merge']}, w_split={g['weight_split']})",
206                     f" NET verdict: {'PASS' if g['passes'] else 'BLOCK'} ",
207                     f"(net cost {g['base_net']:.3f} ? {g['cand_net']:.3f}, ?
{g['net_delta']:+.3f}; "
208                     f"no_merge_rate_regress={g['no_merge_rate_regress']})",
209                     f" strict per-video rule (R2 §2.3.3, for contrast): "
210                     f"{'pass' if g['strict_passes'] else 'BLOCK'} "
211                     f"(count increases ? merges:{len(g['new_merges'])})
splits:{len(g['new_splits'])})"]
212             if g["merge_rate_regress"]:
213                 names = ", ".join(f"{r['name']}({r['base']:.2f}?{r['cand']:.2f})"
214                                   for r in g["merge_rate_regress"])
215                 lines.append(f" ? merge_rate REGRESSED on: {names}")
216             return "\n".join(lines)
217
218
219         def format_report(result: Dict[str, Any]) -> str:
220             p = result["preset"]
221             bounded = []
222             if p.get("max_window_duration"):
223                 bounded.append(f"cap={p['max_window_duration']} + ("(hard)" if
p.get("hard_cap") else ""))
224             if p.get("inactivity_end"):
225
bounded.append(f"inact_end={p['inactivity_end']}/rt={p['inactivity_rally_thresh']}")
226                 f"/md={p['inactivity_min_density']}")
227             if p.get("blob_fallback"):
228                 bounded.append("blob_fallback")
229             extra = (" + " + ", ".join(bounded)) if bounded else ""
230             lines = [f"# Rally-segmentation eval ? windowing '{p['name']}' "

```

```

231         f"(vt={p['velocity_thresh']], merge_gap={p['merge_gap']], "
232         f"min_win={p['min_window_duration']}{extra}), IoU={result['iou']})", "")
233     lines.append("| video | F1 | F1@0.25 | F1@0.5 | merges | merge_rate | seg_ratio |
pred/gt |")
234     lines.append("|---|---|---|---|---|---|---|")
235     for r in result["rows"]:
236         lines.append(f"| {r['name']} | {r['f1']:.3f} | {r['f1@0.25']:.3f} |
{r['f1@0.5']:.3f} | "
237                     f"{int(r['merges'])} | {r['merge_rate']:.2f} |
{r['segment_count_ratio']:.2f} | "
238                     f"{r['num_pred']}/{r['num_gt']} |")
239     a = result["aggregate"]
240     lines += [f"***Aggregate (n={a['n']}, mean [95% CI]):** "
241             f"F1 {_ci(a['f1'])} · F1@0.25 {_ci(a['f1@0.25'])} · F1@0.5
{_ci(a['f1@0.5'])} · "
242             f"merge_rate {_ci(a['merge_rate'])} · seg_ratio
{_ci(a['segment_count_ratio'])}"]
243     # --- R2 (tolerance-match) block ? the task-faithful headline (tIoU above is
secondary) ---
244     rows = result["rows"]
245     if rows and "tol_f1" in rows[0]:
246         ts, te = rows[0]["tau_start"], rows[0]["tau_end"]
247         lines += [f"### R2 ? tolerance-match (?_start={ts}s, ?_end={te}s, strict
1:1)", "",
248                 "| video | tolF1 | tolP | tolR | |?start| | |?end| | split | merge |",
249                 "|---|---|---|---|---|---|---|"]
250     for r in rows:
251         lines.append(f"| {r['name']} | {r['tol_f1']:.3f} | {r['tol_precision']:.2f}
| "
252                     f"{r['tol_recall']:.2f} | {r['dstart_abs_median']:.2f} | "
253                     f"{r['dend_abs_median']:.2f} | {int(r['split_count'])} |
{int(r['merge_count'])} |")
254     lines += [f"***R2 aggregate (n={a['n']}):** tolF1 {_ci(a['tol_f1'])} · "
255             f"tolP {_ci(a['tol_precision'])} · tolR {_ci(a['tol_recall'])} · "
256             f"|?start| {_ci(a['dstart_abs_median'])} · |?end|
{_ci(a['dend_abs_median'])} "
257             f"(start?solved / end=the gap)"]
258     return "\n".join(lines)
259
260
261 def _resolve_preset(name: str) -> WindowingPreset:
262     if name in PRESETS:
263         return PRESETS[name]
264     raise SystemExit(f"unknown preset '{name}'; choices: {sorted(PRESETS)}")
265
266
267 def main() -> None:
268     ap = argparse.ArgumentParser(description="Rally-segmentation eval vs golden labels
(offline).")
269     ap.add_argument("--manifest", default=DEFAULT_MANIFEST)
270     ap.add_argument("--features-dir", default=DEFAULT_FEATURES_DIR)
271     ap.add_argument("--preset", default=SERVED.name, help="windowing preset (default:
served)")
272     ap.add_argument("--vs", default=None, help="second preset to diff against --preset")
273     ap.add_argument("--iou", type=float, default=DEFAULT_IOU)
274     ap.add_argument("--min-crossings", type=int, default=0,
275                     help="net-crossings rally-state gate on --preset (0=off; W2
Gate-A)")
276     ap.add_argument("--vs-min-crossings", type=int, default=0,
277                     help="net-crossings gate on --vs (default: same as --min-
crossings)")
278     ap.add_argument("--max-window-duration", type=float, default=None,
279                     help="override bounded-windowing cap (s) on BOTH presets (W1; e.g.
12)")
280     ap.add_argument("--vs-max-window-duration", type=float, default=None,

```

```

281             help="cap (s) on --vs ONLY (default: same as --max-window-duration);
lets a "
282             "BASELINE (no cap) vs CANDIDATE (capped) promotion check run in
one shot")
283         ap.add_argument("--hard-cap", action="store_true",
284             help="enforce the cap as a HARD cap on --preset (chop continuously-
active "
285             "windows at the cap when no inactivity gap exists; W1 hard-cap
fallback)")
286         ap.add_argument("--vs-hard-cap", action="store_true",
287             help="hard-cap fallback on --vs (default: same as --hard-cap)")
288         ap.add_argument("--blob-fallback", action="store_true",
289             help="R5 blob-fallback on --preset (after R4, force-chop any group
still over the "
290             "cap at its midpoint + flag/log it; the continuously-active
blob last resort)")
291         ap.add_argument("--vs-blob-fallback", action="store_true",
292             help="R5 blob-fallback on --vs (default: same as --blob-fallback)")
293         ap.add_argument("--blob-factor", type=float, default=None,
294             help="R5 magnitude gate (both presets): chop only groups > this ×
cap (default 4.0)")
295         ap.add_argument("--inactivity-end", type=float, default=None,
296             help="R4 rally-end inactivity trim (s) on --preset (0=off; trims
post-rally drift tail)")
297         ap.add_argument("--vs-inactivity-end", type=float, default=None,
298             help="R4 inactivity trim on --vs (default: same as --inactivity-
end)")
299         ap.add_argument("--inactivity-rally-thresh", type=float, default=None,
300             help="R4 velocity above which motion counts as 'rally' (both
presets; default 0.08)")
301         ap.add_argument("--inactivity-min-density", type=float, default=None,
302             help="R4 min trailing fast-fraction to stay 'in rally' (both
presets; default 0.34)")
303         ap.add_argument("--tau-start", type=float, default=None,
304             help="R2 tolerance-match start window (s); default 2.0 (forgives the
service window)")
305         ap.add_argument("--tau-end", type=float, default=None,
306             help="R2 tolerance-match end window (s); default 1.5 (keeps the
rally-end honest)")
307         ap.add_argument("--guard-weight-merge", type=float, default=None,
308             help="R1 over-seg guard: merge cost weight (default 2.0; merge hides
a rally)")
309         ap.add_argument("--guard-weight-split", type=float, default=None,
310             help="R1 over-seg guard: split cost weight (default 1.0; split only
fragments)")
311         ap.add_argument("--guard-counts", action="store_true",
312             help="R1 over-seg guard scores raw counts instead of normalized
rates "
313             "(rates are the default ? comparable across a structural
mega?bounded change)")
314         ap.add_argument("--json-out", default=None, help="optional path to write the full
result JSON")
315         args = ap.parse_args()
316
317         from dataclasses import replace
318         golden = load_golden(args.manifest, args.features_dir)
319         def _apply_inactivity(preset, end_val):
320             if end_val is not None:
321                 preset = replace(preset, inactivity_end=end_val)
322             if args.inactivity_rally_thresh is not None:
323                 preset = replace(preset,
inactivity_rally_thresh=args.inactivity_rally_thresh)
324             if args.inactivity_min_density is not None:
325                 preset = replace(preset, inactivity_min_density=args.inactivity_min_density)
326             return preset

```



```

327
328     base = _resolve_preset(args.preset)
329     if args.max_window_duration is not None:
330         base = replace(base, max_window_duration=args.max_window_duration)
331     if args.hard_cap:
332         base = replace(base, hard_cap=True)
333     if args.blob_fallback:
334         base = replace(base, blob_fallback=True)
335     if args.blob_factor is not None:
336         base = replace(base, blob_factor=args.blob_factor)
337     base = _apply_inactivity(base, args.inactivity_end)
338     if args.vs:
339         vs_mc = args.vs_min_crossings or args.min_crossings
340         cand = _resolve_preset(args.vs)
341         cand_cap = (args.vs_max_window_duration if args.vs_max_window_duration is not
None
342                     else args.max_window_duration)
343         if cand_cap is not None:
344             cand = replace(cand, max_window_duration=cand_cap)
345         if args.vs_hard_cap or args.hard_cap:
346             cand = replace(cand, hard_cap=True)
347         if args.vs_blob_fallback or args.blob_fallback:
348             cand = replace(cand, blob_fallback=True)
349         if args.blob_factor is not None:
350             cand = replace(cand, blob_factor=args.blob_factor)
351         cand = _apply_inactivity(cand, args.vs_inactivity_end if args.vs_inactivity_end
is not None
352                                else args.inactivity_end)
353         result = compare(golden, base, cand, args.iou,
354                         base_min_crossings=args.min_crossings,
cand_min_crossings=vs_mc,
355                         tau_start=args.tau_start, tau_end=args.tau_end,
356                         guard_weight_merge=args.guard_weight_merge,
357                         guard_weight_split=args.guard_weight_split,
358                         guard_use_rates=not args.guard_counts)
359         print(format_report(result["base"]))
360         print("\n" + format_report(result["cand"]))
361         d = result["delta"]
362         print(f"\n## ? ({args.vs} mc={vs_mc} ? {args.preset} mc={args.min_crossings}), "
363               f"n_paired={result['n_paired']}")
364         for k, dd in d.items():
365             st = dd["sign_test"]
366             print(f" ?{k}: {dd['mean_delta']:+.3f}
[{dd['ci_low']:+.3f},{dd['ci_high']:+.3f}] "
367               f"CI?0={dd['ci_excludes_zero']} sign(W/L)={st['wins']}/{st['losses']}
p={st['p_value']:.3f}")
368         print(_format_guard(result["over_seg_guard"]))
369     else:
370         result = evaluate(golden, base, args.iou, args.min_crossings,
371                         tau_start=args.tau_start, tau_end=args.tau_end)
372         print(format_report(result))
373     if args.json_out:
374         with open(args.json_out, "w", encoding="utf-8") as fh:
375             json.dump(result, fh, indent=2)
376         print(f"\n[written {args.json_out}]")
377
378
379     if __name__ == "__main__":
380         main()
381

```

9. user (2026-06-18T01:28:01.376Z)

```

1     """Regression runner ? verify the production tool against golden-set ground truth.
2

```

```

3     Implements the explicit ask: *"given a video, get a golden set created and verify
4     its precision and recall against the production tool."* This is deliberately a thin
5     wrapper over machinery that already exists:
6
7         ground truth ? AnnotationProvider (file / vendor / oracle) [backend.annotations]
8         predictions  ? the production pipeline's stored DB rows    [backend.eval.harness]
9         comparison   ? interval matching + P/R/F1                  [backend.eval.metrics]
10        baseline     ? a per-video snapshot of "known good" numbers [this module]
11
12    A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted
13    baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI),
14    later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an
15    independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.
16    ""
17    import os
18    import json
19    import hashlib
20    import logging
21    from dataclasses import dataclass, asdict
22    from typing import Dict, List, Optional
23
24    from backend.infrastructure.database import Database
25    from backend.eval import metrics
26    from backend.eval.harness import get_predictions
27    from backend.annotations import annotation_registry
28
29    logger = logging.getLogger(__name__)
30
31    # Tracked location (NOT under the gitignored output/), so a fresh checkout / CI run can
32    # actually load a committed baseline to compare against.
33    DEFAULT_BASELINE_PATH = os.path.join("eval_baselines", "regression_baseline.json")
34    DEFAULT_TOLERANCE = 0.05
35
36
37    def gt_hash(ground_truth: List[metrics.Interval]) -> str:
38        """Stable short hash of the GT intervals ? detects a baseline taken against
39        different labels."""
40        norm = sorted((round(float(s), 3), round(float(e), 3)) for s, e in ground_truth)
41        return hashlib.sha1(json.dumps(norm).encode()).hexdigest()[:12]
42
43    @dataclass
44    class RegressionResult:
45        video_id: str
46        stage: str
47        iou_threshold: float
48        precision: float
49        recall: float
50        f1: float
51        # Baseline values compared against (None when no baseline existed yet).
52        baseline_precision: Optional[float]
53        baseline_recall: Optional[float]
54        tolerance: float
55        regressed: bool
56        reasons: List[str]
57        # True only when a baseline existed to compare against. Distinguishes a genuine PASS
58        # from "nothing to compare" so the CLI can refuse to silently green-light the
59        latter.
60        has_baseline: bool = False
61        gt_hash: Optional[str] = None
62
63        def as_dict(self) -> dict:
64            return asdict(self)
65

```

```

66 # ----- baseline store
67
68 def load_baselines(path: str = DEFAULT_BASELINE_PATH) -> Dict[str, dict]:
69     """Load the per-video baseline snapshot file (returns {} if absent)."""
70     if not os.path.isfile(path):
71         return {}
72     with open(path) as f:
73         return json.load(f)
74
75
76 def _baseline_key(video_id: str, stage: str) -> str:
77     return f"{video_id}::{stage}"
78
79
80 def save_baseline(video_id: str, stage: str, precision: float, recall: float,
81                  f1: float, path: str = DEFAULT_BASELINE_PATH,
82                  iou_threshold: Optional[float] = None, detector: Optional[str] = None,
83                  gt_fingerprint: Optional[str] = None) -> None:
84     """Snapshot a video/stage's current numbers as the new 'known good' baseline.
85
86     Records the iou_threshold, detector, and a GT fingerprint alongside the metrics so a
87     later run computed under different settings (or against different labels) is
88     as incommensurable rather than silently compared.
89     """
90     baselines = load_baselines(path)
91     baselines[_baseline_key(video_id, stage)] = {
92         "video_id": video_id, "stage": stage,
93         "precision": precision, "recall": recall, "f1": f1,
94         "iou_threshold": iou_threshold, "detector": detector, "gt_hash": gt_fingerprint,
95     }
96     os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
97     with open(path, "w") as f:
98         json.dump(baselines, f, indent=2, sort_keys=True)
99     logger.info("Saved baseline for %s (precision=%.3f recall=%.3f)",
100 _baseline_key(video_id, stage), precision, recall)
101
102 # ----- the runner
103
104 def regress(db: Database, video_id: str, ground_truth: List[metrics.Interval],
105            stage: str = "final", iou_threshold: float = 0.5,
106            detector: Optional[str] = None,
107            tolerance: float = DEFAULT_TOLERANCE,
108            baseline_path: str = DEFAULT_BASELINE_PATH) -> RegressionResult:
109     """Score stored predictions for one video against ground truth and compare to
110     baseline.
111
112     `ground_truth` is a list of (start, end) intervals ? typically obtained from an
113     AnnotationProvider (see `regress_with_provider`). A regression is flagged when
114     precision OR recall falls more than `tolerance` below the snapshotted baseline.
115     If no baseline exists yet, `regressed` is False (nothing to compare to) and the
116     caller can choose to snapshot the current numbers via `save_baseline`.
117     """
118     preds = get_predictions(db, video_id, stage, detector=detector)
119     s = metrics.score(preds, ground_truth, iou_threshold)
120     fp = gt_hash(ground_truth)
121
122     baselines = load_baselines(baseline_path)
123     b = baselines.get(_baseline_key(video_id, stage))
124
125     reasons: List[str] = []
126     regressed = False
127     has_baseline = b is not None
128     base_p = base_r = None

```

```

128         if b is not None:
129             base_p, base_r = b["precision"], b["recall"]
130             # Incommensurable comparison guard: a baseline taken at a different IoU/detector
or
131             # against different labels is not a valid reference ? flag it instead of
trusting it.
132             b_iou, b_det, b_hash = b.get("iou_threshold"), b.get("detector"),
b.get("gt_hash")
133             if b_iou is not None and abs(float(b_iou) - iou_threshold) > 1e-9:
134                 regressed = True
135                 reasons.append(f"baseline IoU {b_iou} != run IoU {iou_threshold}
(incommensurable)")
136             if b_det is not None and b_det != detector:
137                 regressed = True
138                 reasons.append(f"baseline detector {b_det!r} != run detector {detector!r}
(incommensurable)")
139             if b_hash is not None and b_hash != fp:
140                 regressed = True
141                 reasons.append(f"baseline GT hash {b_hash} != run GT hash {fp} (labels
changed)")
142             if s.precision < base_p - tolerance:
143                 regressed = True
144                 reasons.append(f"precision {s.precision:.3f} < baseline {base_p:.3f} -
{tolerance}")
145             if s.recall < base_r - tolerance:
146                 regressed = True
147                 reasons.append(f"recall {s.recall:.3f} < baseline {base_r:.3f} -
{tolerance}")
148         else:
149             reasons.append("no baseline recorded for this video/stage ? nothing to compare")
150
151         return RegressionResult(
152             video_id=video_id, stage=stage, iou_threshold=iou_threshold,
153             precision=s.precision, recall=s.recall, f1=s.f1,
154             baseline_precision=base_p, baseline_recall=base_r,
155             tolerance=tolerance, regressed=regressed, reasons=reasons,
156             has_baseline=has_baseline, gt_hash=fp,
157         )
158
159
160     def regress_with_provider(db: Database, video_id: str, video_ref: str,
161                             tier: str = "file", stage: str = "final",
162                             iou_threshold: float = 0.5,
163                             detector: Optional[str] = None,
164                             tolerance: float = DEFAULT_TOLERANCE,
165                             baseline_path: str = DEFAULT_BASELINE_PATH,
166                             guideline: Optional[str] = None,
167                             sport: str = "badminton",
168                             provider_kwargs: Optional[dict] = None) -> RegressionResult:
169         """End-to-end: obtain ground truth from the `tier` annotation provider, then
regress.
170
171         `tier` is an annotation-provider registry key ("file" today; "vendor"/"auto"
once those providers land). `video_ref` is whatever that provider expects (a CSV
172         path for FileProvider, a video_id/URL for a vendor). `video_id` keys the stored
173         predictions in the DB (the file MD5).
174         """
175
176         provider = annotation_registry.get(tier, **(provider_kwargs or {}))
177         gt = provider.annotate(video_ref, guideline=guideline, sport=sport)
178         return regress(db, video_id, gt, stage=stage, iou_threshold=iou_threshold,
179                       detector=detector, tolerance=tolerance, baseline_path=baseline_path)
180
181
182     def format_result(r: RegressionResult) -> str:
183         verdict = "REGRESSED" if r.regressed else "ok"

```

```

184     head = (f"[{verdict}] {r.video_id} ({r.stage}) "
185             f"precision={r.precision:.3f} recall={r.recall:.3f} f1={r.f1:.3f}")
186     if r.baseline_precision is not None:
187         head += f" | baseline p={r.baseline_precision:.3f} r={r.baseline_recall:.3f}
tol={r.tolerance}"
188     return head + "".join(f"\n    - {why}" for why in r.reasons)
189

```

10. assistant (2026-06-18T01:28:04.305Z)

Now let me read the remaining files: experiment.py, calibration.py, per_user_gate.py, and the supporting metrics/segmentation_metrics.

11. user (2026-06-18T01:28:05.275Z)

```

1     """Quantify rally-segmentation improvement ? and guard against overfitting.
2
3     This is the measurement layer for per-video / per-corpus calibration. It answers
4     two questions honestly:
5
6     1. """Did precision improve, by how much?""" ? `improvement_report` scores a
7       baseline config vs a tuned config on the same ground truth and reports the
8       before/after metrics and the % delta.
9
10    2. """Is that gain real or overfit?""" ? calibration must be measured on data the
11       tuning never saw, otherwise you're just memorising. Two protocols:
12       - `cross_validate` ? k-fold over one video's rallies: pick the config on the
13         train folds, score on the held-out fold. The generalization_gap
14         (train ? test) is the overfit alarm.
15       - `leave_one_video_out` ? the protocol that matters as more videos arrive:
16         pick the config on all other videos, score on the held-out video. Reports
17         per-video held-out scores + mean±std. This is the number to trust.
18
19    Everything is segmenter-agnostic: callers pass a `predict_fn(config) ->
20    List[Interval]` (e.g. WASB trajectory ? windows at those thresholds), so the same
21    harness works for WASB, yolo_hybrid, or anything else. Pure + unit-tested; no I/O.
22    """
23    from __future__ import annotations
24
25    import statistics as _st
26    from typing import Callable, Dict, List, Sequence, Tuple, Any
27
28    from backend.eval.metrics import score, Score, Interval
29
30    Config = Any                                # opaque to this module (e.g. a thresholds
dict)
31    PredictFn = Callable[[Config], List[Interval]]
32
33
34    def pct_improvement(before: float, after: float) -> float:
35        """Percentage change from `before` to `after` (e.g. 0.6 -> 0.75 == +25.0)."""
36        if before <= 0:
37            return float("inf") if after > 0 else 0.0
38        return (after - before) / before * 100.0
39
40
41    def evaluate(preds: List[Interval], gts: List[Interval], iou_threshold: float = 0.5) ->
Score:
42        """Precision / recall / F1 / mean-IoU of predicted vs ground-truth intervals."""
43        return score(preds, gts, iou_threshold)
44
45
46    def kfold_indices(n: int, k: int) -> List[Tuple[List[int], List[int]]]:
47        """Deterministic k-fold split of indices 0..n-1 into (train_idx, test_idx)."""
48        if n <= 0:

```

```

49         return []
50         k = max(2, min(k, n))                # need >=2 folds; cap at n
51         folds = [list(range(i, n, k)) for i in range(k)] # round-robin ? balanced,
deterministic
52         splits = []
53         for f in range(k):
54             test = folds[f]
55             train = [i for i in range(n) if i not in set(test)]
56             if test and train:
57                 splits.append((train, test))
58         return splits
59
60
61     def select_best(predict_fn: PredictFn, configs: Sequence[Config], gts: List[Interval],
62                     metric: str = "f1", iou_threshold: float = 0.5) -> Tuple[Config, float]:
63         """Pick the config whose predictions maximise `metric` against `gts`."""
64         best_cfg, best_val = None, -1.0
65         for cfg in configs:
66             val = getattr(score(predict_fn(cfg), gts, iou_threshold), metric)
67             if val > best_val:
68                 best_cfg, best_val = cfg, val
69         return best_cfg, best_val
70
71
72     def cross_validate(predict_fn: PredictFn, configs: Sequence[Config], gts:
List[Interval],
73                       k: int = 5, metric: str = "recall", iou_threshold: float = 0.5) ->
Dict[str, Any]:
74         """k-fold threshold calibration on ONE video's rallies (within-video overfit guard).
75
76         Per fold: choose the best config on the train rallies, score it on the held-out
77         rallies. A large `generalization_gap` (train >> test) means the thresholds are
78         overfitting that video's quirks.
79
80         Default metric is **recall** on purpose: predictions are global (whole-video
81         windows), so scoring them against a *fold subset* of GT would unfairly count the
82         other rallies' windows as false positives ? precision/F1 are not well-defined
83         per-fold. Recall ("did the tuned thresholds still detect the held-out rallies?")
84         is the honest within-video generalization signal. For precision/F1, use
85         `leave_one_video_out` (each video scored whole) or `improvement_report` on a
86         held-out video.
87         """
88         splits = kfold_indices(len(gts), k)
89         if not splits:
90             return {"error": "not enough rallies for cross-validation", "n": len(gts)}
91         train_scores, test_scores, picked = [], [], []
92         for train_idx, test_idx in splits:
93             gts_tr = [gts[i] for i in train_idx]
94             gts_te = [gts[i] for i in test_idx]
95             cfg, tr = select_best(predict_fn, configs, gts_tr, metric, iou_threshold)
96             te = getattr(score(predict_fn(cfg), gts_te, iou_threshold), metric)
97             train_scores.append(tr)
98             test_scores.append(te)
99             picked.append(cfg)
100         mean_test = _st.mean(test_scores)
101         mean_train = _st.mean(train_scores)
102         return {
103             "metric": metric,
104             "folds": len(splits),
105             "mean_test": mean_test,
106             "std_test": _st.pstdev(test_scores) if len(test_scores) > 1 else 0.0,
107             "mean_train": mean_train,
108             "generalization_gap": mean_train - mean_test, # >~0.1 = likely overfitting
109             "picked_configs": picked,
110         }

```

```

111
112
113     def leave_one_video_out(videos: List[Dict[str, Any]], configs: Sequence[Config],
114                             metric: str = "f1", iou_threshold: float = 0.5) -> Dict[str,
Any]:
115         """Leave-one-video-out CV ? the honest metric as the corpus grows.
116
117         `videos`: list of {"name": str, "predict_fn": PredictFn, "gts": [Interval]}.
118         For each held-out video: pick the config that's best *pooled across the other
119         videos*, then score it on the held-out one. The mean held-out score is the
120         "precision on unseen videos" number; per-video spread shows stability.
121         """
122         if len(videos) < 2:
123             return {"error": "need >=2 videos for leave-one-video-out", "n": len(videos)}
124         per_video = []
125         for i, held in enumerate(videos):
126             others = [v for j, v in enumerate(videos) if j != i]
127
128             def pooled_score(cfg: Config, others: List[dict] = others) -> float:
129                 vals = [getattr(score(v["predict_fn"])(cfg), v["gts"], iou_threshold),
metric)
130                             for v in others]
131                 return _st.mean(vals)
132
133             best_cfg, best_train = max(((c, pooled_score(c)) for c in configs), key=lambda
t: t[1])
134             held_score = getattr(score(held["predict_fn"])(best_cfg), held["gts"],
iou_threshold), metric)
135             per_video.append({"video": held.get("name", f"video_{i}"),
136                             "held_out": held_score, "train_pooled": best_train, "config":
best_cfg})
137         held_scores = [p["held_out"] for p in per_video]
138         train_scores = [p["train_pooled"] for p in per_video]
139         return {
140             "metric": metric,
141             "mean_held_out": _st.mean(held_scores),
142             "std_held_out": _st.pstdev(held_scores) if len(held_scores) > 1 else 0.0,
143             "generalization_gap": _st.mean(train_scores) - _st.mean(held_scores),
144             "per_video": per_video,
145         }
146
147
148     def improvement_report(predict_fn: PredictFn, baseline_config: Config, tuned_config:
Config,
149                             gts: List[Interval], iou_threshold: float = 0.5) -> Dict[str,
Any]:
150         """The "+X%" precision headline: baseline vs tuned config on the same GT.
151
152         NB: for an *honest* headline, pass GT the tuning never saw (a held-out video),
153         or read the cross-validated `mean_test` from `cross_validate` instead.
154         """
155         b = score(predict_fn(baseline_config), gts, iou_threshold)
156         t = score(predict_fn(tuned_config), gts, iou_threshold)
157         return {
158             "baseline": {"precision": b.precision, "recall": b.recall, "f1": b.f1,
"mean_iou": b.mean_iou},
159             "tuned": {"precision": t.precision, "recall": t.recall, "f1": t.f1, "mean_iou":
t.mean_iou},
160             "precision_delta_pts": t.precision - b.precision,
161             "precision_delta_pct": pct_improvement(b.precision, t.precision),
162             "f1_delta_pct": pct_improvement(b.f1, t.f1),
163         }
164

```

```

1      """Per-user promotion gate ? the safety rail for flag-gated personalization.
2
3      The owner adopted personalization as a flag-gated feature on a generalization-
first
4      baseline: a cue/variant B may be flipped ON for one user only when that user's own
5      held-out evidence honestly clears the same small-N bar the global eval uses ? otherwise
we
6      fall back to the shipped baseline (variant A). This module is the decision logic for
that
7      flip; it is pure (no I/O) and reuses backend.eval.eval_stats for every statistic
8      (paired_delta_ci + paired_sign_test) so the math is never reinvented.
9
10     The counter-case this guards against (and its mitigations, implemented here):
11
12     1. Minimum-N floor. Personalizing off 1-2 of a user's videos is the small-N-lies
regime
13     (see eval_stats C1). Refuse to promote below min_n videos ? point estimates
from
14     too few held-out videos are noise, not signal.
15     2. Honest lift, not a single mean. Promote only when the paired ?F1 95% bootstrap CI
16     excludes 0 and the paired sign test agrees (majority of videos win,  $p \leq$ 
threshold).
17     This is the global promotion bar applied per user.
18     3. Structural guards are never disabled on "no lift". A structural guard (e.g.
Gate-A
19     net-crossings ? the held-shuttle false-positive killer) earns its keep through
failure
20     modes a given user's corpus may not contain yet. Measuring "no per-user lift" on a
corpus
21     that simply lacks those failure cases must NEVER turn the guard OFF. For
structural=True
22     the decision is therefore always "keep ON", independent of the measured delta.
23
24     Below the floor / insufficient evidence ? fall back to the shipped baseline (the
conservative
25     default), never an unsafe promotion. All additive, pure, deterministic (the bootstrap
seed is
26     threaded through). Nothing here changes existing eval behaviour; callers opt in.
27     """
28     from __future__ import annotations
29
30     from dataclasses import dataclass, field
31     from typing import Any, Dict, Sequence
32
33     from backend.eval.eval_stats import paired_delta_ci, paired_sign_test
34
35     __all__ = ["PromotionDecision", "should_promote_for_user"]
36
37
38     @dataclass
39     class PromotionDecision:
40         """Structured result of a per-user promotion decision.
41
42         - promote ? flip variant B ON for this user (only ever True for a non-
structural
43         cue that cleared the full bar).
44         - keep_on ? the cue/guard should be ON for this user. For a structural guard
this is
45         always True (it is never disabled on "no lift"); for a non-structural cue it
equals
46         promote.
47         - reason ? short human-readable explanation of the decision.
48         - the remaining fields surface the evidence (n, mean delta, CI bounds and the two
tests)
49         so a caller can log/audit exactly why a user was or wasn't personalized.

```



```

50     """
51
52     promote: bool
53     keep_on: bool
54     reason: str
55     n: int
56     mean_delta: float
57     ci_low: float
58     ci_high: float
59     ci_excludes_zero: bool
60     sign_test: Dict[str, float] = field(default_factory=dict)
61     floor_met: bool = False
62     structural_protected: bool = False
63
64     def as_dict(self) -> Dict[str, Any]:
65         """Plain-dict view (for JSON logging / telemetry)."""
66         return {
67             "promote": self.promote,
68             "keep_on": self.keep_on,
69             "reason": self.reason,
70             "n": self.n,
71             "mean_delta": self.mean_delta,
72             "ci_low": self.ci_low,
73             "ci_high": self.ci_high,
74             "ci_excludes_zero": self.ci_excludes_zero,
75             "sign_test": self.sign_test,
76             "floor_met": self.floor_met,
77             "structural_protected": self.structural_protected,
78         }
79
80
81     def should_promote_for_user(
82         a_scores: Sequence[float],
83         b_scores: Sequence[float],
84         *,
85         structural: bool = False,
86         min_n: int = 5,
87         p_threshold: float = 0.05,
88         confidence: float = 0.95,
89         n_boot: int = 2000,
90         seed: int = 0,
91         eps: float = 0.0,
92     ) -> PromotionDecision:
93         """Decide whether to flip variant B ON for ONE user from that user's held-out
94         scores.
95
96         ``a_scores`` / ``b_scores`` are the user's **per-video** held-out F1 (or any score)
97         for
98         variant A (control / shipped baseline) and variant B (with the cue), aligned by
99         video
100         (same order). The decision rules (faithful to the module docstring):
101
102         1. ``structural=True`` ? the guard is **never disabled on "no lift"**: return
103            ``keep_on=True, promote=False`` regardless of the measured per-user delta. Its
104            value is
105            in failure modes the user's corpus may not yet contain.
106
107         2. Otherwise promote **iff ALL** hold ? the minimum-N floor (``n >= min_n``), the
108            paired
109            ?F1 bootstrap CI **excludes 0**, and the paired sign test agrees (majority of
110            videos
111            win, ``p_value <= p_threshold``). Any miss ? fall back to the shipped baseline.
112
113            Statistics come straight from ``eval_stats`` (``paired_delta_ci`` /
114            ``paired_sign_test``);
115            deterministic given ``seed``. Returns a :class:`PromotionDecision`.

```

```

108     """
109     n = min(len(a_scores), len(b_scores))
110     floor_met = n >= min_n
111
112     # --- Evidence (reuse eval_stats; never reinvent the statistics)
-----
113     if n == 0:
114         # No paired videos at all ? nothing to test. Surface an empty-but-honest result.
115         sign_test: Dict[str, float] = paired_sign_test([], eps)
116         mean_delta, ci_low, ci_high, ci_excludes_zero = 0.0, 0.0, 0.0, False
117     else:
118         delta = paired_delta_ci(
119             a_scores, b_scores,
120             confidence=confidence, n_boot=n_boot, seed=seed, eps=eps,
121         )
122         mean_delta = float(delta["mean_delta"])
123         ci_low = float(delta["ci_low"])
124         ci_high = float(delta["ci_high"])
125         ci_excludes_zero = bool(delta["ci_excludes_zero"])
126         sign_test = delta["sign_test"]
127
128     sign_p = float(sign_test.get("p_value", 1.0))
129     sign_wins = float(sign_test.get("wins", 0.0))
130     sign_losses = float(sign_test.get("losses", 0.0))
131     # The sign test "agrees" with a promotion only if it is significant AND the majority
132     # leans the *winning* way (more B-wins than B-losses) ? a significant net *loss*
must
133     # never satisfy the promotion bar.
134     sign_agrees = (sign_p <= p_threshold) and (sign_wins > sign_losses)
135
136     # --- Rule 1: structural guards are never disabled on "no lift"
-----
137     if structural:
138         return PromotionDecision(
139             promote=False,          # nothing to "promote" ? it ships ON by design
140             keep_on=True,          # ALWAYS kept ON, regardless of measured per-user
lift
141             reason=(
142                 "structural guard kept ON regardless of measured per-user lift "
143                 "(value is in failure modes this user's corpus may not yet contain)"
144             ),
145             n=n,
146             mean_delta=mean_delta,
147             ci_low=ci_low,
148             ci_high=ci_high,
149             ci_excludes_zero=ci_excludes_zero,
150             sign_test=sign_test,
151             floor_met=floor_met,
152             structural_protected=True,
153         )
154
155     # --- Rule 2: minimum-N floor (refuse before even looking at lift)
-----
156     if not floor_met:
157         return PromotionDecision(
158             promote=False,
159             keep_on=False,          # fall back to the shipped baseline (variant A)
160             reason=(
161                 f"below minimum-N floor (n={n} < min_n={min_n}); "
162                 "falling back to shipped baseline"
163             ),
164             n=n,
165             mean_delta=mean_delta,
166             ci_low=ci_low,
167             ci_high=ci_high,

```

```

168         ci_excludes_zero=ci_excludes_zero,
169         sign_test=sign_test,
170         floor_met=False,
171         structural_protected=False,
172     )
173
174     # --- Rule 2 (cont.): honest lift ? CI excludes 0 AND sign test agrees
175     -----
176     if ci_excludes_zero and sign_agrees:
177         promote = True
178         reason = (
179             f"promoted: n={n} >= {min_n}, ?F1 95% CI "
180             f"[{ci_low:.4f}, {ci_high:.4f}] excludes 0, "
181             f"sign test {int(sign_wins)}-{int(sign_losses)} (p={sign_p:.4f} <=
182             {p_threshold})"
183         )
184     else:
185         promote = False
186         misses = []
187         if not ci_excludes_zero:
188             misses.append(f"?F1 95% CI [{ci_low:.4f}, {ci_high:.4f}] spans 0")
189         if not sign_agrees:
190             misses.append(
191                 f"sign test does not agree ({int(sign_wins)}-{int(sign_losses)}), "
192                 f"p={sign_p:.4f}"
193             )
194         reason = (
195             "insufficient evidence (" + "; ".join(misses) + "); "
196             "falling back to shipped baseline"
197         )
198
199     return PromotionDecision(
200         promote=promote,
201         keep_on=promote,          # non-structural: keep_on tracks promote exactly
202         reason=reason,
203         n=n,
204         mean_delta=mean_delta,
205         ci_low=ci_low,
206         ci_high=ci_high,
207         ci_excludes_zero=ci_excludes_zero,
208         sign_test=sign_test,
209         floor_met=True,
210         structural_protected=False,
211     )

```

13. assistant (2026-06-18T01:28:07.607Z)

Now let me read experiment.py and the supporting metrics/segmentation_metrics files.

14. user (2026-06-18T01:28:09.084Z)

```

1     """Experiment harness ? A/B + shadow + side-by-side for rally detection.
2
3     This is the thin **variant/experiment layer** that `docs/EXPERIMENT_HARNESS.md`
4     identifies as the one missing piece: ~80% of the machinery (scoring, LOVO, the
5     no-regression gate, the pinnable classifier artifact, the persisted-candidate
6     substrate) already exists ? this module composes it. **No new scoring math lives
7     here**; everything defers to:
8
9     - `backend.eval.metrics`      ? `score`, `match_intervals`, `interval_iou`
10    - `backend.eval.training`      ? `label_candidates` (y by the same IoU matcher)
11    - `backend.eval.classifier`    ? `LogisticRegression` (the pinnable JSON model)
12    - `backend.eval.gemini_refine` ? `merge_overlaps` (the blessed window merge)
13    - `backend.eval.regression`    ? `gt_hash` (label-set fingerprint)

```

```

14     - `backend.eval.calibration`      ? `pct_improvement`
15
16 The thesis (`EXPERIMENT_HARNESS.md` §2): separate the expensive, risky DETECTION
17 (per-frame signals ? run once on the GPU/WSL box, persisted into
18 `candidate_segments.stats`) from the cheap, swappable DECISION (given those
19 signals, where are the rallies). A `Variant` is a declarative re-scoring recipe;
20 given persisted candidate features it produces `List[Interval]` deterministically,
21 with no GPU and zero production risk. So most experiments are pure offline
22 re-scoring of stored data and never touch the served pipeline.
23
24 Three modes (`EXPERIMENT_HARNESS.md` §5):
25     - `compare()`      ? labeled A/B vs golden labels: per-video + LOVO-honest
26       aggregate deltas. Mirrors `distill_local`'s honest train-on-others/predict-
27       held-out loop, but variant-parameterised.
28     - `compare_unlabeled()` ? SxS on NEW footage with no ground truth: where do two
29       variants agree / diverge (A-only, B-only, agreed). The divergences are what a
30       human spot-checks (and what two highlight reels would show side by side).
31     - the promotion gate stays `run_regression.py` + `regression_baseline.json`
32       (exit 0/1/3); rollback = flip the variant/config, never `git revert`.
33
34 Pure + offline + unit-tested. The only DB-touching helper is
35 `load_video_candidates`, which reads persisted candidates via
36 `Database.query_candidate_segments`.
37 """
38 from __future__ import annotations
39
40 import json
41 import logging
42 import os
43 from dataclasses import dataclass, field
44 from datetime import datetime, timezone
45 from hashlib import sha1
46 from typing import Any, Callable, Dict, List, Optional, Sequence
47
48 from backend.eval.calibration import pct_improvement
49 from backend.eval.classifier import LogisticRegression
50 from backend.eval.gemini_refine import merge_overlaps
51 from backend.eval.metrics import Interval, match_intervals, score
52 from backend.eval.regression import gt_hash
53 from backend.eval.training import label_candidates
54 from backend.eval import eval_stats as _stats      # C1: CIs + paired sign test
55 from backend.eval import segmentation_metrics as _segm # C4: F1@k + segment-count
56 ratio
57
58 from backend.eval.score_calibration import fit_isotonic, fit_platt # C2: calibrate cue
59 scores
60
61 logger = logging.getLogger(__name__)
62
63 # Where a comparison run appends one reproducible record. Tracked location (NOT under
64 # the gitignored output/) so the leaderboard survives a clean checkout, mirroring
65 # regression.DEFAULT_BASELINE_PATH.
66 DEFAULT_LEADERBOARD_PATH = os.path.join("eval_baselines", "experiment_leaderboard.json")
67 _METRICS = ("precision", "recall", "f1", "mean_iou")
68
69
70
71
72
73
74
75
76
77
78
79
80
81
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84
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86
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88
89
90
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92
93
94
95
96
97
98
99
100

```

```

77     class Variant:
78         """A declarative re-scoring recipe ? NOT a code branch (`EXPERIMENT_HARNESS.md` §4).
79
80         A variant turns one video's persisted candidate windows + features into rally
81         intervals. Every knob defaults to the control behaviour (no gate, no learned
82         filter), so a variant that merely carries a new, disabled cue is byte-identical
83         to control ? the core no-regression guarantee.
84
85         Fields:
86         - ``segmenter`` / ``config_overrides`` / ``cue_flags`` ? informational provenance
87           for the leaderboard and for selecting the served path (the existing
88           `segmenter_registry` + `IndexingConfig` do the actual selection in production).
89           New cues are recorded here default-OFF.
90         - ``min_crossings`` ? decision-gate Gate A: keep only windows whose persisted
91           ``net_crossings`` feature is >= this. 0 = off. This is the eval-only gate from
92           `rally_gate.py` expressed as a re-scoring knob (kills service-feed / held-
shuttle
93           windows ? see `MULTI_SIGNAL_FUSION_PLAN.md` §3.1).
94         - ``min_players`` ? decision-gate Gate B (the future person cue): keep windows
95           whose persisted ``players_in_court`` is >= this. 0 = off (no-op until the person
96           cue persists that feature).
97         - ``classifier_model`` ? path to a frozen `LogisticRegression` JSON. When set,
98           `compare()` scores videos with the SHIPPED model as-is (no per-fold training);
99           required for `compare_unlabeled` on a learned variant.
100        - ``feature_names`` ? the persisted features the learned filter consumes. When set
101          WITHOUT ``classifier_model``, `compare()` trains a fresh model per LOVO fold
102          (honest) ? the recipe, not a pinned artifact.
103        - ``threshold`` ? classifier probability cutoff. ``merge_gap`` ? post-filter
104          overlap-merge gap (seconds), the same `gemini_refine.merge_overlaps` default.
105        """
106
107        name: str
108        segmenter: str = "wasb_hybrid"
109        config_overrides: Dict[str, Any] = field(default_factory=dict)
110        cue_flags: Dict[str, bool] = field(default_factory=dict)
111        min_crossings: int = 0
112        min_players: int = 0
113        classifier_model: Optional[str] = None
114        feature_names: Optional[List[str]] = None
115        threshold: float = 0.5
116        merge_gap: float = 1.0
117        # C2 / threshold-in-LOVO (default OFF ? unchanged behaviour). When set, the
operating
118        # point is chosen on the TRAINING fold, never the held-out one ? fixing the
first-A/B
119        # failure where a fixed 0.5 zeroed out a small-candidate video.
120        tune_threshold: bool = False          # pick the F1-best threshold on the train fold
121        calibrate: Optional[str] = None        # None | "platt" | "isotonic" ? calibrate
proba first
122
123        def is_learned(self) -> bool:
124            """True if this variant applies a learned classifier (pinned model or
recipe)."""
125            return self.classifier_model is not None or bool(self.feature_names)
126
127        def to_dict(self) -> dict:
128            return {
129                "name": self.name,
130                "segmenter": self.segmenter,
131                "config_overrides": self.config_overrides,
132                "cue_flags": self.cue_flags,
133                "min_crossings": self.min_crossings,
134                "min_players": self.min_players,
135                "classifier_model": self.classifier_model,
136                "feature_names": self.feature_names,

```

```

137         "threshold": self.threshold,
138         "merge_gap": self.merge_gap,
139         "tune_threshold": self.tune_threshold,
140         "calibrate": self.calibrate,
141     }
142
143     @classmethod
144     def from_dict(cls, d: dict) -> "Variant":
145         known = {f for f in cls.__dataclass_fields__} # type: ignore[attr-defined]
146         return cls(**{k: v for k, v in d.items() if k in known})
147
148     def spec_hash(self) -> str:
149         """Stable 12-char hash of the recipe ? identifies the variant in the leaderboard
150         and makes a result reproducible. The pinned **control** variant always hashes
151         the
152         same, so a regression is always traceable to a recipe change, never to drift."""
153         blob = json.dumps(self.to_dict(), sort_keys=True, default=str)
154         return sha1(blob.encode()).hexdigest()[:12]
155
156 #: The pinned, frozen baseline: candidates as detected, merged, no gate, no learned
157 #: filter. Always the comparison reference; "rollback" = make this the served variant.
158 CONTROL = Variant(name="control")
159
160
161 # ----- persisted candidates
162
163
164 @dataclass
165 class VideoCandidates:
166     """One video's persisted detection, ready for offline re-scoring.
167
168     ``intervals`` are the candidate windows; ``features`` is column-major
169     (``feature_name -> per-candidate value``, each list aligned to ``intervals``);
170     ``gts`` are golden labels (None for new/unlabeled footage). Build one from the DB
171     with ``load_video_candidates``, or directly in tests.
172     """
173
174     name: str
175     intervals: List[Interval]
176     features: Dict[str, List[float]] = field(default_factory=dict)
177     gts: Optional[List[Interval]] = None
178
179
180     def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
181         """Persisted feature column, defaulting missing/short columns to 0.0 (matches
182         ``training.extract_yolo_features``, which lets a sparse/old row still vectorise)."""
183         col = vc.features.get(name)
184         n = len(vc.intervals)
185         if col is None:
186             logger.warning("variant references feature %r absent from %s ? treating as 0.0",
187                             name, vc.name)
188             return [0.0] * n
189         if len(col) != n:
190             raise ExperimentError(
191                 f"feature {name!r} has {len(col)} values but {vc.name} has {n} candidates")
192         return [float(x) for x in col]
193
194
195     def _feature_matrix(vc: VideoCandidates, feature_names: Sequence[str]) ->
196     List[List[float]]:
197         cols = [_feature_column(vc, name) for name in feature_names]
198         return [list(row) for row in zip(*cols)] if cols else [[] for _ in vc.intervals]
199

```

```

200     def _gate_mask(variant: Variant, vc: VideoCandidates) -> List[bool]:
201         """Boolean keep-mask from the decision gates (Gate A net-crossings, Gate B
202         players)."""
203         n = len(vc.intervals)
204         mask = [True] * n
205         if variant.min_crossings > 0:
206             col = _feature_column(vc, "net_crossings")
207             mask = [m and (col[i] >= variant.min_crossings) for i, m in enumerate(mask)]
208         if variant.min_players > 0:
209             col = _feature_column(vc, "players_in_court")
210             mask = [m and (col[i] >= variant.min_players) for i, m in enumerate(mask)]
211         return mask
212
213     def predict(variant: Variant, vc: VideoCandidates,
214                 model: Optional[LogisticRegression] = None,
215                 threshold: Optional[float] = None,
216                 calibrator: Optional[Callable[[float], float]] = None) -> List[Interval]:
217         """Variant prediction: gate by persisted features, optionally filter by a learned
218         model, then merge. Pure and deterministic.
219
220         ``model`` (an already-fitted `LogisticRegression`) is supplied by `compare` per LOVO
221         fold; if omitted and the variant pins ``classifier_model``, that frozen model is
222         loaded. A learned variant with neither a supplied nor a pinned model raises ? there
223         is nothing to score with. ``threshold``/``calibrator`` (both fitted on the TRAINING
224         fold) override the variant defaults when the C2 / threshold-in-LOVO path supplies
225         them.
226
227         """
228         mask = _gate_mask(variant, vc)
229         if variant.feature_names:
230             if model is None:
231                 if variant.classifier_model is None:
232                     raise ExperimentError(
233                         f"variant {variant.name!r} is learned but has no model to predict "
234                         f"with (pass a fitted model or set classifier_model)")
235                 model = LogisticRegression.load(variant.classifier_model)
236             proba = [float(p) for p in model.predict_proba(_feature_matrix(vc,
237 variant.feature_names))]
238             if calibrator is not None:
239                 proba = [calibrator(p) for p in proba]
240             thr = variant.threshold if threshold is None else threshold
241             mask = [m and (proba[i] >= thr) for i, m in enumerate(mask)]
242             kept = [iv for iv, keep in zip(vc.intervals, mask) if keep]
243             return merge_overlaps(kept, gap_tol=variant.merge_gap)
244
245     def _calibrate_and_tune(variant: Variant, model: LogisticRegression,
246                             train_videos: Sequence[VideoCandidates], iou: float
247                             ) -> "tuple[Optional[Callable[[float], float]],
248 Optional[float]]":
249         """Fit a calibrator and/or pick the operating threshold on the TRAINING fold only
250         (C2 + threshold-in-LOVO). Returns ``(calibrator, threshold)`` ? either may be None
251         (use the variant default). Honest: never looks at the held-out video.
252
253         """
254         calibrator: Optional[Callable[[float], float]] = None
255         if variant.calibrate:
256             raw: List[float] = []
257             y: List[int] = []
258             for vc in train_videos:
259                 raw.extend(float(p) for p in
260                             model.predict_proba(_feature_matrix(vc, variant.feature_names or
261 [])))
262                 y.extend(label_candidates(vc.intervals, vc.gts or [], iou))
263             if variant.calibrate == "platt":
264                 calibrator = fit_platt(raw, y)

```

```

260         elif variant.calibrate == "isotonic":
261             calibrator = fit_isotonic(raw, y)
262         else:
263             raise ExperimentError(f"unknown calibrate={variant.calibrate!r} (use
'platt'/'isotonic')")
264         threshold: Optional[float] = None
265         if variant.tune_threshold:
266             best_t, best_f1 = variant.threshold, -1.0
267             for k in range(1, 20): # grid 0.05..0.95 on the train
fold's rally F1
268                 t = round(0.05 * k, 2)
269                 f1s = [score(predict(variant, vc, model=model, threshold=t,
calibrator=calibrator),
270                             vc.gts or [], iou).f1 for vc in train_videos]
271                 mean_f1 = sum(f1s) / len(f1s) if f1s else 0.0
272                 if mean_f1 > best_f1:
273                     best_f1, best_t = mean_f1, t
274             threshold = best_t
275         return calibrator, threshold
276
277
278     # ----- variant scoring
279
280
281     def _train(variant: Variant, videos: Sequence[VideoCandidates], iou: float) ->
LogisticRegression:
282         """Fit the variant's classifier on the pooled candidates of ``videos`` (labels via
the
283         evaluator's own IoU matcher). The honest-LOVO training step from `distill_local`. """
284         X: List[List[float]] = []
285         y: List[int] = []
286         for vc in videos:
287             if vc.gts is None:
288                 raise ExperimentError(f"cannot train: {vc.name} has no golden labels")
289             X.extend(_feature_matrix(vc, variant.feature_names or []))
290             y.extend(label_candidates(vc.intervals, vc.gts, iou))
291         if not X:
292             raise ExperimentError("cannot train a learned variant on zero candidates")
293         return LogisticRegression().fit(X, y, variant.feature_names)
294
295
296     def evaluate_variant(videos: Sequence[VideoCandidates], variant: Variant,
297                         iou: float = 0.5, honest: bool = True) -> Dict[str, Any]:
298         """Score a variant on a labeled corpus. Returns per-video Scores + predictions + a
299         mean aggregate.
300
301         Prediction policy:
302         - **static variant** (no learned filter): predict each video directly ? no
training,
303         so no leakage; per-video scoring is already honest.
304         - **learned, pinned model** (``classifier_model`` set): score every video with the
SHIPPED model. ? optimistic if those videos were in its training set ? use this
305         to
306         report "how the shipped artifact does here", not for the promotion decision.
307         - **learned recipe** (``feature_names``, no pinned model) + ``honest=True``: LOVO
?
308         train on the OTHER videos, predict the held-out one. The number to trust.
309         - learned recipe + ``honest=False``: train on all, predict each (overfit; sanity
only).
310         """
311         for vc in videos:
312             if vc.gts is None:
313                 raise ExperimentError(f"{vc.name} has no golden labels; use
compare_unlabeled")
314

```



```

315     per_video: List[Dict[str, Any]] = []
316     predictions: Dict[str, List[Interval]] = {}
317
318     recipe_lovo = bool(variant.feature_names) and variant.classifier_model is None
319     pinned_model = (LogisticRegression.load(variant.classifier_model)
320                     if variant.classifier_model is not None else None)
321     all_model = None
322     if recipe_lovo and not honest:
323         all_model = _train(variant, videos, iou)
324
325     for i, vc in enumerate(videos):
326         cal: Optional[Callable[[float], float]] = None
327         thr: Optional[float] = None
328         if recipe_lovo and honest:
329             others = [v for j, v in enumerate(videos) if j != i]
330             if not others:
331                 raise ExperimentError("LOVO needs >=2 videos; pass honest=False or a
pinned model")
332             model: Optional[LogisticRegression] = _train(variant, others, iou)
333             if variant.tune_threshold or variant.calibrate: # operating point from
the train fold
334                 assert model is not None # _train never returns
None
335                 cal, thr = _calibrate_and_tune(variant, model, others, iou)
336             elif recipe_lovo: # honest=False ? one model over all
videos
337                 model = all_model
338             else: # static variant or pinned model
339                 model = pinned_model
340             preds = predict(variant, vc, model=model, threshold=thr, calibrator=cal)
341             assert vc.gts is not None # guarded at function top
342             sc = score(preds, vc.gts, iou)
343             per_video.append({"video": vc.name, "num_pred": len(preds), **{m: getattr(sc, m)
for m in _METRICS}})
344             predictions[vc.name] = preds
345
346         aggregate = {m: (sum(p[m] for p in per_video) / len(per_video) if per_video else
0.0)
347                     for m in _METRICS}
348     return {
349         "variant": variant.name,
350         "spec_hash": variant.spec_hash(),
351         "is_learned": variant.is_learned(),
352         "honest_lovo": recipe_lovo and honest,
353         "per_video": per_video,
354         "aggregate": aggregate,
355         "predictions": predictions,
356     }
357
358
359     def _ci_block(eval_v: Dict[str, Any]) -> Dict[str, Any]:
360         """Per-metric mean + bootstrap CI over the per-video scores (C1 ? never a bare
mean)."""
361         return {m: _stats.mean_with_ci([p[m] for p in eval_v["per_video"]]) for m in
_METRICS}
362
363
364     def _segmentation_block(videos: Sequence[VideoCandidates], eval_v: Dict[str, Any]) ->
Dict[str, Any]:
365         """Mean F1@k + segment-count ratio across videos (C4 ? the over-segmentation tIoU
hides)."""
366         gts_by_name = {v.name: (v.gts or []) for v in videos}
367         overlaps = _segm.DEFAULT_OVERLAPS
368         f1k: Dict[float, List[float]] = {ov: [] for ov in overlaps}
369         ratios: List[float] = []

```

```

370     for name, preds in eval_v.get("predictions", {}).items():
371         gts = gts_by_name.get(name, [])
372         got = _segm.f1_at_overlaps(preds, gts, overlaps)
373         for ov in overlaps:
374             f1k[ov].append(got[ov])
375         ratios.append(_segm.segment_count_ratio(preds, gts))
376
377     def _mean(xs: List[float]) -> float:
378         return sum(xs) / len(xs) if xs else 0.0
379     out: Dict[str, Any] = {f"f1@{int(ov * 100)}": _mean(vals) for ov, vals in
380 f1k.items()}
381     out["segment_count_ratio"] = _mean(ratios)
382     return out
383
384     def honest_summary(videos: Sequence[VideoCandidates], eval_a: Dict[str, Any],
385                       eval_b: Dict[str, Any]) -> Dict[str, Any]:
386         """C1 + C4 honest reporting layered over a compare() pair (additive,
387 EXPERIMENT_HARNESS §6).
388         ``per_video`` lists are video-aligned (``evaluate_variant`` iterates ``videos`` in
389 order),
390 so ``paired_delta_f1`` pairs B?A by video ? the promotion-grade view: a lift is real
391 when
392 the ?F1 CI clears 0 *and* the sign test agrees, not when a bare mean ticked up.
393 """
394 a_f1 = [p["f1"] for p in eval_a["per_video"]]
395 b_f1 = [p["f1"] for p in eval_b["per_video"]]
396 return {
397     "variant_a_ci": _ci_block(eval_a),
398     "variant_b_ci": _ci_block(eval_b),
399     "paired_delta_f1": _stats.paired_delta_ci(a_f1, b_f1),
400     "segmentation": {"variant_a": _segmentation_block(videos, eval_a),
401                      "variant_b": _segmentation_block(videos, eval_b)},
402 }
403
404     def compare(videos: Sequence[VideoCandidates], variant_a: Variant, variant_b: Variant,
405               iou: float = 0.5, honest: bool = True, honest_stats: bool = True) ->
406 Dict[str, Any]:
407         """Offline labeled A/B (EXPERIMENT_HARNESS.md §5.1). Scores both variants on the
408 same
409 golden corpus and reports the **B ? A** deltas, per video and in aggregate.
410
411 The deltas use the existing `metrics.score` (per video) +
412 `calibration.pct_improvement`;
413 learned variants are scored LOVO-honestly. The result is a self-contained record fit
414 for
415 `record_experiment` ? variant specs + hashes + the label-set fingerprint travel with
416 it.
417
418 ``honest_stats`` (default True) **adds** a ``"honest"`` block ? per-metric bootstrap
419 CIs, a
420 paired ?F1 CI + exact sign test (C1), and F1@k + segment-count ratio (C4) ?
421 *alongside* the
422 existing bare-mean fields (additive: nothing existing changes). Set False for the
423 legacy
424 shape. This is the measurement-honesty wiring (ANALYZERS_AND_RECONCILERS.md
425 §13/C1+C4): a
426 bare LOVO mean at n?6 is not trustworthy on its own.
427 """
428 if len(videos) < 1:
429     raise ExperimentError("compare needs at least one labeled video")
430 eval_a = evaluate_variant(videos, variant_a, iou, honest)
431 eval_b = evaluate_variant(videos, variant_b, iou, honest)

```

```

422
423     delta = {m: eval_b["aggregate"][m] - eval_a["aggregate"][m] for m in _METRICS}
424     delta_pct = {m: pct_improvement(eval_a["aggregate"][m], eval_b["aggregate"][m]) for
m in _METRICS}
425
426     by_video_a = {p["video"]: p for p in eval_a["per_video"]}
427     per_video_delta = [
428         {"video": p["video"], **{m: p[m] - by_video_a[p["video"]][m] for m in _METRICS}}
429         for p in eval_b["per_video"]
430     ]
431
432     # One fingerprint over the whole label set ? detects "the deltas were measured
against
433     # different labels" the same way regression.gt_hash guards the no-regression
baseline.
434     all_gts: List[Interval] = [iv for vc in videos for iv in (vc.gts or [])]
435
436     result: Dict[str, Any] = {
437         "iou": iou,
438         "label_set_hash": gt_hash(all_gts),
439         "num_videos": len(videos),
440         "variant_a": eval_a,
441         "variant_b": eval_b,
442         "delta": delta,
443         "delta_pct": delta_pct,
444         "per_video_delta": per_video_delta,
445     }
446     if honest_stats:
447         result["honest"] = honest_summary(videos, eval_a, eval_b)
448     return result
449
450
451     # ----- SxS on unlabeled video
452
453
454     def compare_unlabeled(video: VideoCandidates, variant_a: Variant, variant_b: Variant,
455         agree_iou: float = 0.5) -> Dict[str, Any]:
456         """Side-by-side on NEW footage with no ground truth (`EXPERIMENT_HARNESS.md` §5.2).
457
458         With no labels there is no lift to measure ? instead this surfaces where the two
459         variants **agree vs diverge**, which is exactly what a human reviews (and what two
460         highlight reels would show side by side):
461
462         - ``agreed`` ? windows both variants found (matched at ``agree_iou``);
463         - ``a_only`` ? A fired, B suppressed (B's added precision, if A was over-firing);
464         - ``b_only`` ? B fired, A did not (B's new detections ? real recall or new FPs:
465           the human / a later golden label adjudicates).
466
467         Both variants must be predictable without labels: a learned variant therefore needs
a
468         pinned ``classifier_model`` (you cannot train on an unlabeled video). Reuses
469         ``metrics.match_intervals`` ? no new matching logic.
470         """
471         preds_a = predict(variant_a, video)
472         preds_b = predict(variant_b, video)
473         mr = match_intervals(preds_a, preds_b, agree_iou)
474
475         agreed = [{"a": preds_a[m.pred_index], "b": preds_b[m.gt_index], "iou": m.iou}
476                 for m in mr.matches]
477         agree_iou = [m.iou for m in mr.matches]
478         a_only = [preds_a[i] for i in mr.unmatched_pred_indices]
479         b_only = [preds_b[i] for i in mr.unmatched_gt_indices]
480
481     def _dur(ivs: List[Interval]) -> float:
482         return sum(e - s for s, e in ivs)

```

```

483
484     return {
485         "video": video.name,
486         "agree_iou": agree_iou,
487         "variant_a": variant_a.name,
488         "variant_b": variant_b.name,
489         "num_a": len(preds_a),
490         "num_b": len(preds_b),
491         "num_agreed": len(agreed),
492         "mean_agree_iou": (sum(agree_ious) / len(agree_ious)) if agree_ious else 0.0,
493         "duration_a": _dur(preds_a),
494         "duration_b": _dur(preds_b),
495         "agreed": agreed,
496         "a_only": a_only,
497         "b_only": b_only,
498     }
499
500
501 # ----- leaderboard / DB
502
503
504 def record_experiment(result: Dict[str, Any], path: str = DEFAULT_LEADERBOARD_PATH,
505                      timestamp: Optional[str] = None) -> Dict[str, Any]:
506     """Append a compact, reproducible record of a `compare()` result to the JSON
507     leaderboard (`EXPERIMENT_HARNESS.md` §6: experiments are records). Generalises
508     `regression_baseline.json`; deliberately the size of a baseline file, not a service
509     (§9). Returns the row written. ``timestamp`` is injectable for deterministic tests.
510     """
511     row: Dict[str, Any] = {
512         "timestamp": timestamp or datetime.now(timezone.utc).isoformat(),
513         "iou": result.get("iou"),
514         "label_set_hash": result.get("label_set_hash"),
515         "num_videos": result.get("num_videos"),
516         "variant_a": {"name": result["variant_a"]["variant"],
517                      "spec_hash": result["variant_a"]["spec_hash"],
518                      "aggregate": result["variant_a"]["aggregate"]},
519         "variant_b": {"name": result["variant_b"]["variant"],
520                      "spec_hash": result["variant_b"]["spec_hash"],
521                      "aggregate": result["variant_b"]["aggregate"]},
522         "delta": result.get("delta"),
523     }
524     # Record the honest ?F1 (CI + sign test) when present, so the leaderboard carries
525     the
526     # promotion-grade evidence, not just the bare-mean delta (additive ? C1).
527     honest = result.get("honest") or {}
528     if honest.get("paired_delta_f1"):
529         row["honest_delta_f1"] = honest["paired_delta_f1"]
530     entries: List[Dict[str, Any]] = []
531     if os.path.isfile(path):
532         with open(path, encoding="utf-8") as f:
533             entries = json.load(f)
534     entries.append(row)
535     os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
536     with open(path, "w", encoding="utf-8") as f:
537         json.dump(entries, f, indent=2)
538     logger.info("recorded experiment %s vs %s (?f1=%.3f) -> %s",
539               row["variant_a"]["name"], row["variant_b"]["name"],
540               (result.get("delta") or {}).get("f1", 0.0), path)
541     return row
542
543 def load_video_candidates(db: Any, video_id: str, *, detector: Optional[str] = None,
544                          gts: Optional[List[Interval]] = None,
545                          feature_names: Sequence[str] = ()) -> VideoCandidates:
546     """Bridge persisted detection ? `VideoCandidates` via

```

```

`Database.query_candidate_segments`.
547
548     ``feature_names`` selects which persisted features to lift out of each row's
``stats``
549     JSON (``"duration"`` is derived from the window bounds). This is the only DB-
touching
550     function; the comparison itself is pure and offline. Exercised owner-side on a real
DB,
551     but works against any `Database` (including an in-memory one in tests).
552     """
553     rows = db.query_candidate_segments(video_id, detector=detector)
554     intervals: List[Interval] = [(float(r["start_time"]), float(r["end_time"])) for r in
rows]
555     features: Dict[str, List[float]] = {}
556     for name in feature_names:
557         if name == "duration":
558             features[name] = [float(r["end_time"]) - float(r["start_time"]) for r in
rows]
559         else:
560             features[name] = [float((r.get("stats") or {}).get(name, 0.0)) for r in
rows]
561     return VideoCandidates(name=video_id, intervals=intervals, features=features,
gts=gts)
562

```

15. assistant (2026-06-18T01:28:12.754Z)

Now let me look at the supporting modules referenced by these ? metrics.py and segmentation_metrics.py ? since the scoring depends on them, and to validate the `\_hungarian\_match\_1to1` logic I want to check the test expectations. Let me read metrics.py and segmentation_metrics.py.

16. user (2026-06-18T01:28:13.718Z)

```

1     """Pure temporal interval-matching metrics for rally evaluation.
2
3     No I/O, no DB, no global state ? every function here is deterministic and unit
4     tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5     start < end. We score a list of *predicted* rally intervals against a list of
6     *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7     """
8
9     from dataclasses import dataclass, field
10    from typing import List, Tuple, Optional
11
12    Interval = Tuple[float, float]
13
14
15    def interval_iou(a: Interval, b: Interval) -> float:
16        """Temporal intersection-over-union of two intervals.
17
18        Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19        overlap / union, both measured as durations on the timeline.
20        """
21        a_start, a_end = a
22        b_start, b_end = b
23        if a_end <= a_start or b_end <= b_start:
24            return 0.0
25        inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26        if inter <= 0.0:
27            return 0.0
28        union = (a_end - a_start) + (b_end - b_start) - inter
29        return inter / union if union > 0 else 0.0
30
31

```

```

32 @dataclass
33 class Match:
34     """One matched predicted?ground-truth pair."""
35     pred_index: int
36     gt_index: int
37     iou: float
38
39
40 @dataclass
41 class MatchResult:
42     """Outcome of matching predictions to ground truth at a given IoU threshold."""
43     matches: List[Match] = field(default_factory=list)
44     unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45     unmatched_gt_indices: List[int] = field(default_factory=list)   # false negatives
(misses)
46
47
48 def match_intervals(preds: List[Interval], gts: List[Interval],
49                     iou_threshold: float = 0.5) -> MatchResult:
50     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52     Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53     are claimed highest-IoU first, and each prediction/GT can be used once. This
54     keeps a single prediction from being credited for two ground-truth rallies
55     (and vice versa).
56     """
57     candidates = []
58     for pi, p in enumerate(preds):
59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             # Require real overlap: at iou_threshold=0.0 a plain `iou >= threshold`
would
62             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
63             if iou > 0.0 and iou >= iou_threshold:
64                 candidates.append((iou, pi, gi))
65             # Highest IoU first; ties broken by indices for determinism.
66             candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
67
68     used_pred, used_gt = set(), set()
69     matches: List[Match] = []
70     for iou, pi, gi in candidates:
71         if pi in used_pred or gi in used_gt:
72             continue
73         used_pred.add(pi)
74         used_gt.add(gi)
75         matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
76
77     unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]
78     unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
79     return MatchResult(matches=matches,
80                        unmatched_pred_indices=unmatched_pred,
81                        unmatched_gt_indices=unmatched_gt)
82
83
84 @dataclass
85 class Score:
86     """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
87     iou_threshold: float
88     num_pred: int
89     num_gt: int
90     true_positives: int
91     false_positives: int
92     false_negatives: int
93     precision: float
94     recall: float

```

```

95         f1: float
96         mean_iou: float           # over matched pairs (0.0 if none)
97         mean_start_error: float   # mean |pred_start - gt_start| over matches, seconds
98         mean_end_error: float     # mean |pred_end - gt_end| over matches, seconds
99
100        def as_dict(self) -> dict:
101            return self.__dict__.copy()
102
103
104        def score(preds: List[Interval], gts: List[Interval],
105                  iou_threshold: float = 0.5,
106                  match_result: Optional[MatchResult] = None) -> Score:
107            """Compute precision/recall/F1, mean matched IoU, and boundary errors.
108
109            Pass a precomputed `match_result` to avoid re-matching (e.g. when also
110            extracting the failure lists); otherwise matching is done here.
111            """
112            mr = match_result if match_result is not None else match_intervals(preds, gts,
113            iou_threshold)
114
115            tp = len(mr.matches)
116            fp = len(mr.unmatched_pred_indices)
117            fn = len(mr.unmatched_gt_indices)
118
119            precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
120            recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
121            f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
122            else 0.0
123
124            if mr.matches:
125                mean_iou = sum(m.iou for m in mr.matches) / tp
126                mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
127                mr.matches) / tp
128                mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
129                mr.matches) / tp
130            else:
131                mean_iou = mean_start_err = mean_end_err = 0.0
132
133            return Score(
134                iou_threshold=iou_threshold,
135                num_pred=len(preds), num_gt=len(gts),
136                true_positives=tp, false_positives=fp, false_negatives=fn,
137                precision=precision, recall=recall, f1=f1,
138                mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
139            )
140
141
142        def pr_curve(preds: List[Interval], gts: List[Interval],
143                    thresholds: Optional[List[float]] = None) -> List[Score]:
144            """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).
145
146            Lets you see strict-vs-lenient behaviour rather than a single operating point.
147            """
148            if thresholds is None:
149                thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
150            return [score(preds, gts, iou_threshold=t) for t in thresholds]

```

17. user (2026-06-18T01:28:14.237Z)

```

1        """Segmentation-aware metrics (course-correction C4) ? make over-segmentation visible.
2
3        Temporal IoU (`metrics.score`) is nearly blind to *fragmentation*: a single rally
4        chopped into three predicted pieces, or two rallies merged across a dead-time gap, can
5        still post a moderate mean-IoU. The temporal-action-segmentation / -localization

```

```

6     literature grades those failure modes with **F1@k** (segmental F1 at several overlap
7     thresholds) and **mAP@tIoU** (ranked, confidence-aware average precision over a tIoU
8     sweep). Reporting these alongside temporal-IoU is what lets a reconciler's split/merge/
9     tighten win actually show up ? and stops an IoU-chasing rule from shredding rally
10    structure invisibly. See ``docs/ANALYZERS_AND_RECONCILERS.md`` §13/C4.
11
12    This is **additive**: new functions over the existing ``metrics`` matcher; nothing here
13    changes ``score``/``compare``/LOVO behaviour. Pure (stdlib only), deterministic, no I/O
14    ?
15    so it runs in the GPU-free / numpy-free lanes too.
16
17    Conventions match ``metrics``: an ``Interval`` is a ``(start, end)`` tuple of seconds.
18    A ``ScoredInterval`` is ``(interval, confidence)`` ? the confidence is whatever the
19    scorer
20    emits per predicted rally (e.g. the classifier probability), used only to *rank* for AP.
21
22    Note on the segmental *edit* score: the classic TAS edit/Levenshtein score operates on
23    the
24    ordered sequence of segment *labels* and is degenerate for a single action class ?
25    rally/
26    background segments strictly alternate, so the edit distance collapses to (twice) the
27    segment-count difference, adding nothing over ``segment_count_ratio``. So we
28    intentionally
29    omit it and instead make over-segmentation explicit with ``merge_split_report`` (a
30    bipartite
31    overlap-graph breakdown into merge/split/matched/missed/spurious ? see below), alongside
32    ``f1_at_overlaps`` (F1@k) + ``mean_average_precision`` (mAP@tIoU). Those, not a single-
33    class
34    edit score, are the over-segmentation-sensitive metrics for this binary rally setting.
35    """
36    from __future__ import annotations
37
38    from typing import Dict, List, Sequence, Tuple
39
40    from backend.eval.metrics import Interval, interval_iou, score
41
42    #: A predicted interval plus the confidence used to rank it for average precision.
43    ScoredInterval = Tuple[Interval, float]
44
45    #: Default overlap thresholds for F1@k ? the TAS convention F1@{10,25,50}.
46    DEFAULT_OVERLAPS: Tuple[float, ...] = (0.1, 0.25, 0.5)
47
48    #: Default tIoU sweep for mean average precision ? the action-detection convention.
49    DEFAULT_TIOUS: Tuple[float, ...] = (0.3, 0.4, 0.5, 0.6, 0.7)
50
51    def f1_at_overlaps(preds: List[Interval], gts: List[Interval],
52                      overlaps: Sequence[float] = DEFAULT_OVERLAPS) -> Dict[float, float]:
53        """Segmental F1 at each overlap threshold (``{overlap: f1}``) ? the TAS F1@k.
54
55        Reuses the canonical greedy one-to-one matcher, so a fragmented prediction (3 pieces
56        for 1 rally) scores extra false positives and a merged prediction misses a rally ?
57        both drag F1 down where temporal-IoU alone would not flinch.
58        """
59        return {ov: score(preds, gts, iou_threshold=ov).f1 for ov in overlaps}
60
61    def average_precision(scored_preds: Sequence[ScoredInterval], gts: List[Interval],
62                          iou_threshold: float = 0.5) -> float:
63        """Average precision at a single tIoU (all-points / VOC-2010 interpolation).
64
65        Predictions are ranked by descending confidence; each is a true positive iff it
66        claims a still-unmatched ground-truth rally at IoU >= ``iou_threshold`` (highest
67        available IoU wins, one GT per prediction). Returns AP in ``[0, 1]``. ``0.0`` when
68        there are no ground-truth rallies or no predictions reach the threshold.
69        """

```



```

64     n_gt = len(gts)
65     if n_gt == 0 or not scored_preds:
66         return 0.0
67
68     order = sorted(range(len(scored_preds)),
69                     key=lambda i: (-scored_preds[i][1], i)) # conf desc, stable by
index
70     matched_gt = set()
71     precisions: List[float] = []
72     recalls: List[float] = []
73     cum_tp = 0
74     cum_fp = 0
75     for i in order:
76         interval, _conf = scored_preds[i]
77         best_iou, best_gi = 0.0, -1
78         for gi, g in enumerate(gts):
79             if gi in matched_gt:
80                 continue
81             iou = interval_iou(interval, g)
82             if iou >= iou_threshold and iou > best_iou:
83                 best_iou, best_gi = iou, gi
84         if best_gi >= 0:
85             matched_gt.add(best_gi)
86             cum_tp += 1
87         else:
88             cum_fp += 1
89         precisions.append(cum_tp / (cum_tp + cum_fp))
90         recalls.append(cum_tp / n_gt)
91
92     # Monotone precision envelope (take the max precision to the right), then integrate
93     # over recall steps: AP = sum_k (recall_k - recall_{k-1}) * precision_env_k.
94     for i in range(len(precisions) - 2, -1, -1):
95         precisions[i] = max(precisions[i], precisions[i + 1])
96     ap = 0.0
97     prev_recall = 0.0
98     for p, r in zip(precisions, recalls):
99         if r > prev_recall:
100             ap += p * (r - prev_recall)
101             prev_recall = r
102     return ap
103
104
105 def mean_average_precision(scored_preds: Sequence[ScoredInterval], gts: List[Interval],
106                             iou_thresholds: Sequence[float] = DEFAULT_TIOUS) -> float:
107     """Mean of ``average_precision`` over a tIoU sweep (the standard mAP@tIoU)."""
108     tious = list(iou_thresholds)
109     if not tious:
110         return 0.0
111     return sum(average_precision(scored_preds, gts, t) for t in tious) / len(tious)
112
113
114 def segment_count_ratio(preds: List[Interval], gts: List[Interval]) -> float:
115     """Predicted-to-GT segment-count ratio ? a crude fragmentation diagnostic.
116
117     ``> 1`` hints over-segmentation (more predicted pieces than rallies), ``< 1`` hints
118     merging/misses. Only meaningful with ground truth present; returns ``0.0`` if
119     ``gts``
120     is empty (use F1@k / mAP for the graded picture).
121     """
122     if not gts:
123         return 0.0
124     return len(preds) / len(gts)
125
126 def _overlaps(a: Interval, b: Interval) -> bool:

```

```

127     """True iff two intervals share any positive temporal extent."""
128     return min(a[1], b[1]) - max(a[0], b[0]) > 0.0
129
130
131 def merge_split_report(preds: List[Interval], gts: List[Interval]) -> Dict[str, float]:
132     """Explicit over-/under-segmentation breakdown via the bipartite overlap graph.
133
134     Where ``segment_count_ratio`` only compares *counts* and ``f1_at_overlaps`` blends
135     merge
136     and split into one F1, this names the two failure modes directly ? the diagnostic
137     that
138     surfaced our local-segmenter over-MERGE (one giant window swallowing many rallies).
139     Build
140     the bipartite graph (edge = any positive temporal overlap between a pred and a gt),
141     take its
142     connected components, and classify each by ``(#preds, #gts)``:
143
144     - ``(1,1)`` matched    ``(1,0)`` spurious (pred, no gt)    ``(0,1)`` missed (gt, no
145     pred)
146     - ``(1,>1)`` **merge** ? one predicted window covers ?2 rallies (over-MERGE; our
147     failure)
148     - ``(>1,1)`` **split** ? one rally cut across ?2 predictions (over-segmentation)
149     - ``(>1,>1)`` complex (tangled many-to-many)
150
151     Returns counts plus ``merge_rate`` (fraction of GT rallies swallowed inside a merge
152     group)
153     and ``split_rate`` (fraction of GT rallies cut by ?2 predictions) ? the two headline
154     over-segmentation rates. Pure, deterministic, stdlib-only.
155     """
156     n, m = len(preds), len(gts)
157     p_adj: List[List[int]] = [[] for _ in range(n)]
158     g_adj: List[List[int]] = [[] for _ in range(m)]
159     for i, p in enumerate(preds):
160         for j, g in enumerate(gts):
161             if _overlaps(p, g):
162                 p_adj[i].append(j)
163                 g_adj[j].append(i)
164
165     seen_p: set = set()
166     seen_g: set = set()
167     comps: List[Tuple[set, set]] = []
168     for start in range(n):
169         if start in seen_p:
170             continue
171         ps: set = set()
172         gs: set = set()
173         stack: List[Tuple[str, int]] = [("p", start)]
174         while stack:
175             side, k = stack.pop()
176             if side == "p":
177                 if k in ps:
178                     continue
179                 ps.add(k)
180                 seen_p.add(k)
181                 stack.extend(("g", j) for j in p_adj[k] if j not in gs)
182             else:
183                 if k in gs:
184                     continue
185                 gs.add(k)
186                 seen_g.add(k)
187                 stack.extend(("p", i) for i in g_adj[k] if i not in ps)
188         comps.append((ps, gs))
189     comps.extend((set(), {j}) for j in range(m) if j not in seen_g) # gts with no
190     overlapping pred
191

```

```

184         counts = {"matched": 0, "merges": 0, "splits": 0, "spurious": 0, "missed": 0,
"complex": 0}
185         gts_merged = gts_split = 0
186         for ps, gs in comps:
187             a, b = len(ps), len(gs)
188             if (a, b) == (1, 1):
189                 counts["matched"] += 1
190             elif (a, b) == (1, 0):
191                 counts["spurious"] += 1
192             elif (a, b) == (0, 1):
193                 counts["missed"] += 1
194             elif a == 1 and b > 1:
195                 counts["merges"] += 1
196                 gts_merged += b
197             elif b == 1 and a > 1:
198                 counts["splits"] += 1
199                 gts_split += 1
200             else:
201                 counts["complex"] += 1
202         out: Dict[str, float] = {k: float(v) for k, v in counts.items()}
203         out["merge_rate"] = (gts_merged / m) if m else 0.0
204         out["split_rate"] = (gts_split / m) if m else 0.0
205         return out
206
207
208     def segmentation_report(scored_preds: Sequence[ScoredInterval], gts: List[Interval],
209                             overlaps: Sequence[float] = DEFAULT_OVERLAPS,
210                             iou_thresholds: Sequence[float] = DEFAULT_TIOUS) -> Dict[str,
object]:
211         """One-call bundle: F1@k + mAP@tIoU + per-tIoU AP + count ratio + merge/split
breakdown.
212
213         ``scored_preds`` carry confidences for AP; F1@k + the merge/split breakdown use the
bare
214         intervals. Returns a plain dict (JSON-friendly, string keys) so it can ride next to
a
215         ``compare`` result or land in the experiment leaderboard.
216         """
217         preds = [iv for iv, _conf in scored_preds]
218         return {
219             "f1_at_overlaps": {f"{ov:g}": v for ov, v in f1_at_overlaps(preds, gts,
overlaps).items()}},
220             "ap_at_tiou": {f"{t:g}": average_precision(scored_preds, gts, t) for t in
iou_thresholds},
221             "mean_ap": mean_average_precision(scored_preds, gts, iou_thresholds),
222             "segment_count_ratio": segment_count_ratio(preds, gts),
223             "merge_split": merge_split_report(preds, gts),
224             "num_pred": len(preds),
225             "num_gt": len(gts),
226         }
227

```

18. assistant (2026-06-18T01:28:20.983Z)

Now I have a good understanding. Let me examine some specific edge cases more carefully. Let me focus on the `over\_seg\_guard` alignment logic and the `\_aligned` helper in tolerance_metrics, plus the `paired\_sign\_test` for the `n=0` case.

Let me look at the `over\_seg\_guard` `\_aligned` function again. There's a potential `KeyError`/`cmap[b["name"]]` issue. Let me trace it carefully:

```

```python
def _aligned():
 if base and cand and all("name" in b for b in base) and all("name" in c for c in cand):
 cmap = {c["name"]: c for c in cand}

```

```
... return [(b, cmap[b["name"]]) for b in base if b["name"] in cmap]
```

This is guarded ? the comprehension filters `if b["name"] in cmap`. OK, safe.

Now let me check the `boundary\_error\_report` `max(..., default=None)` with a `key` over an empty `preds`. When `preds` is empty, `max([], key=..., default=None)` returns `None`. That's handled. Good.

Let me check the calibration's `select\_best` returning `best\_cfg=None` when configs is empty or all metrics are negative. Let me trace `cross\_validate` and `leave\_one\_video\_out` for edge cases, and check `paired\_sign\_test` with the `math.comb` (Python 3.8+).

#### 19. user (2026-06-18T01:28:23.154Z)

Python 3.13.13

10

#### 20. assistant (2026-06-18T01:28:26.122Z)

Now let me look at the tests for these modules to understand the contracts, especially the tolerance\_metrics greedy vs hungarian agreement claim and the percentile function. Let me find the relevant test files.

#### 21. user (2026-06-18T01:28:26.836Z)

```
.claude\worktrees\r2-impl\tests\test_tolerance_metrics.py
.claude\worktrees\r4-endrule\tests\test_tolerance_metrics.py
tests\test_tolerance_metrics.py
.claude\worktrees\pr-stitch\tests\test_tolerance_metrics.py
.claude\worktrees\boxhill17\tests\test_tolerance_metrics.py
.claude\worktrees\hunt-base\tests\test_tolerance_metrics.py
```

#### 22. user (2026-06-18T01:28:27.325Z)

```
tests\test_eval_stats.py
.claude\worktrees\bhi-execpolicy\tests\test_eval_stats.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_eval_stats.py
.claude\worktrees\local-noai\tests\test_eval_stats.py
.claude\worktrees\docs-rally-quality\tests\test_eval_stats.py
.claude\worktrees\tier0-eval\tests\test_eval_stats.py
.claude\worktrees\tier1-windowing\tests\test_eval_stats.py
.claude\worktrees\w2-netcross\tests\test_eval_stats.py
.claude\worktrees\w2-safeguard\tests\test_eval_stats.py
.claude\worktrees\hardchop\tests\test_eval_stats.py
.claude\worktrees\docs-today\tests\test_eval_stats.py
.claude\worktrees\docs-golden\tests\test_eval_stats.py
.claude\worktrees\r2-impl\tests\test_eval_stats.py
.claude\worktrees\r4-endrule\tests\test_eval_stats.py
.claude\worktrees\boxhill17\tests\test_eval_stats.py
.claude\worktrees\pr-stitch\tests\test_eval_stats.py
.claude\worktrees\hunt-base\tests\test_eval_stats.py
```

#### 23. user (2026-06-18T01:28:27.450Z)

```
tests\test_calibration.py
tests\test_score_calibration.py
.claude\worktrees\bhi-execpolicy\tests\test_calibration.py
.claude\worktrees\bhi-execpolicy\tests\test_score_calibration.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_calibration.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_score_calibration.py
.claude\worktrees\local-noai\tests\test_calibration.py
.claude\worktrees\local-noai\tests\test_score_calibration.py
.claude\worktrees\docs-rally-quality\tests\test_calibration.py
```

```
.claude\worktrees\docs-rally-quality\tests\test_score_calibration.py
.claude\worktrees\tier0-eval\tests\test_calibration.py
.claude\worktrees\tier0-eval\tests\test_score_calibration.py
.claude\worktrees\tier1-windowing\tests\test_calibration.py
.claude\worktrees\tier1-windowing\tests\test_score_calibration.py
.claude\worktrees\w2-netcross\tests\test_calibration.py
.claude\worktrees\w2-netcross\tests\test_score_calibration.py
.claude\worktrees\w2-safeguard\tests\test_calibration.py
.claude\worktrees\w2-safeguard\tests\test_score_calibration.py
.claude\worktrees\hardchop\tests\test_calibration.py
.claude\worktrees\hardchop\tests\test_score_calibration.py
.claude\worktrees\docs-today\tests\test_calibration.py
.claude\worktrees\docs-today\tests\test_score_calibration.py
.claude\worktrees\docs-golden\tests\test_calibration.py
.claude\worktrees\docs-golden\tests\test_score_calibration.py
.claude\worktrees\r2-impl\tests\test_calibration.py
.claude\worktrees\r2-impl\tests\test_score_calibration.py
.claude\worktrees\r4-endrule\tests\test_calibration.py
.claude\worktrees\r4-endrule\tests\test_score_calibration.py
.claude\worktrees\boxhill17\tests\test_calibration.py
.claude\worktrees\boxhill17\tests\test_score_calibration.py
.claude\worktrees\pr-stitch\tests\test_calibration.py
.claude\worktrees\pr-stitch\tests\test_score_calibration.py
.claude\worktrees\hunt-base\tests\test_calibration.py
.claude\worktrees\hunt-base\tests\test_score_calibration.py
```

#### 24. assistant (2026-06-18T01:28:37.847Z)

Let me examine the `over_seg_guard` more carefully, and the ``_hungarian_match_1to1`` claim about agreeing with greedy. First let me focus on something I noticed: the ``_greedy_match_1to1`` tiebreak. Let me re-examine the Hungarian vs greedy for max-cardinality.

The key claim in `tolerance_metrics` docstring is the Hungarian gives "max-cardinality (then min total boundary error)". But the cost matrix uses ``big=1e6`` for ineligible pairs, and eligible costs are ``abs(...)+abs(...)`` which is at most ``tau_start+tau_end = 3.5``. So ``linear_sum_assignment`` minimizes total cost. Since ``big`` (1e6) dominates, it will prefer to use as many eligible (low cost) pairs as possible ? that does maximize cardinality. Fine.

Let me look closer at a potential bug in ``over_seg_guard``: when ``use_rates=False`` (counts mode), the per-video cost uses ``merge_count`/`split_count``, but ``merge_rate`` may not be in the rows. The ``no_merge_rate_regress`` check is guarded by ``if use_rates and ...``. Good. But ``passes = net_cost_ok and no_merge_rate_regress``. When ``use_rates=False``, ``merge_rate_regress`` is empty list ? ``no_merge_rate_regress=True``. OK.

Let me check the ``tolerance_sweep`` "Monotonic non-decreasing in ?" claim and a possible issue in ``_greedy_match_1to1`` where larger ? could actually reduce matches due to greedy reassignment. But that's a quality claim, not a crash. Let me focus on concrete reachable defects. Let me re-examine ``bootstrap_ci`` percentile indexing and the ``paired_sign_test``.

Let me look at the test files to confirm contracts.

#### 25. user (2026-06-18T01:28:38.852Z)

```
1 """Tests for backend.eval.eval_stats (course-corrections C1 + C3).
2
3 Pure/offline + deterministic (the bootstrap takes a fixed seed). Pins: CIs bracket the
4 mean and degrade sanely at tiny n; the sign test's exact p-value; leave-one-group-out
5 folds; and that out-of-fold predictions are genuinely out of fold (no leakage).
6 """
7 import pytest
8
9 from backend.eval.eval_stats import (
10 bootstrap_ci,
11 grouped_folds,
12 mean_with_ci,
```

```

13 out_of_fold_predictions,
14 paired_delta_ci,
15 paired_sign_test,
16)
17
18
19 def test_bootstrap_ci_is_deterministic_and_brackets_mean():
20 vals = [0.5, 0.6, 0.7, 0.55, 0.65, 0.6]
21 lo1, hi1 = bootstrap_ci(vals, seed=0)
22 lo2, hi2 = bootstrap_ci(vals, seed=0)
23 assert (lo1, hi1) == (lo2, hi2) # same seed ? identical
24 assert lo1 <= sum(vals) / len(vals) <= hi1
25
26
27 def test_bootstrap_ci_degenerate():
28 assert bootstrap_ci([]) == (0.0, 0.0)
29 assert bootstrap_ci([0.42]) == (0.42, 0.42)
30 assert bootstrap_ci([0.5, 0.5, 0.5]) == (0.5, 0.5) # no spread ? point interval
31
32
33 def test_mean_with_ci_shape():
34 out = mean_with_ci([0.4, 0.6, 0.5], seed=1)
35 assert out["mean"] == pytest.approx(0.5)
36 assert out["n"] == 3.0
37 assert out["ci_low"] <= out["mean"] <= out["ci_high"]
38
39
40 def test_sign_test_all_wins_exact_p():
41 res = paired_sign_test([0.1, 0.2, 0.05, 0.3, 0.15, 0.2]) # 6 wins
42 assert res["wins"] == 6.0 and res["losses"] == 0.0
43 assert res["p_value"] == pytest.approx(2 * 0.5 ** 6) # 0.03125
44
45
46 def test_sign_test_ties_and_mixed():
47 res = paired_sign_test([0.0, 0.0, 0.0])
48 assert res["ties"] == 3.0 and res["n_effective"] == 0.0 and res["p_value"] == 1.0
49 mixed = paired_sign_test([0.1, -0.1, 0.2])
50 assert mixed["wins"] == 2.0 and mixed["losses"] == 1.0
51
52
53 def test_paired_delta_ci_detects_consistent_improvement():
54 a = [0.50, 0.55, 0.60, 0.52, 0.58, 0.54]
55 b = [0.60, 0.66, 0.69, 0.61, 0.70, 0.63] # B beats A every fold
56 out = paired_delta_ci(a, b, seed=0)
57 assert out["mean_delta"] > 0.0
58 assert out["sign_test"]["wins"] == 6.0
59 assert out["ci_excludes_zero"] is True
60
61
62 def test_grouped_folds_leave_one_group_out():
63 groups = ["m1", "m1", "m2", "m2", "m3"]
64 folds = grouped_folds(groups)
65 assert len(folds) == 3
66 train0, test0 = folds[0] # first-seen group m1 held out
67 assert test0 == [0, 1] and train0 == [2, 3, 4]
68 assert grouped_folds(["only"]) == [] # single group ? no fold
69
70
71 def test_out_of_fold_predictions_are_out_of_fold():
72 groups = ["a", "a", "b", "b"]
73 y = [0, 0, 1, 1]
74
75 def fit_predict(train_idx, test_idx):
76 # "Model" = mean label of the training groups; returned for each test item.
77 m = sum(y[i] for i in train_idx) / len(train_idx)

```

```

78 return [m] * len(test_idx)
79
80 oof = out_of_fold_predictions(groups, fit_predict)
81 # Group a's OOF uses only group b (mean 1.0) and vice-versa ? never its own labels.
82 assert oof == [1.0, 1.0, 0.0, 0.0]
83

```

## 26. assistant (2026-06-18T01:28:50.213Z)

Now let me examine the `bootstrap\_ci` percentile index computation carefully ? there's a subtle issue with how it picks indices. Let me look at the math:

```

```python
last = len(means) - 1
lo = means[max(0, int(math.floor(alpha * last)))]
hi = means[min(last, int(math.ceil((1.0 - alpha) * last)))]
```

```

For confidence=0.95, alpha=0.025. With n\_boot=2000, last=1999. `lo = means[floor(0.025\*1999)] = means[floor(49.975)] = means[49]`. `hi = means[ceil(0.975\*1999)] = means[ceil(1949.025)] = means[1950]`. That's reasonable for a percentile bootstrap. No bug.

Now let me look more carefully at the `calibration.select\_best` returning `None` config. If `configs` is empty, `best\_cfg=None` is returned, then `predict\_fn(None)` would be called. But that requires an empty configs list which is a caller contract issue. Let me check `cross\_validate`/`leave\_one\_video\_out` ? they pass `configs` through. Not clearly a bug if callers always pass configs.

Let me focus on a more promising lead. Let me re-examine `over\_seg\_guard`'s `\_aligned` when names are present in base but NOT in cand. The list comprehension `[(b, cmap[b["name"]]) for b in base if b["name"] in cmap]` filters correctly. But then the per-video regression loop iterates `for b, c in pairs` ? using the aligned pairs. Good.

Now, let me look at the `paired\_delta\_ci` when `a\_values`/`b\_values` have different lengths ? it uses `n = min(...)` and zips by index. Fine.

Let me re-examine a subtle issue in `experiment.py`'s `\_calibrate\_and\_tune` and the `tune\_threshold` path. And check the `compare\_unlabeled`'s `b\_only` using `mr.unmatched\_gt\_indices` to index `preds\_b`. In `match\_intervals(preds\_a, preds\_b, ...)` , `preds\_a` is "preds" and `preds\_b` is "gts". So `unmatched\_gt\_indices` indexes into `preds\_b`. Correct.

Let me dig into the `windowing.py` and `gt\_loader.py` for IO key access bugs, and check `per\_user\_eval.py` / `per\_user\_regression.py` / `promotion.py` which are also in eval core.

## 27. user (2026-06-18T01:28:51.167Z)

```

1 """Promotion gate with the eval?serve guard: a cue is promotable ONLY if it does not
regress
2 at the SERVED windowing.
3
4 **Why this exists.** ``per_user_gate.should_promote_for_user`` already grades a B?A lift
5 honestly (min-N floor + ?F1 CI excludes 0 + paired sign test). But it grades whatever
scores
6 it is handed ? and the foundational bug was that those scores came from the *recall-
oriented
7 PROPOSAL* windowing (``backend.eval.windowing.PROPOSAL``), which production never
serves. So a
8 cue could clear the bar on a windowing nobody runs (Gate-A: proposal ?F1 **+0.074**)
while
9 regressing on the served one (**?0.008**). Grading harder isn't the fix; **measuring on
the
10 served windowing** is.
11
12 This module composes ? it adds no new statistics. It layers ONE rule on top of

```

```

13 ``should_promote_for_user``:
14
15 A cue is promotable only if, at the **SERVED** windowing, it WINS or at least DOES
NOT
16 REGRESS (its served paired ?F1 mean ? ``-served_regress_eps``, and ? when its served
CI is
17 estimable ? that CI's lower bound does not sit clearly in regression territory).
18
19 A proposal-windowing win is still *reported* (it's useful signal for where to look
next), but it
20 can never carry a promotion on its own. The decision returns BOTH the served and the
proposal
21 verdicts so the eval?serve contrast is explicit and auditable.
22
23 Pure, deterministic, stdlib-only over ``eval_stats``/``per_user_gate``. Additive:
existing eval
24 paths are untouched until a caller routes its A/B through
:func:`promote_if_served_safe`.
25 """
26 from __future__ import annotations
27
28 from dataclasses import dataclass, field
29 from typing import Any, Dict, Optional, Sequence
30
31 from backend.eval.eval_stats import paired_delta_ci
32 from backend.eval.per_user_gate import PromotionDecision, should_promote_for_user
33
34 __all__ = ["ServedGateResult", "promote_if_served_safe", "served_regresses"]
35
36
37 def served_regresses(a_scores: Sequence[float], b_scores: Sequence[float], *,
38 eps: float = 0.0, n_boot: int = 2000, seed: int = 0) -> Dict[str,
Any]:
39 """Does variant B REGRESS variant A at the served windowing? (the eval?serve guard
core).
40
41 ``a_scores``/``b_scores`` are per-video held-out scores **measured at the SERVED
windowing**
42 (aligned by video). Returns the paired B?A summary (mean ?F1 + bootstrap CI + sign
test) plus
43 a single ``regresses`` boolean:
44
45 - ``regresses=True`` when the served mean ?F1 is below ``-eps`` (a real served
drop), OR
46 the served ?F1 CI lies **entirely** below ``-eps`` (a confidently-negative
band).
47 - ``regresses=False`` when the cue holds the line at the served operating point
(mean ?
48 ``-eps`` and the CI is not a confident net-negative).
49
50 ``eps`` is the regression tolerance: 0.0 = "must not drop at all on the served
mean". A tiny
51 positive ``eps`` lets noise-level wobble through (e.g. 0.005 ? half an F1 point at
n?6).
52 """
53 delta = paired_delta_ci(a_scores, b_scores, n_boot=n_boot, seed=seed)
54 mean_delta = float(delta["mean_delta"])
55 ci_low = float(delta["ci_low"])
56 ci_high = float(delta["ci_high"])
57 # Regression if the served mean dropped beyond tolerance, OR the whole CI is net-
negative.
58 mean_regresses = mean_delta < -eps
59 ci_confidently_negative = ci_high < -eps
60 regresses = bool(mean_regresses or ci_confidently_negative)
61 return {

```



```

62 "regresses": regresses,
63 "mean_delta": mean_delta,
64 "ci_low": ci_low,
65 "ci_high": ci_high,
66 "ci_excludes_zero": bool(delta["ci_excludes_zero"]),
67 "sign_test": delta["sign_test"],
68 "n": int(delta["n"]),
69 "eps": float(eps),
70 }
71
72
73 @dataclass
74 class ServedGateResult:
75 """A promotion decision that BOTH measures the cue on the served windowing AND
reports the
76 proposal-windowing view, so the eval?serve contrast is never hidden.
77
78 - ``promote`` ? final verdict. ``True`` only when the underlying honest gate would
promote
79 *and* the served check did not veto (the cue does not regress at the served
windowing).
80 - ``served_safe`` ? the eval?serve guard's verdict in isolation (cue does not
regress served).
81 - ``vetoed_by_served`` ? ``True`` when the proposal/base gate WOULD have promoted
but the
82 served regression check overruled it (the exact Gate-A failure mode this module
prevents).
83 - ``base_decision`` ? the ``PromotionDecision`` from ``should_promote_for_user`` on
the
84 *promotion-grade* scores (served by default ? see :func:`promote_if_served_safe`).
85 - ``served`` / ``proposal`` ? the paired B?A summary at each windowing (means + CIs
+ sign
86 tests), so a reviewer sees "+0.074 proposal / ?0.008 served" side by side.
87 - ``reason`` ? short human-readable explanation of the final verdict.
88 """
89
90 promote: bool
91 served_safe: bool
92 vetoed_by_served: bool
93 reason: str
94 base_decision: PromotionDecision
95 served: Dict[str, Any]
96 proposal: Optional[Dict[str, Any]] = field(default=None)
97
98 def as_dict(self) -> Dict[str, Any]:
99 return {
100 "promote": self.promote,
101 "served_safe": self.served_safe,
102 "vetoed_by_served": self.vetoed_by_served,
103 "reason": self.reason,
104 "base_decision": self.base_decision.as_dict(),
105 "served": self.served,
106 "proposal": self.proposal,
107 }
108
109
110 def promote_if_served_safe(
111 served_a: Sequence[float],
112 served_b: Sequence[float],
113 *,
114 proposal_a: Optional[Sequence[float]] = None,
115 proposal_b: Optional[Sequence[float]] = None,
116 structural: bool = False,
117 min_n: int = 5,
118 p_threshold: float = 0.05,

```

```

119 served_regress_eps: float = 0.0,
120 n_boot: int = 2000,
121 seed: int = 0,
122) -> ServedGateResult:
123 """Promote a cue ONLY if it does not regress at the SERVED windowing (the
foundational rule).
124
125 The honest bar (``should_promote_for_user``: min-N floor + ?F1 CI excludes 0 + sign
test) is
126 applied to the **served** scores ? the operating point production runs ? so a
proposal-only
127 win can never carry a promotion. The optional ``proposal_`` scores are scored too
and
128 reported alongside, purely to surface the eval?serve contrast (they NEVER promote on
their
129 own).
130
131 Args:
132 served_a/served_b: per-video held-out scores for variant A (control) and B (the
cue),
133 measured at the **SERVED** windowing. These drive the decision.
134 proposal_a/proposal_b: the same scores at the PROPOSAL windowing (optional).
Reported for
135 contrast; the served regression guard still has the final say.
136 structural: forwarded to ``should_promote_for_user`` ? a structural guard (e.g.
Gate-A vs
137 held-shuttle FPs) is never *disabled* on "no lift", but it is ALSO not
auto-*promoted* to
138 the served default if it regresses served. So for ``structural=True`` we report
139 ``keep_on`` honestly but ``promote`` reflects the served-safety check.
140 served_regress_eps: served regression tolerance (0.0 = "no served-mean drop
allowed").
141
142 Returns a :class:`ServedGateResult` with the final verdict + both windowings'
evidence.
143 """
144 # 1) The honest gate, applied to the SERVED scores (never the proposal ones).
145 base = should_promote_for_user(
146 served_a, served_b,
147 structural=structural, min_n=min_n, p_threshold=p_threshold,
148 n_boot=n_boot, seed=seed,
149)
150
151 # 2) The eval?serve guard: does B regress A at the served windowing?
152 served = served_regresses(served_a, served_b, eps=served_regress_eps,
153 n_boot=n_boot, seed=seed)
154 served_safe = not served["regresses"]
155
156 # 3) Optional proposal-windowing view (reported, never promoting on its own).
157 proposal: Optional[Dict[str, Any]] = None
158 if proposal_a is not None and proposal_b is not None:
159 proposal = served_regresses(proposal_a, proposal_b, eps=served_regress_eps,
160 n_boot=n_boot, seed=seed)
161
162 # 4) Compose the verdict. The served guard can only ever REMOVE a promotion, never
add one.
163 if structural:
164 # A structural guard ships ON regardless of measured lift (base.keep_on stays
True), but
165 # it is only auto-promoted to the SERVED default when it is served-safe. If it
regresses
166 # served, promote=False (don't bake an eval-only win into production) while
keep_on holds.
167 promote = served_safe
168 vetoed = False

```

```

169 if served_safe:
170 reason = ("structural guard kept ON; served-safe (does not regress at the
served "
171 f"windowing, ?F1 {served['mean_delta']:+.4f}) ? eligible for
served default")
172 else:
173 reason = ("structural guard kept ON, but NOT promoted to the served default:
it "
174 f"REGRESSES at the served windowing (?F1
{served['mean_delta']:+.4f}); "
175 "an eval-only win must not be served")
176 else:
177 would_promote = base.promote
178 if would_promote and not served_safe:
179 promote = False
180 vetoed = True
181 reason = ("VETOED by the served-windowing guard: the cue clears the honest
bar but "
182 f"REGRESSES at the served windowing (served ?F1
{served['mean_delta']:+.4f}")
183 f", CI [{served['ci_low']:+.4f}, {served['ci_high']:+.4f}]). A win
on a "
184 "windowing production doesn't serve cannot promote.")
185 elif would_promote:
186 promote = True
187 vetoed = False
188 reason = ("promoted: clears the honest bar AND is served-safe (served ?F1 "
189 f"{served['mean_delta']:+.4f}). " + base.reason)
190 else:
191 promote = False
192 vetoed = False
193 reason = base.reason
194
195 return ServedGateResult(
196 promote=promote,
197 served_safe=served_safe,
198 vetoed_by_served=vetoed,
199 reason=reason,
200 base_decision=base,
201 served=served,
202 proposal=proposal,
203)
204

```

## 28. user (2026-06-18T01:28:51.975Z)

```

1 """Canonical ground-truth interval loader ? THE one GT/golden-CSV reader (ADR-008
prereq).
2
3 The repo historically carried two incompatible loaders:
4 - ``annotations.load_annotations``: positional ``start,end`` (the collector's
5 ``curate export`` format), raises on bad rows;
6 - ``calibrate_wasb.load_golden_gts``: DictReader requiring ``start_time/end_time``
7 (the rally-annotator golden format), silently skipping bad rows ? fed 7 modules.
8
9 A file in one convention fed to the other loader yielded zero/garbage intervals with no
10 error. This module accepts BOTH conventions (header-mapped, else positional), so every
11 quantity the ship-gate depends on (gt_hash, LOVO scores, the future consent fingerprint)
12 flows through one reader. Both legacy entry points now delegate here.
13
14 Accepted shapes (one rally per row; blank lines and ``#`` comments ignored):
15 - header with ``start``/``end`` OR ``start_time``/``end_time`` columns (any other
16 columns ? rally_number, ending_reason, sport, ? ? are ignored);
17 - no header: columns 0,1 positionally.
18 Timestamps: decimal seconds or ``MM:SS`` / ``HH:MM:SS``

```

```

(``annotations.parse_timestamp``).
19 """
20 import csv
21 import logging
22 from typing import List, Optional, Tuple
23
24 from backend.eval.annotations import parse_timestamp
25
26 logger = logging.getLogger(__name__)
27
28 Interval = Tuple[float, float]
29
30 _START_NAMES = ("start", "start_time")
31 _END_NAMES = ("end", "end_time")
32
33
34 def _header_columns(row: List[str]) -> Optional[Tuple[int, int]]:
35 """If `row` is a header naming start/end columns, return their indices, else
None."""
36 cells = [(c or "").strip().lower() for c in row]
37 start_idx = next((i for i, c in enumerate(cells) if c in _START_NAMES), None)
38 end_idx = next((i for i, c in enumerate(cells) if c in _END_NAMES), None)
39 if start_idx is not None and end_idx is not None:
40 return start_idx, end_idx
41 return None
42
43
44 def load_gt_intervals(csv_path: str, *, strict: bool = True) -> List[Interval]:
45 """Load GT rally intervals from either golden-CSV convention.
46
47 strict=True (corpus/gate use): raise ValueError with file:line on any malformed row
48 (bad timestamp, end <= start, negative start) so labeling mistakes fail loudly.
49 strict=False (legacy tuning-driver behavior): skip bad rows, but LOG how many were
50 skipped ? a silently shrinking GT set flatters precision.
51
52 Returns intervals sorted by start; overlapping GT intervals are logged as a warning
53 (they break one-to-one matching and usually indicate a labeling error).
54 """
55 intervals: List[Interval] = []
56 skipped = 0
57 cols: Optional[Tuple[int, int]] = None # (start_idx, end_idx); None until detected
58 saw_header = False
59
60 with open(csv_path, newline="", encoding="utf-8-sig") as f:
61 for lineno, row in enumerate(csv.reader(f), start=1):
62 if not row or not (row[0] or "").strip() or row[0].strip().startswith("#"):
63 continue
64 if cols is None and not saw_header:
65 header = _header_columns(row)
66 if header is not None:
67 cols = header
68 saw_header = True
69 continue # consume the header row
70 cols = (0, 1) # headerless: positional start,end
71 assert cols is not None # every branch above assigns it
72 si, ei = cols
73 try:
74 if len(row) <= max(si, ei):
75 raise ValueError(f"expected at least {max(si, ei) + 1} columns, got
{row!r}")
76 start = parse_timestamp(row[si])
77 end = parse_timestamp(row[ei])
78 if start < 0:
79 raise ValueError(f"negative start ({start})")
80 if start >= end:

```

```

81 raise ValueError(f"start ({start}) >= end ({end})")
82 except ValueError as e:
83 if strict:
84 raise ValueError(f"{csv_path}:{lineno}: {e}") from e
85 skipped += 1
86 continue
87 intervals.append((start, end))
88
89 if skipped:
90 logger.warning("%s: skipped %d malformed GT row(s) (strict=False)", csv_path,
91 skipped)
92
93 intervals.sort(key=lambda iv: iv[0])
94 for (s1, e1), (s2, e2) in zip(intervals, intervals[1:]):
95 if s2 < e1:
96 logger.warning("%s: overlapping GT intervals (%.3f-%.3f vs %.3f-%.3f) ? "
97 "labeling error? Overlaps break one-to-one matching.",
98 csv_path, s1, e1, s2, e2)
99 return intervals

```

## 29. assistant (2026-06-18T01:28:58.045Z)

Let me look at `per_user_regression.py` and `per_user_eval.py`, which weren't read yet, plus `windowing.py` for IO/key issues.

## 30. user (2026-06-18T01:28:58.850Z)

```

1 """Per-user NO-REGRESSION guard ? the missing per-user analogue of `regression.py`.
2
3 The shared golden set has a disciplined no-regression baseline (`regression.py` +
4 ``regression_baseline.json``): a candidate is gated so a global change can never
5 silently
6 make the *shared* corpus worse. Personalization has no such rail ? a deliberately-tuned
7 per-user model "has no shared baseline to regress against", so a per-user retrain or a
8 baseline upgrade could quietly make ONE user worse than the model they're already
9 running.
10 This module builds that missing per-user regression baseline, mirroring the shared
11 discipline.
12
13 The decision it makes is deliberately narrow and CONSERVATIVE ? it is a *guarantee
14 against
15 regression*, not a promotion decision (that is `per_user_gate.should_promote_for_user`):
16
17 * **CONTROL** = the user's *frozen incumbent* ? the variant/config currently active
18 FOR THAT
19 USER. For a brand-new user with nothing personalized yet, the incumbent simply IS
20 the
21 shipped baseline. This is the thing we must never regress below.
22 * **CANDIDATE** = the proposed replacement (a per-user retrain, or a global baseline
23 upgrade
24 rolled out to this user).
25
26 Both are evaluated on the user's own held-out videos (`experiment.evaluate_variant`,
27 LOVO-honest
28 for a learned recipe ? no new scoring math), paired by video, and summarised with the
29 shipped
30 honest statistics (`eval_stats.paired_delta_ci` ? bootstrap ?F1 CI + exact sign test).
31 Then:
32
33 * **regression** ? the candidate is *significantly worse*: the paired ?F1 (candidate ?
34 control)
35 95% CI excludes 0 on the WRONG side (`ci_high < 0`). ? REJECT, keep the incumbent.
36 * **clear improvement** ? the candidate is *significantly better*: the CI excludes 0
37 on the

```

```

26 right side AND the sign test agrees (candidate wins on strictly more videos than it
loses).
27 ? ACCEPT the swap.
28 * **tie / insufficient evidence** ? the CI spans 0 (the small-N near-null regime):
there is no
29 evidence the swap helps, so we never take the risk. ? KEEP THE INCUMBENT.
30
31 So we only ACCEPT a swap we can *defend* and only ever toward a measured improvement;
everything
32 ambiguous keeps the incumbent. That is strictly stronger than "don't regress" (it also
refuses
33 risk-neutral churn), which is exactly the safety the personalization feature was
missing.
34
35 A small per-user baseline record is keyed by a ``user_id::stage`` string (``user_id``
nullable ?
36 ``None`` = the local single-user deployment; design-for-north-star, implement-for-
current) and
37 saved/loaded as JSON, a sibling to `regression.py`'s `regression_baseline.json`
approach.
38
39 Pure-CPU, deterministic given ``seed`` (the only randomness is the CI bootstrap); no GPU
/ torch /
40 cv2 / file I/O beyond the explicit baseline save/load. See `docs/EXPERIMENT_HARNESS.md`
+
41 `docs/ANALYZERS_AND_RECONCILERS.md` §13/C1.
42 ""
43 from __future__ import annotations
44
45 import json
46 import logging
47 import os
48 from dataclasses import asdict, dataclass, field
49 from typing import Any, Dict, List, Optional, Sequence
50
51 from backend.eval import eval_stats as _stats
52 from backend.eval import experiment
53 from backend.eval.experiment import Variant, VideoCandidates
54
55 logger = logging.getLogger(__name__)
56
57 __all__ = [
58 "DEFAULT_PER_USER_BASELINE_PATH",
59 "PerUserRegressionResult",
60 "PerUserBaseline",
61 "baseline_key",
62 "load_baselines",
63 "save_baseline",
64 "evaluate_no_regression",
65]
66
67 # Tracked location (NOT under the gitignored output/), sibling to
68 # regression.DEFAULT_BASELINE_PATH so a fresh checkout / CI can load a committed per-
user
69 # baseline. NULL user_id (local single-user) keys as the literal "local".
70 DEFAULT_PER_USER_BASELINE_PATH = os.path.join("eval_baselines",
"per_user_baseline.json")
71
72
73 # ----- result
74
75 @dataclass
76 class PerUserRegressionResult:
77 """The per-user no-regression decision plus the evidence behind it.
78

```

```

79 - ``accept`` ? swap the candidate in for this user (only ever ``True`` on a
defended
80 improvement: the ?F1 CI clears 0 on the right side AND the sign test agrees).
81 - ``keep_incumbent`` ? the user keeps their frozen incumbent (control). Always the
exact
82 complement of ``accept`` ? a swap is taken or it is not.
83 - ``regressed`` ? the candidate is *significantly worse* than the incumbent (CI
excludes 0
84 on the wrong side). A True here always implies ``accept=False``.
85 - ``reason`` ? short human-readable explanation, for logging / audit.
86 - the remaining fields surface the paired evidence (per-variant mean held-out F1,
mean ?F1,
87 CI bounds, and the sign test) so a caller can audit exactly why the swap was/
wasn't taken.
88 """
89
90 accept: bool
91 keep_incumbent: bool
92 reason: str
93 n: int
94 control_f1: float
95 candidate_f1: float
96 mean_delta: float
97 ci_low: float
98 ci_high: float
99 ci_excludes_zero: bool
100 regressed: bool
101 sign_test: Dict[str, float] = field(default_factory=dict)
102
103 def as_dict(self) -> Dict[str, Any]:
104 """Plain-dict view (for JSON logging / telemetry)."""
105 return asdict(self)
106
107
108 # ----- per-user baseline
store
109
110 @dataclass
111 class PerUserBaseline:
112 """A snapshotted per-user incumbent record ? the per-user analogue of a regression
baseline.
113
114 Keyed by ``user_id::stage``. ``user_id`` is nullable (``None`` ? the local single-
user
115 deployment, keyed as the literal ``"local"``). ``control_spec_hash`` pins WHICH
incumbent
116 variant the recorded numbers were measured for, so a later check against a different
117 incumbent is detectable rather than silently trusted (mirrors regression.py's
118 iou/detector/gt fingerprint guards).
119 """
120
121 user_id: Optional[str]
122 stage: str
123 control_f1: float
124 control_spec_hash: str
125 n: int
126 iou: float
127
128 def as_dict(self) -> Dict[str, Any]:
129 return asdict(self)
130
131
132 def baseline_key(user_id: Optional[str], stage: str) -> str:
133 """Stable ``user_id::stage`` key. A NULL ``user_id`` (local single-user) keys as
``local``."""

```

```

134 return f"{user_id if user_id is not None else 'local'}::{stage}"
135
136
137 def load_baselines(path: str = DEFAULT_PER_USER_BASELINE_PATH) -> Dict[str, dict]:
138 """Load the per-user baseline file (returns ``{}`` if absent)."""
139 if not os.path.isfile(path):
140 return {}
141 with open(path, encoding="utf-8") as f:
142 return json.load(f)
143
144
145 def save_baseline(baseline: PerUserBaseline,
146 path: str = DEFAULT_PER_USER_BASELINE_PATH) -> str:
147 """Snapshot a user's incumbent numbers as the new per-user baseline, keyed
148 ``user_id::stage``.
149
150 Round-trips through ``load_baselines``; returns the key written. Mirrors
151 ``regression.save_baseline`` (atomic-ish read-modify-write of the one JSON file).
152 """
153 baselines = load_baselines(path)
154 key = baseline_key(baseline.user_id, baseline.stage)
155 baselines[key] = baseline.as_dict()
156 os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
157 with open(path, "w", encoding="utf-8") as f:
158 json.dump(baselines, f, indent=2, sort_keys=True)
159 logger.info("Saved per-user baseline %s (control_f1=%.3f n=%d)",
160 key, baseline.control_f1, baseline.n)
161 return key
162
163 # ----- the guard
164
165 def _mean(xs: Sequence[float]) -> float:
166 return sum(xs) / len(xs) if xs else 0.0
167
168
169 def evaluate_no_regression(
170 videos: Sequence[VideoCandidates],
171 control: Variant,
172 candidate: Variant,
173 *,
174 user_id: Optional[str] = None,
175 stage: str = "final",
176 iou: float = 0.5,
177 honest: bool = True,
178 p_threshold: float = 0.05,
179 confidence: float = 0.95,
180 n_boot: int = 2000,
181 seed: int = 0,
182 eps: float = 0.0,
183 baseline_path: Optional[str] = None,
184) -> PerUserRegressionResult:
185 """Guard a per-user swap: ACCEPT only if ``candidate`` does NOT regress vs the
186 ``control``.
187
188 ``control`` is the user's FROZEN incumbent (their currently-active variant ? for a
189 brand-new user this is just the shipped baseline). ``candidate`` is the proposed
190 replacement (a per-user retrain or a baseline upgrade). Both are scored on the
191 user's own
192 ``videos`` via ``experiment.evaluate_variant`` (LOVO-honest for a learned recipe ? no
193 new
194 scoring math), the per-video held-out F1s are paired by video, and the paired ?F1
195 (candidate ? control) is summarised with ``eval_stats.paired_delta_ci`` (bootstrap CI
196 +
197 exact sign test). The decision (faithful to the module docstring):

```



```

194
195 * **regression** (`ci_high < 0` ? candidate significantly worse): REJECT, keep
incumbent.
196 * **clear improvement** (CI excludes 0 on the right side AND the sign test agrees
?
197 candidate wins on strictly more videos than it loses): ACCEPT the swap.
198 * **tie / insufficient evidence** (CI spans 0): KEEP THE INCUMBENT (conservative;
never
199 accept a risky swap).
200
201 ``user_id``/``stage`` key the baseline record (`user_id=None` ? local single-
user). When
202 ``baseline_path`` is given the incumbent's measured numbers are snapshotted there
(the swap
203 is then evaluated as usual). Deterministic given ``seed``. Raises `ExperimentError`
(from
204 `evaluate_variant`) on unlabeled videos or a learned recipe with too few videos for
LOVO.
205 """
206 eval_control = experiment.evaluate_variant(videos, control, iou, honest)
207 eval_candidate = experiment.evaluate_variant(videos, candidate, iou, honest)
208
209 # Per-video held-out F1, video-aligned (evaluate_variant iterates ``videos`` in
order, so
210 # both lists are pairable by position ? exactly how honest_summary pairs B ? A).
211 control_f1s: List[float] = [p["f1"] for p in eval_control["per_video"]]
212 candidate_f1s: List[float] = [p["f1"] for p in eval_candidate["per_video"]]
213 n = min(len(control_f1s), len(candidate_f1s))
214
215 control_mean = _mean(control_f1s)
216 candidate_mean = _mean(candidate_f1s)
217
218 if n == 0:
219 # No paired videos at all ? no evidence either way ? keep the incumbent.
220 sign_test: Dict[str, float] = _stats.paired_sign_test([], eps)
221 result = PerUserRegressionResult(
222 accept=False, keep_incumbent=True,
223 reason="no held-out videos to evaluate; keeping incumbent",
224 n=0, control_f1=control_mean, candidate_f1=candidate_mean,
225 mean_delta=0.0, ci_low=0.0, ci_high=0.0, ci_excludes_zero=False,
226 regressed=False, sign_test=sign_test,
227)
228 else:
229 delta = _stats.paired_delta_ci(
230 control_f1s, candidate_f1s,
231 confidence=confidence, n_boot=n_boot, seed=seed, eps=eps,
232)
233 mean_delta = float(delta["mean_delta"])
234 ci_low = float(delta["ci_low"])
235 ci_high = float(delta["ci_high"])
236 ci_excludes_zero = bool(delta["ci_excludes_zero"])
237 sign_test = delta["sign_test"]
238 sign_p = float(sign_test.get("p_value", 1.0))
239 sign_wins = float(sign_test.get("wins", 0.0))
240 sign_losses = float(sign_test.get("losses", 0.0))
241
242 # A regression = the candidate is significantly WORSE: the ?F1 CI excludes 0
entirely
243 # on the wrong side (its whole high end is below 0). Never accept such a swap.
244 regressed = ci_excludes_zero and ci_high < 0.0
245 # A defended improvement = the CI clears 0 on the RIGHT side AND the sign test
agrees
246 # (more candidate-wins than losses, significant) ? the same promotion-grade bar
the
247 # global eval uses, applied per user. Everything else is a tie / insufficient

```

```

evidence.
248 sign_agrees = (sign_p <= p_threshold) and (sign_wins > sign_losses)
249 improved = ci_excludes_zero and ci_low > 0.0 and sign_agrees
250
251 if regressed:
252 accept = False
253 reason = (
254 f"REGRESSION: candidate ?F1 95% CI [{ci_low:.4f}, {ci_high:.4f}] "
255 f"is entirely below 0 (n={n}); keeping incumbent"
256)
257 elif improved:
258 accept = True
259 reason = (
260 f"accepted: candidate ?F1 95% CI [{ci_low:.4f}, {ci_high:.4f}] excludes
0 "
261 f"(improvement), sign test {int(sign_wins)}-{int(sign_losses)} "
262 f"(p={sign_p:.4f} <= {p_threshold}); swapping in candidate"
263)
264 else:
265 accept = False
266 reason = (
267 f"tie / insufficient evidence: candidate ?F1 95% CI "
268 f"[{ci_low:.4f}, {ci_high:.4f}] spans 0 (n={n}); keeping incumbent "
269 f"(no defended improvement, never a risky swap)"
270)
271
272 result = PerUserRegressionResult(
273 accept=accept, keep_incumbent=not accept, reason=reason,
274 n=n, control_f1=control_mean, candidate_f1=candidate_mean,
275 mean_delta=mean_delta, ci_low=ci_low, ci_high=ci_high,
276 ci_excludes_zero=ci_excludes_zero, regressed=regressed, sign_test=sign_test,
277)
278
279 # Snapshot the incumbent's measured numbers as this user's baseline when asked ? the
280 # per-user analogue of regression.save_baseline. Does not change the decision above.
281 if baseline_path is not None:
282 save_baseline(
283 PerUserBaseline(
284 user_id=user_id, stage=stage,
285 control_f1=control_mean, control_spec_hash=control.spec_hash(),
286 n=n, iou=iou,
287),
288 path=baseline_path,
289)
290
291 return result
292

```

### 31. assistant (2026-06-18T01:29:05.948Z)

Now let me look at windowing.py, which is critical to rally\_seg\_eval, and check the `boundary\_error\_report` claim about percentile. Actually, let me revisit the `\_percentile` function in tolerance\_metrics more carefully ? it had a subtle potential issue but let me verify.

```

```python
def _percentile(xs, p):
    if not xs: return 0.0
    s = sorted(xs)
    k = (len(s) - 1) * p / 100.0
    lo = int(k)
    hi = min(lo + 1, len(s) - 1)
    return s[lo] + (s[hi] - s[lo]) * (k - lo)
```

```

For a single element `xs=[5.0]`, `p=75`: `k=0`, `lo=0`, `hi=0`, returns `5.0`. Fine. No bug.

Let me now look at windowing.py ? it's part of the eval pipeline that rally\_seg\_eval depends on heavily and could have edge-case bugs in the inactivity/blob logic.

### 32. user (2026-06-18T01:29:06.839Z)

```
1 """Single-sourced candidate-windowing presets ? the eval?serve foundation fix.
2
3 **The bug this closes.** The offline eval substrate (`distill_local.build_candidates`
4 ?
5 `fusion_golden` ? the golden feature JSONs the ablation/compare path re-scores) was
6 built
7 on a *recall-oriented PROPOSAL* windowing (`vt=0.005, merge_gap=1.0, min_window=0.5`):
8 deliberately permissive so a learned scorer/gate can filter the junk. But the SERVED
9 pipeline
10 (`TrajectoryHybridSegmenter.process_video`) windows COARSELY (`vt=0.02,
11 merge_gap=2.0,
12 min_window=2.0` ? the `config.indexing` defaults). A cue measured on the proposal set
13 is
14 measured on candidates **production never serves**, so a "win" need not transfer. This
15 is
16 exactly how Gate-A scored **+0.074** on the proposal set yet **?0.008** on the served
17 set
18 (`scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md` vs `output/gateA_served_verify/`).
19
20 **The fix.** One module owns the two named windowings as :class:`WindowingPreset`s, and
21 the
22 SERVED preset is **single-sourced from the same Pydantic config defaults production
23 reads**
24 (:class:`backend.config.models.IndexingConfig`) ? eval and serve can no longer drift.
25 Helpers
26 build candidates under either preset, and :func:`served_windowing_from_config` lifts the
27 *actual* served windowing out of a live config (overrides included), so an A/B can
28 target the
29 operating point a given deployment truly serves.
30
31 The companion rule ? *a cue is promotable only if it does NOT regress at the SERVED
32 windowing* ? lives in :mod:`backend.eval.promotion` (which composes `eval_stats` +
33 `per_user_gate`). This module is the **what windowing** ; that one is the **promotion
34 gate**.
35
36 Pure, offline, stdlib + config only. Nothing here changes existing eval behaviour until
37 a
38 caller opts in (`distill_local` re-points its module-level presets here, byte-
39 identically).
40
41 """
42 from __future__ import annotations
43
44 from dataclasses import dataclass
45 from typing import Any, Dict, List, Mapping, Tuple
46
47 from backend.config.models import IndexingConfig
48
49 Interval = Tuple[float, float]
50
51 __all__ = [
52 "WindowingPreset",
53 "SERVED",
54 "PROPOSAL",
55 "PRESETS",
56 "served_windowing_from_config",
57 "build_windows",
58 "windows_to_intervals",
59]
```

```

45
46
47 @dataclass(frozen=True)
48 class WindowingPreset:
49 """A named (velocity_thresh, merge_gap, min_window_duration) windowing operating
point.
50
51 ``velocity_thresh`` ? normalised per-second shuttle velocity above which a frame
counts as
52 "in flight"; ``merge_gap`` ? seconds within which active frames coalesce into one
window;
53 ``min_window_duration`` ? windows shorter than this are dropped. These are exactly
the three
54 knobs ``TrackNetRunner.trajectory_to_action_windows`` takes, so a preset *is* the
windowing.
55 """
56
57 name: str
58 velocity_thresh: float
59 merge_gap: float
60 min_window_duration: float
61 #: 0 = OFF (default). When > 0, windows longer than this are split at their largest
internal
62 #: inactivity gap ? the bounded-windowing over-merge fix (W1,
RALLY_QUALITY_RESEARCH.md §6).
63 max_window_duration: float = 0.0
64 #: When True, ``max_window_duration`` is a HARD cap ? an over-long window with no
internal
65 #: inactivity gap is chopped at its midpoint until ? the cap (continuously-active
fix). OFF by default.
66 hard_cap: bool = False
67 #: R4 rally-END inactivity trim (s, 0 = OFF): trim a window's post-rally slow-drift
tail back to
68 #: the last frame whose trailing window stays dense with FAST motion. The measured
rally-end fix.
69 inactivity_end: float = 0.0
70 #: velocity above which motion counts as "rally" (vs slow drift) for the R4 trim;
min fast-fraction.
71 inactivity_rally_thresh: float = 0.08
72 inactivity_min_density: float = 0.34
73 #: R5 blob-fallback (default OFF). After the cap split + R4 trim, midpoint-chop any
group still
74 #: longer than ``blob_factor`` × ``max_window_duration`` (the gap-less, R4-survived
blob ?
75 #: mahadevpura-1's ~11× residual) and flag it. Runs AFTER R4 (vs ``hard_cap``
before) and the
76 #: factor keeps it surgical (a normal long rally is left alone).
77 blob_fallback: bool = False
78 blob_factor: float = 4.0
79
80 def as_tuple(self) -> Tuple[float, float, float]:
81 """``(velocity_thresh, merge_gap, min_window_duration)`` ? the legacy tuple
shape the
82 ``trajectory_to_action_windows`` / ``distill_local`` call sites already speak.
(The
83 bounded-windowing knob ``max_window_duration`` is passed separately by
``build_windows``.)"""
84 return (self.velocity_thresh, self.merge_gap, self.min_window_duration)
85
86 def to_dict(self) -> Dict[str, Any]:
87 return {
88 "name": self.name,
89 "velocity_thresh": self.velocity_thresh,
90 "merge_gap": self.merge_gap,
91 "min_window_duration": self.min_window_duration,

```

```

92 "max_window_duration": self.max_window_duration,
93 "hard_cap": self.hard_cap,
94 "inactivity_end": self.inactivity_end,
95 "inactivity_rally_thresh": self.inactivity_rally_thresh,
96 "inactivity_min_density": self.inactivity_min_density,
97 "blob_fallback": self.blob_fallback,
98 "blob_factor": self.blob_factor,
99 }
100
101
102 def _served_preset_from_defaults() -> WindowingPreset:
103 """The SERVED windowing, lifted from the SAME Pydantic config defaults production
104 reads.
105 ``TrajectoryHybridSegmenter.process_video`` reads exactly these three:
106 - velocity: ``idx_cfg[<family>].velocity_threshold`` (default 0.02 ?
107 wasb/fusion/tracknet)
108 - merge_gap: ``idx_cfg.dead_time_merge_gap`` (default 2.0)
109 - min window: ``idx_cfg.min_action_window_duration`` (default 2.0)
110
111 Sourcing them from :class:`IndexingConfig`'s defaults (not a hand-typed tuple) is
112 the whole
113 point: change a served default in ``config/models.py`` and the eval SERVED preset
114 follows,
115 so the two can never silently diverge again.
116 """
117 idx = IndexingConfig() # construct with defaults ? the served operating point
118 return WindowingPreset(
119 name="served",
120 velocity_thresh=float(idx.wasb.velocity_threshold), # == fusion/tracknet
121 default 0.02
122 merge_gap=float(idx.dead_time_merge_gap),
123 min_window_duration=float(idx.min_action_window_duration),
124 max_window_duration=float(idx.max_window_duration), # W1 bounded-windowing
125 (default 0=OFF)
126 hard_cap=bool(idx.window_hard_cap), # W1 hard-cap fallback
127 (default OFF)
128 inactivity_end=float(idx.window_inactivity_end), # R4 rally-end trim
129 (default 0=OFF)
130 inactivity_rally_thresh=float(idx.inactivity_rally_thresh),
131 inactivity_min_density=float(idx.inactivity_min_density),
132 blob_fallback=bool(idx.window_blob_fallback), # R5 blob-fallback
133 (default OFF)
134 blob_factor=float(idx.window_blob_factor),
135)
136
137
138 #: The COARSE windowing production actually serves (config defaults; ~0.02/2.0/2.0). A
139 cue is
140 #: only promotable if it holds up HERE ? the operating point real users get.
141 SERVED: WindowingPreset = _served_preset_from_defaults()
142
143 #: The recall-oriented PROPOSAL windowing the offline substrate is built on
144 (deliberately
145 #: permissive: propose many, let the classifier/gate filter). NOT what production serves
146 ? a
147 #: win here must be re-confirmed on SERVED before promotion. Mirrors the historical
148 #: ``distill_local.PROPOSAL`` tuple (0.005/1.0/0.5) so the existing golden JSONs stay
149 valid.
150 PROPOSAL: WindowingPreset = WindowingPreset(
151 name="proposal", velocity_thresh=0.005, merge_gap=1.0, min_window_duration=0.5
152)
153
154 #: Name ? preset, so a CLI / config string ("served" | "proposal") selects one.
155 PRESETS: Dict[str, WindowingPreset] = {SERVED.name: SERVED, PROPOSAL.name: PROPOSAL}

```

```

144
145
146 def served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset:
147 """The SERVED windowing a *specific* live config serves (overrides honoured).
148
149 Reproduces ``TrajectoryHybridSegmenter.process_video``'s knob reads exactly, against
150 an
151 arbitrary config (the typed :class:`AppConfig` or a plain dict) and segmenter
152 ``family``
153 (``"wasb"`` / ``"fusion"`` / ``"tracknet"``). Use this when a deployment overrode a
154 served
155 default in ``config.json`` ? the module-level :data:`SERVED` is the *default*
156 operating point;
157 this is *this config's* operating point. Falls back to the served defaults for any
158 absent key.
159 """
160 idx = _as_mapping(config, "indexing")
161 family_cfg = _as_mapping(idx, family)
162 return WindowingPreset(
163 name=f"served:{family}",
164 velocity_thresh=float(family_cfg.get("velocity_threshold",
165 SERVED.velocity_thresh)),
166 merge_gap=float(idx.get("dead_time_merge_gap", SERVED.merge_gap)),
167 min_window_duration=float(idx.get("min_action_window_duration",
168 SERVED.min_window_duration)),
169 max_window_duration=float(idx.get("max_window_duration",
170 SERVED.max_window_duration)),
171 hard_cap=bool(idx.get("window_hard_cap", SERVED.hard_cap)),
172 inactivity_end=float(idx.get("window_inactivity_end", SERVED.inactivity_end)),
173 inactivity_rally_thresh=float(idx.get("inactivity_rally_thresh",
174 SERVED.inactivity_rally_thresh)),
175 inactivity_min_density=float(idx.get("inactivity_min_density",
176 SERVED.inactivity_min_density)),
177 blob_fallback=bool(idx.get("window_blob_fallback", SERVED.blob_fallback)),
178 blob_factor=float(idx.get("window_blob_factor", SERVED.blob_factor)),
179)
180
181 def _as_mapping(obj: Any, key: str) -> Mapping[str, Any]:
182 """``obj[key]`` as a plain mapping, whether ``obj`` is a dict or a typed config
183 section
184 (``AppConfig.get`` returns nested sections as dicts). Empty mapping if absent/non-
185 mapping."""
186 if obj is None:
187 return {}
188 if isinstance(obj, Mapping):
189 sub = obj.get(key, {})
190 elif hasattr(obj, "get"):
191 sub = obj.get(key, {})
192 else:
193 sub = getattr(obj, key, {})
194 return sub if isinstance(sub, Mapping) else {}
195
196 def build_windows(points: List[Any], fps: float, frame_width: float,
197 preset: WindowingPreset) -> List[Dict[str, Any]]:
198 """Trajectory ``points`` ? candidate windows under ``preset`` (the one windowing
199 call site).
200
201 A thin wrapper over ``TrackNetRunner.trajectory_to_action_windows`` that takes a
202 named preset
203 instead of three loose floats, so eval candidate-building and the served path
204 reference the
205 SAME knobs by name. Imported lazily to keep this module import-light (no detector
206 deps)."""

```

```

193 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
194
195 vt, mg, mwd = preset.as_tuple()
196 return TrackNetRunner.trajectory_to_action_windows(
197 points, fps=fps, frame_width=frame_width,
198 velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd,
199 max_window_duration=preset.max_window_duration,
200 window_hard_cap=preset.hard_cap,
201 window_inactivity_end=preset.inactivity_end,
202 inactivity_rally_thresh=preset.inactivity_rally_thresh,
203 inactivity_min_density=preset.inactivity_min_density,
204 window_blob_fallback=preset.blob_fallback,
205 window_blob_factor=preset.blob_factor)
206
207
208 def windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]:
209 """`[{start, end, ...}]` ? `[(start, end)]` ? the interval shape the scorer/gate
210 speak."""
211 return [(w["start"], w["end"]) for w in windows]

```

### 33. assistant (2026-06-18T01:29:16.953Z)

Now let me revisit the `aggregate` function in `rally\_seg\_eval.py` more carefully. There's a subtle issue I want to verify:

```

```python
def aggregate(rows):
    out = {"n": len(rows)}
    for k in _AGG_KEYS:
        vals = [r[k] for r in rows]
        out[k] = eval_stats.mean_with_ci(vals) if vals else None
    return out
```

```

This requires every row to have every key in `\_AGG\_KEYS`. If `rows` is empty, `vals=[]`, returns None. Fine. The `format\_report` calls `\_ci(a['f1'])` ? `\_ci(None)` returns "?". Fine.

Now let me look at a more subtle concern in `tolerance\_metrics.over\_seg\_guard`. When `use\_rates=True` but a row is missing `merge\_rate`/`split\_rate`, `\_cost` does `float(row["merge\_rate"])` ? `KeyError`. But the docstring says rows from `rally\_seg\_eval` always emit these. Let me verify the contract holds: `score\_intervals` outputs `merge\_rate`, `split\_rate`, `merge\_count`, `split\_count`. Yes. So when called through `compare`, the rows have these keys. The function is also called directly in tests. Not a production bug.

Let me now investigate a real concern. In `experiment.\_calibrate\_and\_tune`, the `tune\_threshold` grid loop calls `predict(...)` then `score(...).f1`. But notice the `tune\_threshold` path: when `variant.tune\_threshold` is True but the variant has no `feature\_names` (not learned), `\_calibrate\_and\_tune` is only invoked inside `if variant.tune\_threshold or variant.calibrate` within the `recipe\_lovo` and `honest` branch. `recipe\_lovo = bool(variant.feature\_names)` and `classifier\_model` is None. So `\_calibrate\_and\_tune` only runs for learned recipes. Fine.

Let me now look at `batch.py`, `harness.py`, and `eval\_partial\_labels.py`, plus the `annotations.py` parse\_timestamp which gt\_loader uses.

### 34. user (2026-06-18T01:29:18.386Z)

```

1 """Load hand-annotated ground-truth rally timestamps from CSV.
2
3 Annotation file format (one file per video):
4
5 start,end
6 00:01:12,00:01:39
7 102.5,131.0
8

```

```

9 - A header row `start,end` is optional (auto-detected and skipped).
10 - Each timestamp is either decimal seconds (`102.5`) or colon-separated
11 `MM:SS` / `HH:MM:SS` (`00:01:12`). Both forms may be mixed in one file.
12 - Blank lines and `#` comment lines are ignored.
13 """
14
15 import csv
16 import logging
17 from typing import List, Tuple
18
19 logger = logging.getLogger(__name__)
20
21 Interval = Tuple[float, float]
22
23
24 def parse_timestamp(value: str) -> float:
25 """Parse a timestamp string to float seconds.
26
27 Accepts decimal seconds or colon-separated MM:SS / HH:MM:SS.
28 Raises ValueError on anything unparseable so bad annotations fail loudly.
29 """
30 value = value.strip()
31 if not value:
32 raise ValueError("empty timestamp")
33 if ":" in value:
34 parts = value.split(":")
35 if len(parts) > 3:
36 raise ValueError(f"too many ':' segments in timestamp '{value}'")
37 total = 0.0
38 for part in parts:
39 total = total * 60 + float(part)
40 return total
41 return float(value)
42
43
44 def load_annotations(csv_path: str) -> List[Interval]:
45 """Load and validate ground-truth rally intervals from a CSV file.
46
47 Delegates to the canonical loader (gt_loader.py, ADR-008 prereq), which accepts BOTH
48 golden conventions: positional/headered ``start,end`` (the collector export) and the
49 rally-annotator ``rally_number,start_time,end_time,...`` schema. Strict: raises
50 ValueError on malformed rows (bad timestamp, start >= end) so labeling mistakes
51 surface immediately rather than silently skewing scores. Returns intervals sorted by
52 start time.
53 """
54 from backend.eval.gt_loader import load_gt_intervals
55 intervals = load_gt_intervals(csv_path, strict=True)
56 logger.info("Loaded %d ground-truth rallies from %s", len(intervals), csv_path)
57 return intervals
58
59
60 def write_scaffold(csv_path: str) -> None:
61 """Write an empty annotation file with just the header, for hand-filling."""
62 with open(csv_path, "w", newline="") as f:
63 writer = csv.writer(f)
64 writer.writerow(["start", "end"])
65

```

### 35. user (2026-06-18T01:29:18.656Z)

```

1 """Evaluate the rally pipeline against PARTIAL human golden labels.
2
3 The VLC rally-annotator produces a human-rated CSV (rally_number,start_time,end_time,
4 ending_reason,sport). While a video is only partially tagged, the labels cover a
5 PREFIX of the video ? every rally up to some point is marked, and everything after is

```



```

6 simply un-watched (not "no rally there"). Scoring naively against such labels is wrong:
7 the detector's *correct* rallies in the un-labeled tail would be counted as false
8 positives and crush precision.
9
10 This harness fixes that by restricting the evaluation to the LABELED WINDOW
11 [window_start, window_end] (window_end defaults to the last labeled rally's end) and
12 dropping predicted segments that fall outside it. Within that window the human labels
13 ARE complete, so precision/recall/F1 are honest.
14
15 It reuses everything from the existing eval stack (no new windowing/metrics/gate logic):
16 - calibrate_wasb.make_predict_fn / load_golden_gts / BASELINE
17 - pipeline.detectors.rally_gate.apply_gate (net-crossing "gate A")
18 - eval.metrics.match_intervals / score
19 - export_wasb_segments.build_comparison / write_comparison_csv (per-rally detail CSV)
20
21 Run:
22 python -m backend.eval.eval_partial_labels \
23 --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
24 --labels "C:/Users/avidu/Projects/Annotation Setup/Collect/Adarsh and
25 --fps 59.94 --frame-width 1920
26 ""
27 from __future__ import annotations
28
29 import argparse
30 import os.path as osp
31 from typing import List, Optional, Tuple
32
33 from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
34 from backend.eval.metrics import score
35 from backend.eval.export_wasb_segments import build_comparison, write_comparison_csv
36 from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
37 from backend.pipeline.detectors import rally_gate
38
39 Interval = Tuple[float, float]
40 Config = Tuple[float, float, float]
41
42 # Operating points to compare. The tuned config wins on its fit-video but over-segments
43 # and does not transfer; baseline (+ net-crossing gate) is the stronger general point.
44 TUNED: Config = (0.05, 1.0, 1.0)
45
46
47 def restrict_to_window(intervals: List[Interval], w0: float, w1: float) ->
48 List[Interval]:
49 """Keep intervals that overlap the labeled window [w0, w1].
50
51 A predicted segment is evaluated iff it overlaps the labeled region; segments wholly
52 in the un-labeled tail (start >= w1) or before the window (end <= w0) are dropped so
53 they are neither rewarded nor penalized. GTs are assumed already inside the window.
54 """
55 return [(s, e) for (s, e) in intervals if e > w0 and s < w1]
56
57 def _eval_point(predict_fn, cfg: Config, points, fps: float, gts: List[Interval],
58 window: Tuple[float, float], gate_crossings: int, iou: float) -> dict:
59 """Score one (config, gate) operating point inside the labeled window."""
60 preds = predict_fn(cfg)
61 if gate_crossings > 0 and points:
62 preds = rally_gate.apply_gate(points, preds, fps, min_crossings=gate_crossings)
63 preds = restrict_to_window(preds, *window)
64 sc = score(preds, gts, iou)
65 return {"cfg": cfg, "gate": gate_crossings, "n_pred": len(preds), "score": sc,
66 "preds": preds}
67

```

```

68 def run(trajjectory_csv: str, labels_csv: str, fps: float, frame_width: float,
69 window_start: float = 0.0, window_end: Optional[float] = None,
70 gate_crossings: int = 2, iou: float = 0.5,
71 comparison_out: Optional[str] = None) -> dict:
72 gts_all = load_golden_gts(labels_csv)
73 if not gts_all:
74 raise SystemExit(f"no usable human rallies in {labels_csv}")
75 if window_end is None:
76 window_end = max(e for _, e in gts_all)
77 window = (window_start, window_end)
78 gts = restrict_to_window(gts_all, *window)
79
80 predict_fn = make_predict_fn(trajjectory_csv, fps, frame_width)
81 points = TrackNetRunner.parse_trajectory_csv(trajjectory_csv)
82 traj_end = (max(p.frame for p in points) / fps) if points else 0.0
83
84 print("=== Pipeline vs PARTIAL human labels ===")
85 print(f"labels: {labels_csv}")
86 print(f" {len(gts_all)} human rallies; labeled window [{window_start:.1f}s,
{window_end:.1f}s] "
87 f"-> {len(gts)} rallies in-window")
88 print(f"trajectory covers ~0-{traj_end:.1f}s "
89 f"({'covers the window' if traj_end >= window_end - 1 else 'WARNING: shorter
than window'})")
90 print(f"IoU>={iou}; net-crossing gate K={gate_crossings}\n")
91
92 # Compare the standard operating points: baseline / baseline+gate / tuned /
tuned+gate.
93 points_for_gate = points
94 candidates = [
95 ("baseline", BASELINE, 0),
96 (f"baseline+gate{gate_crossings}", BASELINE, gate_crossings),
97 ("tuned", TUNED, 0),
98 (f"tuned+gate{gate_crossings}", TUNED, gate_crossings),
99]
100 results = []
101 header = f"{'operating point':<18} {'pred':>4} {'TP':>3} {'FN':>3} {'FP':>3} " \
102 f"{'prec':>6} {'recall':>6} {'F1':>6} {'mIoU':>6} {'sErr':>6} {'eErr':>6}"
103 print(header)
104 print("-" * len(header))
105 for name, cfg, gate in candidates:
106 r = _eval_point(predict_fn, cfg, points_for_gate, fps, gts, window, gate, iou)
107 sc = r["score"]
108 r["name"] = name
109 results.append(r)
110 print(f"{name:<18} {r['n_pred']:>4} {sc.true_positives:>3}
{sc.false_negatives:>3} "
111 f"{sc.false_positives:>3} {sc.precision:>6.3f} {sc.recall:>6.3f}
{sc.f1:>6.3f} "
112 f"{sc.mean_iou:>6.3f} {sc.mean_start_error:>6.2f}
{sc.mean_end_error:>6.2f}")
113
114 best = max(results, key=lambda r: r["score"].f1)
115 print(f"\nbest operating point by F1: {best['name']} (F1 {best['score'].f1:.3f})")
116
117 # Per-rally detail for the best point: which human rallies matched/missed + extra
fires.
118 comp = build_comparison(best["preds"], gts, iou)
119 comp_out = comparison_out or osp.join("output", "adarsh_avi_vs_human.csv")
120 write_comparison_csv(comp_out, comp)
121 print(f"per-rally detail ({best['name']}) -> {comp_out}")
122
123 return {
124 "window": window, "n_human_in_window": len(gts),
125 "results": [{"name": r["name"], "cfg": r["cfg"], "gate": r["gate"],

```

```

126 "n_pred": r["n_pred"], **r["score"].as_dict()} for r in results],
127 "best": best["name"], "comparison_csv": comp_out,
128 }
129
130
131 # Grid centered on the two levers the human-GT failure analysis flagged:
132 # min_window_duration DOWN (recover short 3-5s rallies) and merge_gap UP (heal
133 # fragmented long rallies). Gate is swept on/off because it can over-prune.
134 DEFAULT_GRID = [
135 (vt, mg, mwd)
136 for vt in (0.01, 0.02, 0.03, 0.05)
137 for mg in (1.0, 2.0, 3.0, 4.0)
138 for mwd in (0.5, 1.0, 1.5, 2.0)
139]
140
141
142 def sweep(trjectory_csv: str, labels_csv: str, fps: float, frame_width: float,
143 window_start: float = 0.0, window_end: Optional[float] = None,
144 gate_options: Tuple[int, ...] = (0, 2), iou: float = 0.5,
145 grid: Optional[List[Config]] = None, top: int = 12) -> dict:
146 """Grid-search (velocity, merge_gap, min_dur) x gate vs human labels, in-window.
147
148 WARNING: this is fit-on-self on ONE video -> optimistic. It shows the achievable
149 ceiling + which levers matter, NOT a transferable config. The honest operating
150 point comes from leave-one-video-out once >=3 videos are labeled.
151 """
152 grid = grid or DEFAULT_GRID
153 gts_all = load_golden_gts(labels_csv)
154 if not gts_all:
155 raise SystemExit(f"no usable human rallies in {labels_csv}")
156 if window_end is None:
157 window_end = max(e for _, e in gts_all)
158 window = (window_start, window_end)
159 gts = restrict_to_window(gts_all, *window)
160 predict_fn = make_predict_fn(trjectory_csv, fps, frame_width)
161 points = TrackNetRunner.parse_trajectory_csv(trjectory_csv)
162
163 rows: List[dict] = []
164 for cfg in grid:
165 for gate in gate_options:
166 r = _eval_point(predict_fn, cfg, points, fps, gts, window, gate, iou)
167 sc = r["score"]
168 rows.append({"cfg": cfg, "gate": gate, "n_pred": r["n_pred"], "f1": sc.f1,
169 "precision": sc.precision, "recall": sc.recall, "mean_iou":
170 sc.mean_iou,
171 "tp": sc.true_positives, "fp": sc.false_positives, "fn":
172 sc.false_negatives})
173 rows.sort(key=lambda x: (x["f1"], x["mean_iou"]), reverse=True)
174
175 print(f"=== Config sweep vs human labels ({len(gts)} rallies in "
176 f"[{window_start:.0f},{window_end:.0f}]s; {len(grid)}x{len(gate_options)} "
177 f"configs) ===")
178 print("NOTE: fit-on-self on ONE video = optimistic; real test is leave-one-video- "
179 f"out.\n")
180 hdr = f"{'#':>3} {'veloc':>6} {'merge':>6} {'minDur':>6} {'gate':>4} " \
181 f"{'F1':>6} {'prec':>6} {'recall':>6} {'mIoU':>6} {'TP':>3} {'FP':>3} "
182 f"{'FN':>3}"
183 print(hdr)
184 print("-" * len(hdr))
185 for i, r in enumerate(rows[:top], 1):
186 vt, mg, mwd = r["cfg"]
187 print(f"{i:>3} {vt:>6.3f} {mg:>6.1f} {mwd:>6.1f} {r['gate']:>4} "
188 f"{r['f1']:>6.3f} {r['precision']:>6.3f} {r['recall']:>6.3f} "
189 f"{r['mean_iou']:>6.3f} "
190 f"{r['tp']:>3} {r['fp']:>3} {r['fn']:>3}")

```

```

185 b = rows[0]
186 print(f"\nbest: velocity={b['cfg'][0]} merge_gap={b['cfg'][1]} min_dur={b['cfg'][2]}
"
187 f"gate={b['gate']} -> F1 {b['f1']:.3f} (P {b['precision']:.3f} / R
{b['recall']:.3f})")
188 return {"window": window, "n_gt": len(gts), "ranked": rows, "best": b}
189
190
191 def main() -> None:
192 ap = argparse.ArgumentParser(description=__doc__)
193 ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV
(Frame,Visibility,X,Y)")
194 ap.add_argument("--labels", required=True, help="human golden CSV
(rally_number,start_time,end_time,...)")
195 ap.add_argument("--fps", type=float, default=59.94)
196 ap.add_argument("--frame-width", type=float, default=1920.0)
197 ap.add_argument("--window-start", type=float, default=0.0)
198 ap.add_argument("--window-end", type=float, default=None,
199 help="end of the labeled region (default: last labeled rally end)")
200 ap.add_argument("--gate", type=int, default=2, help="net-crossing gate K (0=off, 2
recommended)")
201 ap.add_argument("--iou", type=float, default=0.5)
202 ap.add_argument("--comparison-out", default=None)
203 ap.add_argument("--sweep", action="store_true",
204 help="grid-search configs vs labels (fit-on-self; finds the ceiling
+ best levers)")
205 ap.add_argument("--top", type=int, default=12, help="how many sweep rows to print")
206 args = ap.parse_args()
207 if args.sweep:
208 sweep(args.trajectory, args.labels, args.fps, args.frame_width,
209 window_start=args.window_start, window_end=args.window_end,
210 iou=args.iou, top=args.top)
211 else:
212 run(args.trajectory, args.labels, args.fps, args.frame_width,
213 window_start=args.window_start, window_end=args.window_end,
214 gate_crossings=args.gate, iou=args.iou, comparison_out=args.comparison_out)
215
216
217 if __name__ == "__main__":
218 main()
219

```

### 36. assistant (2026-06-18T01:29:27.274Z)

I noticed something important in `parse\_timestamp`. Let me re-examine it carefully:

```

```python
def parse_timestamp(value: str) -> float:
    value = value.strip()
    if not value:
        raise ValueError("empty timestamp")
    if ":" in value:
        parts = value.split(":")
        if len(parts) > 3:
            raise ValueError(...)
        total = 0.0
        for part in parts:
            total = total * 60 + float(part)
        return total
    return float(value)
```

```

This is fine. `gt\_loader` catches `ValueError`. But there's a subtle thing: `parse\_timestamp` doesn't validate that minutes/seconds are < 60, but that's lenient, not a bug.

Now, a key observation about ``gt_loader.load_gt_intervals`` with ``strict=False``. The ``except ValueError`` only catches `ValueError`. But what catches a row where ``parse_timestamp`` is fed something like a string with a non-numeric value? ``float("abc")`` raises `ValueError`. Good. That's handled.

Let me look more carefully at a potential real bug. In ``gt_loader``, the negative-start check is ``if start < 0``. But what about a negative END that's still `> start`? E.g. `start=-5, end=-2`: caught by ``start < 0``. What about `start=0, end` being negative but ``start >= end``? Caught. Fine.

Let me focus now on the ``calibrate_wasb.load_golden_gts`` since ``eval_partial_labels`` calls it. But that's outside my core area. Let me look at the most promising lead: the ``score_calibration.py`` (`fit_platt/fit_isotonic`) which ``experiment.py`` depends on. Let me also check ``classifier.py`` since `experiment` uses `LogisticRegression`.

### 37. user (2026-06-18T01:29:28.149Z)

```
1 """Per-cue score calibration (course-correction C2) ? make ``confidence`` mean
something.
2
3 A CNN softmax, an audio onset score, and an integer net-crossing count live on wildly
4 different scales. Fed raw into the learned arbiter (the ADR-004 logistic regression),
the
5 most over-confident cue dominates and starves complementary cues. Calibrating each
6 producer's score to a probability *before* it becomes a feature column makes the
7 ``(value, confidence)`` semantics real and the arbiter's weights stable. See
8 ``docs/ANALYZERS_AND_RECONCILERS.md`` §13/C2.
9
10 Two calibrators, both standard:
11 - **Platt scaling** ? fit ``p = sigmoid(a*s + b)`` (parametric; good for monotone,
12 roughly-sigmoidal cues; uses Platt's target smoothing to avoid 0/1 overfitting).
13 - **Isotonic regression** (pool-adjacent-violators) ? non-parametric monotone fit; more
14 flexible, needs a bit more data.
15
16 Plus ``brier_score`` and ``expected_calibration_error`` to *measure* calibration. All
17 **additive**, pure (stdlib only), deterministic ? fits run in the numpy-free lanes too.
18 Calibrators are tiny dataclasses (their params are JSON-friendly), so a fitted
calibrator
19 can be pinned alongside a variant the way the classifier is.
20 """
21 from __future__ import annotations
22
23 import bisect
24 import math
25 from dataclasses import dataclass
26 from typing import List, Sequence, Tuple
27
28 _EPS = 1e-12
29
30
31 def _sigmoid(z: float) -> float:
32 if z >= 0:
33 return 1.0 / (1.0 + math.exp(-z))
34 e = math.exp(z)
35 return e / (1.0 + e)
36
37
38 def _clip01(p: float) -> float:
39 return min(1.0 - _EPS, max(_EPS, p))
40
41
42 @dataclass(frozen=True)
43 class PlattCalibrator:
44 """``p = sigmoid(a*score + b)``. Call it on a raw cue score to get a probability."""
45 a: float
46 b: float
```

```

47
48 def __call__(self, s: float) -> float:
49 return _clip01(_sigmoid(self.a * s + self.b))
50
51
52 @dataclass(frozen=True)
53 class IsotonicCalibrator:
54 """Piecewise-constant monotone map from raw score to probability (PAVA fit).
55
56 ``xs`` are sorted training scores, ``ys`` the non-decreasing fitted probabilities;
57 a query returns the fitted value of the largest ``x <= score`` (clipped at the
ends).
58 """
59 xs: List[float]
60 ys: List[float]
61
62 def __call__(self, s: float) -> float:
63 if not self.xs:
64 return 0.5
65 idx = bisect.bisect_right(self.xs, s) - 1
66 if idx < 0:
67 idx = 0
68 return self.ys[idx]
69
70
71 def fit_platt(scores: Sequence[float], labels: Sequence[int],
72 iters: int = 200, lr: float = 0.5) -> PlattCalibrator:
73 """Fit Platt scaling by gradient descent on (smoothed) log-loss.
74
75 Scores are standardised internally for a stable fit, then folded back so the
returned
76 ``a``/``b`` apply to the **raw** score. Platt target smoothing
77 (``t+ = (N+ + 1)/(N+ + 2)``, ``t- = 1/(N- + 2)``) keeps the fit off the 0/1 rails.
78 Degenerate inputs (empty, zero-variance, single class) fall back to a sane constant
map.
79 """
80 n = len(scores)
81 if n == 0 or n != len(labels):
82 return PlattCalibrator(a=0.0, b=0.0) # ? p = 0.5 everywhere
83 mean = sum(scores) / n
84 var = sum((s - mean) ** 2 for s in scores) / n
85 sd = math.sqrt(var)
86 n_pos = sum(1 for y in labels if y >= 0.5)
87 n_neg = n - n_pos
88 if sd <= _EPS or n_pos == 0 or n_neg == 0:
89 # No usable score spread or one class: best constant probability = base rate.
90 base = _clip01(n_pos / n)
91 return PlattCalibrator(a=0.0, b=math.log(base / (1.0 - base)))
92 t_pos = (n_pos + 1.0) / (n_pos + 2.0)
93 t_neg = 1.0 / (n_neg + 2.0)
94 z = [(s - mean) / sd for s in scores]
95 targets = [t_pos if y >= 0.5 else t_neg for y in labels]
96
97 a, b = 1.0, 0.0 # in standardised space
98 for _ in range(max(1, iters)):
99 ga = gb = 0.0
100 for zi, t in zip(z, targets):
101 p = _sigmoid(a * zi + b)
102 d = p - t
103 ga += d * zi
104 gb += d
105 a -= lr * ga / n
106 b -= lr * gb / n
107 # Fold standardisation back to raw score: z = (s-mean)/sd ? a*z+b = (a/sd)*s + (b -
a*mean/sd).

```

```

108 return PlattCalibrator(a=a / sd, b=b - a * mean / sd)
109
110
111 def _pava(ys: Sequence[float]) -> List[float]:
112 """Pool-adjacent-violators: nearest non-decreasing fit to ``ys`` (in given
113 order)."""
114 blocks: List[List[float]] = [] # each [value, weight]
115 for y in ys:
116 blocks.append([float(y), 1.0])
117 while len(blocks) >= 2 and blocks[-2][0] > blocks[-1][0]:
118 v2, w2 = blocks.pop()
119 v1, w1 = blocks.pop()
120 blocks.append([(v1 * w1 + v2 * w2) / (w1 + w2), w1 + w2])
121 fitted: List[float] = []
122 for value, weight in blocks:
123 fitted.extend([value] * int(round(weight)))
124 return fitted
125
126 def fit_isotonic(scores: Sequence[float], labels: Sequence[int]) -> IsotonicCalibrator:
127 """Fit isotonic (monotone) calibration via PAVA on score-sorted labels."""
128 n = len(scores)
129 if n == 0 or n != len(labels):
130 return IsotonicCalibrator(xs=[], ys=[])
131 order = sorted(range(n), key=lambda i: (scores[i], i))
132 xs = [float(scores[i]) for i in order]
133 ys = [_clip01(v) for v in _pava([float(labels[i]) for i in order])]
134 return IsotonicCalibrator(xs=xs, ys=ys)
135
136
137 def brier_score(probs: Sequence[float], labels: Sequence[int]) -> float:
138 """Mean squared error of probabilities vs 0/1 labels (lower = better calibrated)."""
139 n = len(probs)
140 if n == 0 or n != len(labels):
141 return 0.0
142 return sum((p - y) ** 2 for p, y in zip(probs, labels)) / n
143
144
145 def expected_calibration_error(probs: Sequence[float], labels: Sequence[int],
146 n_bins: int = 10) -> float:
147 """Expected calibration error: |accuracy - confidence| per equal-width bin,
148 weighted."""
149 n = len(probs)
150 if n == 0 or n != len(labels):
151 return 0.0
152 bins = max(1, n_bins)
153 bin_sum_p = [0.0] * bins
154 bin_sum_y = [0.0] * bins
155 bin_cnt = [0] * bins
156 for p, y in zip(probs, labels):
157 b = min(bins - 1, int(p * bins))
158 bin_sum_p[b] += p
159 bin_sum_y[b] += y
160 bin_cnt[b] += 1
161 ece = 0.0
162 for i in range(bins):
163 if bin_cnt[i]:
164 conf = bin_sum_p[i] / bin_cnt[i]
165 acc = bin_sum_y[i] / bin_cnt[i]
166 ece += (bin_cnt[i] / n) * abs(acc - conf)
167 return ece
168
169
170 def calibration_report(probs: Sequence[float], labels: Sequence[int],
171 n_bins: int = 10) -> Tuple[float, float]:

```

```

171 """Convenience: ``brier, ece`` for a set of probabilities vs labels."""
172 return brier_score(probs, labels), expected_calibration_error(probs, labels, n_bins)
173

```

### 38. assistant (2026-06-18T01:29:40.438Z)

I found a real concern in `expected\_calibration\_error`. The bin index computation:

```

```python
b = min(bins - 1, int(p * bins))
```

```

If `p` is negative (probabilities should be in  $[0,1]$ , but `probs` could be raw or out-of-range), `int(p\*bins)` could be negative ? `bin\_sum\_p[negative\_index]` wraps around. But the contract is probabilities in  $[0,1]$ . With `\_clip01`, calibrators produce values in  $[_EPS, 1-_EPS]$ . So `int(p\*bins)` ?  $[0, \text{bins}-1]$ . Not reachable with valid input. This is a measurement helper given probabilities. I'll set this aside ? not clearly reachable.

Now let me reconsider the `\_pava` function's `int(round(weight))` reconstruction. This is a subtle correctness issue: weights are accumulated as floats (1.0 each), so weight is always an exact integer-valued float. `int(round(weight))` is safe. Fine.

Let me now go back and look very carefully at a real candidate in `tolerance\_metrics.tolerance\_sweep`'s claim and `boundary\_error\_report`. Actually, let me reconsider the `over\_seg\_guard` `\_aligned` ? the by-name path. The docstring says "Optional name keys align the two lists by video (else by position)". The condition:

```

```python
if base and cand and all("name" in b for b in base) and all("name" in c for c in cand):
```

```

If `base` has names but `cand` doesn't (or vice versa), it falls to positional. But there's an asymmetry issue: the positional fallback `[(base[i], cand[i]) for i in range(min(len))` pairs by position even if the lists are in different video orders. In `compare()`, the rows come from `evaluate()` which iterates `golden` in the same order for both base and cand. Actually no ? `compare` builds `paired` by name already and passes aligned lists. So positional == by-name there. Fine.

Let me now look at the harness.py and batch.py, and then check `calibrate\_wasb.load\_golden\_gts` since it's used by partial labels.

### 39. user (2026-06-18T01:29:41.387Z)

```

1 """Score the pipeline's detected rallies against ground-truth annotations.
2
3 Reads predictions straight from the SQLite DB (no pipeline re-run, no API cost),
4 so scoring is fast, deterministic, and free. Two prediction "stages" can be
5 scored:
6
7 - ``candidates`` ? raw detector windows (``candidate_segments`` table). What
8 the shuttle/person detector found, before AI confirmation.
9 - ``final`` ? confirmed play segments (``play_segments`` table). After the
10 AI handoff trimmed/validated boundaries.
11
12 Comparing the two tells you whether the weak link is *detection* (candidates
13 already miss rallies) or *segmentation* (candidates catch them but final is
14 wrong) ? see docs/EVALUATION.md.
15 """
16
17 import logging
18 from dataclasses import dataclass, field
19 from typing import Any, List, Dict, Optional
20
21 from backend.infrastructure.database import Database
22 from backend.eval import metrics

```



```

23 from backend.eval.metrics import Interval, Score, MatchResult
24
25 logger = logging.getLogger(__name__)
26
27 STAGES = ("candidates", "final")
28
29
30 @dataclass
31 class StageReport:
32 stage: str
33 pred_intervals: List[Interval]
34 score: Score
35 match_result: MatchResult
36 pr_curve: List[Score] = field(default_factory=list)
37
38
39 @dataclass
40 class EvalReport:
41 video_id: str
42 iou_threshold: float
43 gt_intervals: List[Interval]
44 stages: Dict[str, StageReport] = field(default_factory=dict)
45
46
47 def _rows_to_intervals(rows: List[dict]) -> List[Interval]:
48 """Pull (start_time, end_time) out of DB segment rows, sorted by start."""
49 ivs = [(float(r["start_time"]), float(r["end_time"])) for r in rows]
50 ivs.sort(key=lambda iv: iv[0])
51 return ivs
52
53
54 def get_predictions(db: Database, video_id: str, stage: str,
55 detector: Optional[str] = None) -> List[Interval]:
56 """Fetch predicted rally intervals for one stage from the DB."""
57 if stage == "candidates":
58 rows = db.query_candidate_segments(video_id, detector=detector)
59 elif stage == "final":
60 rows = db.query_play_segments(video_id, {})
61 else:
62 raise ValueError(f"unknown stage {stage!r}; expected one of {STAGES}")
63 return _rows_to_intervals(rows)
64
65
66 def evaluate(db: Database, video_id: str, gt_intervals: List[Interval],
67 stages: List[str], iou_threshold: float = 0.5,
68 detector: Optional[str] = None,
69 with_pr_curve: bool = False) -> EvalReport:
70 """Score the requested stages for one video against ground truth."""
71 report = EvalReport(video_id=video_id, iou_threshold=iou_threshold,
72 gt_intervals=gt_intervals)
73 for stage in stages:
74 preds = get_predictions(db, video_id, stage, detector=detector)
75 mr = metrics.match_intervals(preds, gt_intervals, iou_threshold)
76 s = metrics.score(preds, gt_intervals, iou_threshold, match_result=mr)
77 curve = metrics.pr_curve(preds, gt_intervals) if with_pr_curve else []
78 report.stages[stage] = StageReport(stage=stage, pred_intervals=preds,
79 score=s, match_result=mr, pr_curve=curve)
80 return report
81
82
83 # ----- reporting
84
85 def _fmt_ts(seconds: float) -> str:
86 seconds = max(0, int(seconds))
87 h, rem = divmod(seconds, 3600)

```

```

88 m, s = divmod(rem, 60)
89 return f"{h:02d}:{m:02d}:{s:02d}"
90
91
92 def report_to_dict(report: EvalReport) -> dict:
93 """JSON-serialisable view of an EvalReport."""
94 out: Dict[str, Any] = {
95 "video_id": report.video_id,
96 "iou_threshold": report.iou_threshold,
97 "num_ground_truth": len(report.gt_intervals),
98 "stages": {},
99 }
100 for stage, sr in report.stages.items():
101 out["stages"][stage] = {
102 "score": sr.score.as_dict(),
103 "pr_curve": [s.as_dict() for s in sr.pr_curve],
104 }
105 return out
106
107
108 def format_report_text(report: EvalReport, show_failures: bool = False) -> str:
109 """Human-readable console report for one video."""
110 lines = [
111 "=" * 64,
112 f" Rally evaluation ? video '{report.video_id}'",
113 f" Ground-truth rallies: {len(report.gt_intervals)} | IoU threshold:
{report.iou_threshold}",
114 "=" * 64,
115 f" {'Stage':<12} {'Pred':>5} {'TP':>4} {'FP':>4} {'FN':>4} "
116 f" {'Prec':>6} {'Rec':>6} {'F1':>6} {'mIoU':>6} {'dStart':>7} {'dEnd':>7}",
117 "-" * 64,
118]
119 for stage in report.stages:
120 s = report.stages[stage].score
121 lines.append(
122 f" {stage:<12} {s.num_pred:>5} {s.true_positives:>4} {s.false_positives:>4}
"
123 f" {s.false_negatives:>4} {s.precision:>6.2f} {s.recall:>6.2f} {s.f1:>6.2f} "
124 f" {s.mean_iou:>6.2f} {s.mean_start_error:>6.1f} {s.mean_end_error:>6.1f}s"
125)
126 lines.append("=" * 64)
127
128 if show_failures:
129 for stage, sr in report.stages.items():
130 misses = [report.gt_intervals[i] for i in
sr.match_result.unmatched_gt_indices]
131 fps = [sr.pred_intervals[i] for i in sr.match_result.unmatched_pred_indices]
132 lines.append(f"\n [{stage}] missed rallies (false negatives):
{len(misses)}")
133 for iv in misses:
134 lines.append(f" - {_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
135 lines.append(f" [{stage}] spurious detections (false positives):
{len(fps)}")
136 for iv in fps:
137 lines.append(f" - {_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
138
139 return "\n".join(lines)
140

```

#### 40. user (2026-06-18T01:29:41.907Z)

```

1 """Batch evaluation across a collector export manifest.
2
3 Phase 3 of the collector->indexer bridge: walk the `manifest.json` emitted by
4 sports-data-collector's `curate export`, (optionally) process each video through

```

```

5 a segmenter so detections land in the indexer DB, then score every video's
6 detected rallies against its ground-truth CSV and aggregate the results.
7
8 This module holds the *pure* pieces (path resolution, stage selection, metric
9 aggregation) so they can be unit-tested without touching video files, the GPU,
10 or any API. The orchestration/side-effects live in `scripts/run_phase3_eval.py`.
11 """
12
13 import os
14 from typing import Dict, List, Optional
15
16
17 # Segmenters that only ever populate the `final` play_segments table (no raw
18 # candidate stage), so scoring `candidates` for them would be meaninglessly empty.
19 _FINAL_ONLY_SEGMENTERS = {"motion", "gemini"}
20
21
22 def resolve_video_path(local_path: Optional[str], videos_root: str) -> Optional[str]:
23 """Resolve a manifest `local_path` to an absolute path.
24
25 The collector stores paths like ``./data/videos\\shuttleset_1.mp4`` relative
26 to its own repo root and with Windows separators. Normalise separators, strip
27 a leading ``./``, and resolve relative paths against `videos_root` (the
28 collector root). Returns None for a missing/empty path.
29 """
30 if not local_path:
31 return None
32 p = local_path.replace("\\", "/")
33 if p.startswith("./"):
34 p = p[2:]
35 # Cross-platform: treat Windows drive-letter absolute paths (C:/...) as absolute
36 # even when running on Linux (os.path.isabs("C:/...") == False on Unix).
37 # This keeps collector manifest paths working in tests and mixed environments.
38 if (len(p) > 1 and p[1] == ":" and p[0].isalpha()) or os.path.isabs(p):
39 return os.path.normpath(p)
40 return os.path.normpath(os.path.join(videos_root, p))
41
42
43 def default_stages_for(segmenter: str, requested: str = "auto") -> List[str]:
44 """Pick which prediction stages to score.
45
46 ``auto`` scores both stages for two-stage detectors (yolo_hybrid/tracknet_*)
47 but only ``final`` for segmenters that don't emit candidates. An explicit
48 request ('candidates' | 'final' | 'both') overrides the heuristic.
49 """
50 if requested == "both":
51 return ["candidates", "final"]
52 if requested in ("candidates", "final"):
53 return [requested]
54 # auto
55 if segmenter in _FINAL_ONLY_SEGMENTERS:
56 return ["final"]
57 return ["candidates", "final"]
58
59
60 def _safe_div(num: float, den: float) -> float:
61 return num / den if den else 0.0
62
63
64 def aggregate(per_video: List[dict], stages: List[str]) -> Dict[str, dict]:
65 """Aggregate per-video stage scores into corpus-level metrics.
66
67 `per_video` is a list of dicts shaped like `report_to_dict` output (each has
68 `stages[stage][score]` with tp/fp/fn/mean_iou/...). For each stage we
69 compute **micro** precision/recall/F1 (pooling TP/FP/FN across all videos) and

```

```

70 a TP-weighted mean IoU, plus the macro mean of per-video F1.
71 """
72 out: Dict[str, dict] = {}
73 for stage in stages:
74 tp = fp = fn = 0
75 iou_weighted = 0.0
76 f1s: List[float] = []
77 n_videos = 0
78 for rep in per_video:
79 sr = rep.get("stages", {}).get(stage)
80 if not sr:
81 continue
82 s = sr["score"]
83 tp += s["true_positives"]
84 fp += s["false_positives"]
85 fn += s["false_negatives"]
86 iou_weighted += s["mean_iou"] * s["true_positives"]
87 f1s.append(s["f1"])
88 n_videos += 1
89 precision = _safe_div(tp, tp + fp)
90 recall = _safe_div(tp, tp + fn)
91 out[stage] = {
92 "videos": n_videos,
93 "tp": tp, "fp": fp, "fn": fn,
94 "micro_precision": precision,
95 "micro_recall": recall,
96 "micro_f1": _safe_div(2 * precision * recall, precision + recall),
97 "macro_f1": _safe_div(sum(f1s), len(f1s)),
98 "mean_iou": _safe_div(iou_weighted, tp),
99 }
100 return out
101
102
103 def format_aggregate(agg: Dict[str, dict]) -> str:
104 """Human-readable corpus-level summary table."""
105 lines = [
106 "=" * 78,
107 " CORPUS AGGREGATE",
108 "=" * 78,
109 f" {'Stage':<12} {'Vids':>4} {'TP':>5} {'FP':>5} {'FN':>5} "
110 f"{'mPrec':>6} {'mRec':>6} {'microF1':>8} {'macroF1':>8} {'mIoU':>6}",
111 "-" * 78,
112]
113 for stage, a in agg.items():
114 lines.append(
115 f" {stage:<12} {a['videos']:>4} {a['tp']:>5} {a['fp']:>5} {a['fn']:>5} "
116 f"{a['micro_precision']:>6.2f} {a['micro_recall']:>6.2f} "
117 f"{a['micro_f1']:>8.2f} {a['macro_f1']:>8.2f} {a['mean_iou']:>6.2f}"
118)
119 lines.append("=" * 78)
120 return "\n".join(lines)
121

```

#### 41. assistant (2026-06-18T01:29:55.914Z)

These look solid. Now let me revisit a careful trace of the `\_greedy\_match\_1to1` tiebreak claim in `tolerance_metrics`. The sort key:

```

```python
pairs.append((abs(p[0]-g[0]) + abs(p[1]-g[1]), float(i*1e-6 + j*1e-9), i, j))
```

```

The secondary key `i*1e-6 + j*1e-9` is meant for deterministic tiebreak. With large `i/j` this could collide or float-precision issues, but it then also has `i, j` as further tiebreaks in the tuple. So determinism holds via `(cost, secondary, i, j)`. Fine.

Now I want to focus on one thing I flagged in `mean\_with\_ci`: `std = statistics.pstdev(vals) if n > 1 else 0.0`. For n=1, std=0. Fine.

Let me reconsider a genuine logic concern in `paired\_sign\_test`. The two-sided p-value:

```
```python
k = min(wins, losses)
tail = sum(math.comb(n_eff, j) for j in range(k+1)) * (0.5 ** n_eff)
p = min(1.0, 2.0 * tail)
```
```

The exact two-sided sign test p-value should be  $2 * P(X \leq k)$  where  $X \sim \text{Binomial}(n_{\text{eff}}, 0.5)$  and  $k = \min(\text{wins}, \text{losses})$ . When  $\text{wins} = \text{losses}$  ( $k = n_{\text{eff}}/2$ ),  $P(X \leq k)$  includes more than half the mass, so  $2 * \text{tail}$  could exceed 1, but it's capped at 1.0. For the standard case this is the conventional formula. Correct.

Now let me check the `calibrate\_wasb.load\_golden\_gts` used by `eval\_partial\_labels.run` ?  
`window\_end = max(e for \_, e in gts\_all)`. If `gts\_all` is non-empty (guarded). Fine.

Let me look at the most subtle area now ? the `IsotonicCalibrator.\_\_call\_\_` and `\_pava` reconstruction when there are ties in scores. Let me reconsider `fit\_isotonic`: `xs` may contain duplicates. `IsotonicCalibrator.\_\_call\_\_` does `bisect.bisect\_right(self.xs, s) - 1`. With duplicate xs, this finds the rightmost matching. The `ys` has the same length as `xs` (PAVA produces one fitted value per input via `int(round(weight))` expansion).

Wait ? here's a subtle concern. `\_pava` reconstructs via `fitted.extend([value] \* int(round(weight)))`. The total length should equal `len(ys)` input. Each block's weight is the count of pooled elements (sum of 1.0s), always an exact integer. So `len(fitted) == len(ys)`. Good, `xs` and `ys` align. No bug.

Let me now check the `classifier.py` LogisticRegression since experiment.py's `predict\_proba` and `.fit` contracts matter for crash-on-input.

## 42. user (2026-06-18T01:29:57.123Z)

```
1 """A tiny, dependency-free logistic-regression classifier (numpy only).
2
3 The offline learned segmenter (ADR-004) must run locally with no external API and
4 no heavy ML stack ? the project already depends on numpy, and sklearn is not
5 installed. This implements just enough: standardized features, balanced-class
6 gradient descent, sigmoid probabilities, and JSON (de)serialization so a trained
7 model is a small, human-readable, version-controllable file.
8
9 It deliberately stays simple: the rally-window features are few and roughly
10 linearly separable (high player motion + sustained activity => real rally), so a
11 well-regularized linear model is a sensible, inspectable baseline. Swap in a
12 richer model later behind the same fit/predict_proba/save/load interface.
13 """
14
15 import json
16 from typing import List, Optional
17
18 import numpy as np
19
20
21 class LogisticRegression:
22 def __init__(self, lr: float = 0.1, epochs: int = 1000, l2: float = 1e-3):
23 self.lr = lr
24 self.epochs = epochs
25 self.l2 = l2
26 self.w: Optional[np.ndarray] = None
27 self.b: float = 0.0
28 self.mean: Optional[np.ndarray] = None
29 self.std: Optional[np.ndarray] = None
30 self.feature_names: Optional[List[str]] = None
```

```

31
32 @staticmethod
33 def _sigmoid(z: np.ndarray) -> np.ndarray:
34 # Clip for numerical stability (avoid overflow in exp).
35 return 1.0 / (1.0 + np.exp(-np.clip(z, -30, 30)))
36
37 def _standardize(self, X: np.ndarray) -> np.ndarray:
38 return (X - self.mean) / self.std
39
40 def fit(self, X, y, feature_names: Optional[List[str]] = None,
41 class_weight: str = "balanced") -> "LogisticRegression":
42 """Fit with optional balanced class weighting (rally windows are imbalanced
43 ? many spurious candidates per real rally)."""
44 X = np.asarray(X, dtype=float)
45 y = np.asarray(y, dtype=float)
46 if X.ndim != 2 or X.shape[0] == 0:
47 raise ValueError("X must be a non-empty 2D array")
48 self.feature_names = feature_names
49
50 self.mean = X.mean(axis=0)
51 self.std = X.std(axis=0)
52 self.std[self.std == 0] = 1.0 # guard constant features
53 Xs = self._standardize(X)
54 n, d = Xs.shape
55
56 if class_weight == "balanced":
57 n_pos = max(1, int(y.sum()))
58 n_neg = max(1, int((1 - y).sum()))
59 w_pos, w_neg = n / (2.0 * n_pos), n / (2.0 * n_neg)
60 else:
61 w_pos = w_neg = 1.0
62 sw = np.where(y == 1, w_pos, w_neg)
63
64 self.w = np.zeros(d)
65 self.b = 0.0
66 for _ in range(self.epochs):
67 p = self._sigmoid(Xs @ self.w + self.b)
68 err = (p - y) * sw
69 grad_w = Xs.T @ err / n + self.l2 * self.w
70 grad_b = float(err.mean())
71 self.w -= self.lr * grad_w
72 self.b -= self.lr * grad_b
73 return self
74
75 def predict_proba(self, X) -> np.ndarray:
76 if self.w is None:
77 raise RuntimeError("model is not fitted")
78 Xs = self._standardize(np.asarray(X, dtype=float))
79 return self._sigmoid(Xs @ self.w + self.b)
80
81 def predict(self, X, threshold: float = 0.5) -> np.ndarray:
82 return (self.predict_proba(X) >= threshold).astype(int)
83
84 # --- persistence (small JSON, no pickle/joblib) -----
85
86 def to_dict(self) -> dict:
87 return {
88 "model": "logreg",
89 "lr": self.lr, "epochs": self.epochs, "l2": self.l2,
90 "w": self.w.tolist() if self.w is not None else None,
91 "b": self.b,
92 "mean": self.mean.tolist() if self.mean is not None else None,
93 "std": self.std.tolist() if self.std is not None else None,
94 "feature_names": self.feature_names,
95 }

```

```

96
97 @classmethod
98 def from_dict(cls, d: dict) -> "LogisticRegression":
99 m = cls(lr=d.get("lr", 0.1), epochs=d.get("epochs", 1000), l2=d.get("l2", 1e-3))
100 m.w = np.asarray(d["w"], dtype=float) if d.get("w") is not None else None
101 m.b = float(d.get("b", 0.0))
102 m.mean = np.asarray(d["mean"], dtype=float) if d.get("mean") is not None else
None
103 m.std = np.asarray(d["std"], dtype=float) if d.get("std") is not None else None
104 m.feature_names = d.get("feature_names")
105 return m
106
107 def save(self, path: str) -> None:
108 with open(path, "w") as f:
109 json.dump(self.to_dict(), f, indent=2)
110
111 @classmethod
112 def load(cls, path: str) -> "LogisticRegression":
113 with open(path) as f:
114 return cls.from_dict(json.load(f))
115

```

#### 43. assistant (2026-06-18T01:30:11.986Z)

Now I want to dig into a potential real bug in `experiment.py`'s `predict` when a learned variant is given `\_feature\_matrix` with empty feature rows. If `variant.feature\_names` is set but `vc.intervals` is empty, `\_feature\_matrix` returns `[[[] for \_ in vc.intervals] = [[]]`. Then `model.predict\_proba([])` ? `np.asarray([], dtype=float)` is 1D shape (0,). `Xs = self.\_standardize(...)` with mean/std arrays ? `(X - self.mean)` where X is shape (0,) and mean is shape (d,). This would attempt broadcasting (0,) - (d,) which fails... but the `kept` list and zip would handle empty. Actually `predict\_proba([])` with empty list: `np.asarray([])` has shape (0,), 1D. `Xs @ self.w` where Xs is (0,) and w is (d,) ? shape mismatch. But this only matters if `vc.intervals` is empty AND it's a learned variant. In that path, the gate mask is `[]`, and predict proceeds to call predict\_proba on empty matrix. Let me check: would a video with zero candidate intervals reach here? In `evaluate\_variant`, every video is scored. If a video has zero candidates, this could crash. But this is an edge case unlikely in the golden corpus (videos always have candidates). Lower confidence ? set aside.

Let me now focus and re-examine the most concrete potential bug I've seen. Let me re-read the `over\_seg\_guard` net cost handling when `use\_rates=True` but rows lack `merge\_rate`/`split\_rate`. Actually let me re-examine the `compare()` call chain in rally\_seg\_eval more carefully, because there's a potential issue with `tolerance\_sweep` not being used. Let me reconsider.

Let me re-examine `bootstrap\_ci`'s handling when all values are identical but n>1. `test\_bootstrap\_ci\_degenerate` asserts `bootstrap\_ci([0.5,0.5,0.5]) == (0.5,0.5)`. With identical values, every resample mean is 0.5, so lo=hi=0.5. Correct.

Let me look at a more promising area ? the `regression.py` incommensurability guard. There's a subtle issue: when `b\_det` is not None and `b\_det != detector`. If the baseline was saved WITHOUT a detector (`b\_det=None`) but the run specifies a detector, this check is skipped (no flag). Conversely, if baseline had a detector but run passes `detector=None`, it flags. That's an intentional asymmetry (None baseline = legacy, lenient). Not a bug.

Let me verify the actual claim about `\_hungarian\_match\_1to1` and greedy disagreeing. Let me run the tolerance\_metrics tests to see if there are edge cases. But first let me check whether numpy/scipy are available here.

#### 44. user (2026-06-18T01:30:14.432Z)

```

numpy 2.4.6
scipy 1.17.1

```

#### 45. assistant (2026-06-18T01:30:18.894Z)

Both available. Let me run the `tolerance_metrics` and `eval_stats` tests to confirm baseline green, then probe edge cases.

#### 46. user (2026-06-18T01:30:22.368Z)

```
..... [100%]
27 passed in 1.35s
```

#### 47. assistant (2026-06-18T01:30:31.421Z)

Now let me probe a specific edge case I'm suspicious about: the ``over_seg_guard`` ``net_cost_ok`` direction and the ``_aligned`` by-name path when there are duplicate names. But more importantly, let me reconsider a genuine logic bug candidate in ``tolerance_metrics.match_1to1``.

The Hungarian path: when there are forced ineligible assignments, the code drops them with ``cost[i,j] < big``. But there's a subtle issue: the padding cells (when ``n != m``) are also set to ``big``. For a real eligible pair, cost could be exactly 0 (perfect match). The check ``cost[i,j] < big`` correctly keeps eligible (`cost < 1e6`) and drops ineligible/padding. Good.

But wait ? what if an eligible pair has cost computed as ``abs(...) + abs(...)`` and that happens to be... no, max eligible cost is `tau_start+tau_end = 3.5`, far below `1e6`. Fine.

Let me reconsider the greedy vs hungarian determinism. Actually, let me now examine a real concern about ``boundary_error_report``'s ``max(preds, key=lambda p: _overlap(p, g), default=None)``. When multiple preds tie on overlap, ``max`` returns the first. That's deterministic. Fine.

Let me look at the one area I haven't fully traced: ``per_user_eval.py``.

#### 48. user (2026-06-18T01:30:31.990Z)

```
1 """Held-out-user LEARNING-CURVE eval ? "<K videos to superior quality" as a release
gate.
2
3 The personalization feature's promise is operational, not aspirational: *after the user
4 contributes K of their own videos, the personalized (candidate) recipe should beat the
5 generic (control) one on the user's STILL-UNSEEN footage.* This module turns that
promise
6 into a falsifiable criterion by sweeping the corpus size ``k`` and asking, at each
``k``,
7 **does the candidate beat control on a held-out split, honestly?**
8
9 It is the per-user analogue of `fusion_compare`: pure-CPU, no new scoring/stats math.
Every
10 fold reuses the shipped experiment primitives ?
11
12 - `experiment.predict` ? variant ? held-out intervals (static variants need no fit);
13 - `experiment._train` ? fit a learned variant on the corpus split (the same honest
14 train-on-others step `evaluate_variant`'s LOVO uses, here train-on-the-k-corpus);
15 - `metrics.score` ? per-video held-out F1;
16 - `eval_stats.paired_delta_ci` ? the honest paired ?F1 (bootstrap CI + exact sign
test, C1).
17
18 **Split semantics (held-out USER split, NOT leave-one-out).** At each ``k`` the first
``k``
19 videos are the *personalization corpus* (fit/operating-point source) and the remaining
20 `len(videos) - k` are *held-out* (scored, never seen during fit). This mirrors the
product
21 reality ? the user has uploaded ``k`` videos and we want quality on their next one ? and
keeps
22 ``k`` and the evaluation set cleanly disjoint, so a measured lift can't be in-sample.
23
24 The headline `smallest_k_with_lift` is the smallest ``k`` at which the candidate beats
25 control with the **CI clearing 0 AND the sign test agreeing** (B wins on more held-out
```





```

81 """Sweep corpus size ``k`` and report where the candidate first honestly beats
control.
82
83 For one user's ``videos``, for each ``k`` in ``min_k .. len(videos) - 1``: fit/score
both
84 variants on the first ``k`` videos (the personalization corpus) and evaluate on the
remaining
85 held-out videos. Per-held-out-video F1s are paired B?A and summarised with
86 ``eval_stats.paired_delta_ci`` (mean ?F1 + bootstrap CI + exact sign test).
87
88 Returns ``{n_videos, min_k, iou, per_k, smallest_k_with_lift}`` where each ``per_k``
record is::
89
90 {k, n_heldout, control_f1, candidate_f1,
91 mean_delta, ci_low, ci_high, ci_excludes_zero, sign_test}
92
93 ``control_f1`` / ``candidate_f1`` are the mean held-out F1 of each variant at that
``k``.
94 ``smallest_k_with_lift`` is the smallest ``k`` whose record shows a real lift ?
candidate
95 above control, the ?F1 CI clearing 0, AND the sign test agreeing (B wins on strictly
more
96 held-out videos than it loses) ? else ``None`` (the honest "never wins" answer).
97
98 Pure + deterministic given ``seed`` (the only randomness is the CI bootstrap).
Raises
99 `ExperimentError` on unlabeled videos or a too-small corpus (no valid ``k`` to
sweep).
100
101 """
102 n = len(videos)
103 for vc in videos:
104 if vc.gts is None:
105 raise ExperimentError(
106 f"{vc.name} has no golden labels; learning_curve needs labeled held-out
videos")
107
108 if min_k < 1:
109 raise ExperimentError(f"min_k must be >= 1 (got {min_k})")
110 if n <= min_k:
111 # Every k in [min_k, n-1] needs k >= min_k corpus videos AND >= 1 held-out
video, so
112 # n must exceed min_k. Too-few videos ? no sweepable k; report an empty, honest
curve.
113 return {
114 "n_videos": n,
115 "min_k": min_k,
116 "iou": iou,
117 "per_k": [],
118 "smallest_k_with_lift": None,
119 }
120
121 per_k: List[Dict[str, Any]] = []
122 smallest_k_with_lift: Optional[int] = None
123 for k in range(min_k, n):
124 corpus = videos[:k]
125 heldout = videos[k:]
126 control_f1s = _heldout_f1s(control_variant, corpus, heldout, iou)
127 candidate_f1s = _heldout_f1s(candidate_variant, corpus, heldout, iou)
128
129 paired = _stats.paired_delta_ci(control_f1s, candidate_f1s, seed=seed)
130 sign_test = paired["sign_test"]
131 # Honest lift = the candidate is above control AND the CI clears 0 AND the sign
test
132 # agrees (B wins on strictly more held-out videos than it loses) ? the promotion
bar.
133 control_mean = sum(control_f1s) / len(control_f1s) if control_f1s else 0.0

```

```

132 candidate_mean = sum(candidate_f1s) / len(candidate_f1s) if candidate_f1s else
0.0
133 has_lift = (
134 candidate_mean > control_mean
135 and bool(paired["ci_excludes_zero"])
136 and paired["mean_delta"] > 0.0
137 and sign_test["wins"] > sign_test["losses"]
138)
139 per_k.append({
140 "k": k,
141 "n_heldout": len(heldout),
142 "control_f1": control_mean,
143 "candidate_f1": candidate_mean,
144 "mean_delta": paired["mean_delta"],
145 "ci_low": paired["ci_low"],
146 "ci_high": paired["ci_high"],
147 "ci_excludes_zero": bool(paired["ci_excludes_zero"]),
148 "sign_test": sign_test,
149 })
150 if has_lift and smallest_k_with_lift is None:
151 smallest_k_with_lift = k
152
153 return {
154 "n_videos": n,
155 "min_k": min_k,
156 "iou": iou,
157 "per_k": per_k,
158 "smallest_k_with_lift": smallest_k_with_lift,
159 }
160

```

#### 49. assistant (2026-06-18T01:30:48.881Z)

Now I've covered the core stats area thoroughly. Let me reconsider one thing that caught my eye earlier and verify it carefully ? the `paired\_delta\_ci` in `eval\_stats` passes `eps` to the sign test but NOT to the bootstrap. Let me also re-examine `per\_user\_gate.should\_promote\_for\_user` n==0 path more carefully ? it constructs `paired\_sign\_test([], eps)` which is fine.

Let me re-examine the most suspicious thing I want to verify: in `per\_user\_gate`, the `sign\_agrees` and CI logic. When `n < min\_n` but `n > 0`, it computes `delta = paired\_delta\_ci(...)` . With small n (e.g. n=1), `paired\_delta\_ci` ? `mean\_with\_ci([d])` ? bootstrap with n=1 returns `(d, d)` . `ci\_excludes\_zero` = `(d > 0) or (d < 0)` . So for a single positive video, `ci\_excludes\_zero=True` ! But the floor check `not floor\_met` returns first (n=1 < min\_n=5). So it never promotes. Good ? floor guards it.

But wait ? in `per\_user\_regression.evaluate\_no\_regression`, there is NO minimum-N floor. Let me re-examine. With n=1:

- `paired\_delta\_ci` ? bootstrap\_ci([d]) returns `(d, d)` .
- `ci\_excludes\_zero = (d>0) or (d<0)` = True for any nonzero single delta.
- `regressed = ci\_excludes\_zero and ci\_high < 0` ? True if single video has candidate worse.
- `improved = ci\_excludes\_zero and ci\_low > 0 and sign\_agrees` . `sign\_agrees = (sign\_p <= 0.05) and (wins>losses)` . With n\_eff=1, sign test  $p = 2 * \text{comb}(1,0) * 0.5^1 = 2 * 0.5 = 1.0$  . So `sign\_p=1.0 > 0.05` ? `sign\_agrees=False` . So `improved=False` . Good ? can't accept on n=1.

But `regressed` on n=1 with a single worse video ? `regressed=True` , REJECT. That's conservative and acceptable per the docstring (it's a no-regression guard; rejecting a single-video regression is the intent). Not a bug.

Now, the same n=1 reasoning for `promotion.served\_regresses`: `mean\_regresses = mean\_delta < -eps` , `ci\_confidently\_negative = ci\_high < -eps` . With a single video, ci\_high = mean\_delta. Both consistent. Fine.

Let me now carefully look at one more area where I have genuine suspicion: the `bootstrap\_ci` when n>1 but `n\_boot` is small. Not relevant.

Let me take a very close look at the `\_AGG\_KEYS` aggregate and the `mean\_with\_ci`  
`ci\_excludes\_zero` not present. Actually let me reconsider whether `aggregate` could KeyError.  
Every row from `score\_intervals` has all `\_AGG\_KEYS`. The dstart/dend medians are present. Yes.

Let me now run the full eval test suite to be sure my understanding of contracts is correct, and  
check the per\_user and promotion tests.

#### 50. user (2026-06-18T01:31:05.398Z)

```
..... [39%]
..... [78%]
.....SSSSS..... [100%]
179 passed, 5 skipped, 669 deselected in 11.11s
```

#### 51. assistant (2026-06-18T01:31:12.257Z)

All green. Now let me focus hard on finding actual defects rather than confirming correctness.  
Let me reconsider a few specific things that smell off.

Looking back at `regression.py` line 142-147 ? the precision/recall regression check. `base\_p,  
base\_r = b["precision"], b["recall"]`. If the baseline JSON was hand-edited or from an older  
format missing these keys, `b["precision"]` ? KeyError. But baselines are written by  
`save\_baseline` which always includes them. Not reachable in practice.

Let me re-examine `experiment.\_calibrate\_and\_tune` threshold tuning more carefully ? there's a  
subtle issue with the threshold grid and the `score` import. Inside the loop it calls  
`predict(variant, vc, model=model, threshold=t, calibrator=calibrator)`. But `predict` for a  
learned variant requires `variant.feature\_names` set. That's guaranteed in the recipe\_lovo path.  
Fine.

Now let me reconsider the `splits.py` and `training.py` which I haven't read ?  
`training.label\_candidates` is a contract used by experiment.

#### 52. user (2026-06-18T01:31:13.212Z)

```
1 """Build a training set for the offline, API-free learned rally segmenter.
2
3 Phase 4 of the collector->indexer bridge (see docs/ROADMAP.md, ADR-004). The idea:
4 a detector (YOLO player-motion today, optionally WASB shuttle-motion later)
5 produces candidate rally windows with features; the collector's ground-truth
6 rally intervals tell us which windows are real. A lightweight classifier learns
7 that mapping, replacing the paid Gemini refinement with a local model.
8
9 This module is the *data layer* ? pure and unit-tested:
10
11 - `label_candidates` ? generic: label windows positive/negative by whether they
12 match a ground-truth rally (same IoU matching the evaluator uses, so training
13 labels are consistent with how we score).
14 - `extract_yolo_features` ? YOLO-specific feature vector from a candidate row's
15 persisted `stats`. Pluggable: a WASB substrate supplies its own feature_fn.
16 - `build_training_set` ? glue: rows + GT -> (X, y, intervals).
17
18 No model training or I/O here; that lives in the trainer/segmenter once the
19 substrate is chosen from the Phase 3 candidate-recall result.
20 """
21
22 from typing import Callable, Dict, List, Tuple
23
24 from backend.eval.metrics import Interval, match_intervals
25
26 # YOLO Stage-1 persists these per candidate window in `candidate_segments.stats`
27 # (see HybridYoloSegmenter). Duration is derived from the window bounds.
28 YOLO_FEATURE_NAMES = [
29 "duration", "peak_velocity", "mean_velocity", "active_frame_count",
```

```

"track_id_count",
30]
31
32
33 def extract_yolo_features(row: Dict) -> List[float]:
34 """Feature vector for one YOLO candidate row (dict from query_candidate_segments).
35
36 `stats` is already JSON-deserialized by the DB layer. Missing fields default
37 to 0.0 so a sparse/old row still yields a well-formed vector.
38 """
39 stats = row.get("stats") or {}
40 duration = float(row["end_time"]) - float(row["start_time"])
41 return [
42 duration,
43 float(stats.get("peak_velocity", 0.0)),
44 float(stats.get("mean_velocity", 0.0)),
45 float(stats.get("active_frame_count", 0)),
46 float(stats.get("track_id_count", 0)),
47]
48
49
50 def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
51 iou_threshold: float = 0.5) -> List[int]:
52 """Label each candidate window 1 (real rally) or 0 (spurious).
53
54 A window is positive iff it is matched one-to-one to a ground-truth rally at
55 `iou_threshold` ? the *same* matching the evaluator uses for true positives,
56 so the classifier is trained on exactly what we score. Substrate-agnostic:
57 works for YOLO or WASB-derived windows alike.
58 """
59 mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
60 positive = {m.pred_index for m in mr.matches}
61 return [1 if i in positive else 0 for i in range(len(candidate_intervals))]
62
63
64 def build_training_set(
65 candidate_rows: List[Dict],
66 gt_intervals: List[Interval],
67 iou_threshold: float = 0.5,
68 feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
69) -> Tuple[List[List[float]], List[int], List[Interval]]:
70 """Turn one video's candidate rows + ground truth into (X, y, intervals).
71
72 `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
73 features without changing the labeling logic. Returns the feature matrix `X`,
74 binary labels `y`, and the candidate intervals (handy for downstream merging).
75 """
76 intervals: List[Interval] = [
77 (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
78]
79 y = label_candidates(intervals, gt_intervals, iou_threshold)
80 X = [feature_fn(r) for r in candidate_rows]
81 return X, y, intervals
82

```

### 53. user (2026-06-18T01:31:13.669Z)

```

1 """Train/eval split guard (pure-logic core, in-container).
2
3 Spec from BACKLOG.md "Train/eval separation guardrails" / DEFERRED #13
4 and plan M0:
5 - Split unit = match (never clip / file MD5).
6 - load_split(manifest) -> (train_ids, eval_ids) keyed by match id.
7 - assert_disjoint(train_ids, eval_ids) raises on any overlap.
8 - This would have caught train/eval leakage in training/eval paths.

```

```

9 - Cross-repo collector manifest `split` field wiring + wiring into
10 distill_local / run_regression deferred to later PR.
11
12 Usage (once wired):
13 train, eval_ = load_split(manifest)
14 assert_disjoint(train, eval_)
15 """
16 from __future__ import annotations
17
18 import json
19 import os
20 from typing import Any, Dict, Iterable, List, Set, Tuple, Union, cast
21
22
23 def load_split(manifest: Union[str, Iterable[Dict[str, Any]], os.PathLike[str]]) ->
24 Tuple[Set[str], Set[str]]:
25 """Load (train_ids, eval_ids) sets keyed by match id.
26
27 manifest: path to JSON (str/bytes/PathLike) or iterable of dicts. Each entry should
28 have 'match_id'
29 (or 'id' fallback; must be non-empty str for real match key; name/video_id
30 deliberately not used as they are weak/Md5).
31
32 Raises ValueError on:
33 - non-list manifest after load
34 - no real match-level key (empty/falsy after explicit check)
35 - split label not exactly "train" or "eval"
36
37 The collector 'split' field and cross-repo wiring is deferred; this supports
38 synthetic or pre-split manifests for the pure-logic guard and tests.
39 """
40 if isinstance(manifest, (str, bytes, os.PathLike)):
41 with open(manifest, "r", encoding="utf-8-sig") as f:
42 raw = json.load(f)
43 if not isinstance(raw, list):
44 raise ValueError(f"manifest must be a list of entries, got {type(raw)}")
45 data = cast(List[Dict[str, Any]], raw)
46 else:
47 if isinstance(manifest, dict):
48 raise ValueError(f"manifest must be a list of entries, got
49 {type(manifest)}")
50 data = list(manifest)
51 if not isinstance(data, list):
52 raise ValueError(f"manifest must be a list of entries, got {type(data)}")
53
54 train_ids: Set[str] = set()
55 eval_ids: Set[str] = set()
56
57 for entry in data:
58 if not isinstance(entry, dict):
59 continue
60 # Explicit not-None + non-empty for real match key; no silent fallback to
61 name/video_id (weak/Md5)
62 mid = entry.get("match_id")
63 if mid is None or str(mid).strip() == "":
64 mid = entry.get("id")
65 if mid is None or str(mid).strip() == "":
66 raise ValueError(f"no real match-level key (match_id or id) in entry:
67 {entry}")
68 mid = str(mid).strip()
69 sp = str(entry.get("split", "")).strip().lower()
70 if sp not in ("train", "eval"):
71 raise ValueError(f"split label must be exactly 'train' or 'eval', got {sp!r}
72 in entry: {entry}")
73 if sp == "eval":

```

```

67 eval_ids.add(mid)
68 else:
69 train_ids.add(mid)
70
71 return train_ids, eval_ids
72
73
74 def assert_disjoint(train_ids: Iterable[str], eval_ids: Iterable[str]) -> None:
75 """Raise if any match-id overlap between train and eval sets.
76
77 This would have caught train/eval leakage (e.g. a video used for both
78 training a segmenter and for regression/eval baselines).
79 """
80 t = set(str(x) for x in train_ids)
81 e = set(str(x) for x in eval_ids)
82 overlap = t & e
83 if overlap:
84 raise ValueError(
85 f"Train/eval leakage detected on match ids: {sorted(overlap)}. "
86 "This would have caught train/eval leakage."
87)
88

```

#### 54. assistant (2026-06-18T01:31:26.148Z)

These are clean. I've now thoroughly reviewed the core scoring/stats area. Let me do one final careful pass on the one thing that still nags me ? the ``tolerance_metrics.over_seg_guard`` ``net_cost_ok`` semantics and a subtle issue in ``_aligned``: when ``base`` is empty but ``cand`` is non-empty (or vice versa), the ``if base and cand and ...`` is False, so it goes to positional: ``n = min(len(base), len(cand))`` = 0, returns ``[]``. Then ``base_net = 0``, ``cand_net = 0``, ``passes=True``. Edge but harmless.

Let me re-examine one genuinely suspicious item: in ``eval_stats.bootstrap_ci``, the ``confidence`` parameter isn't validated, and ``mean_with_ci`` returns ``"confidence": confidence``. Not a bug.

Let me reconsider the ``paired_delta_ci`` ``eps`` usage. It threads ``eps`` into ``paired_sign_test`` but the ``mean_with_ci`/bootstrap` doesn't use ``eps``. The docstring says the sign test uses ``eps``. This is consistent with intent. Not a bug.

Let me now look very carefully at something in ``calibration.kfold_indices`` that could be a real off-by-something. The round-robin:

```

```python
folds = [list(range(i, n, k)) for i in range(k)]
```

```

With `n=5, k=5`: `folds = `[[0],[1],[2],[3],[4]]``. Each test fold has 1 element, train has 4. Fine. With `n=3, k=5` ? ``k=max(2,min(5,3))=3``. `folds=`[[0],[1],[2]]``. Fine. With `n=2, k=5` ? `k=2`. `folds=`[[0],[1]]``. Fine. No bug.

Now let me carefully reconsider ``cross_validate`` ? ``select_best`` may return ``best_cfg=None`` if ``configs`` is empty (``best_val`` stays -1.0, ``best_cfg`` stays None). Then ``score(predict_fn(None), ...)``. But callers always pass non-empty configs. And if a config produces a metric of exactly -1.0... metrics are in `[0,1]`, never negative, so ``val > best_val`` (``val >= 0 > -1.0``) always picks the first config. So ``best_cfg`` is never None for non-empty configs. Fine.

I've done a thorough adversarial pass across the entire core scoring/stats area. Let me do one more targeted check on a real edge I want to confirm: ``IsotonicCalibrator`` reconstruction length when PAVA pools. Let me verify with a concrete test.

#### 55. user (2026-06-18T01:31:29.832Z)

```

pava [0.5, 0.5, 0.5, 0.5]
xs [0.1, 0.2, 0.3, 0.4]
ys [0.5, 0.5, 0.5, 0.5] len match True

```

call 0.5

**56. assistant (2026-06-18T01:31:34.114Z)**

Aligned correctly.

Let me do one final verification on something I want to be sure about ? the `match\_1to1` greedy vs Hungarian agreement is claimed but the greedy result is re-sorted by `mm[1]` (gt\_idx). Both return matches sorted by gt index. Let me confirm the count (cardinality) agrees between paths in a tricky case where greedy might be suboptimal. Let me construct a case where greedy picks a low-cost pair that blocks a higher-cardinality assignment.

**57. user (2026-06-18T01:31:39.659Z)**

```
greedy [(0, 0, -0.1, -0.09999999999999964), (1, 1, -0.5, -0.5)]
hungarian [(0, 0, -0.1, -0.09999999999999964), (1, 1, -0.5, -0.5)]
cardinality greedy/hung: 2 2
```

**58. assistant (2026-06-18T01:31:51.148Z)**

They agree, as the docstring claims for near-diagonal graphs. Even in cross-eligible cases greedy's "ascending combined error" heuristic generally matches max-cardinality here because rally intervals are temporally ordered.

I've now completed a thorough adversarial review of the entire `backend/eval/` core scoring/stats area:

- `eval\_stats.py` ? bootstrap CI, sign test, paired delta, grouped folds, OOF predictions: all edge cases (n=0/1, ties) handled correctly.
- `tolerance\_metrics.py` ? Hungarian/greedy 1:1 matching, percentile, boundary error, over-seg guard: empty-input guards present, alignment filtered, cost matrix correct.
- `rally\_seg\_eval.py` ? score\_intervals/aggregate/compare/CLI: aggregate handles empty, key contracts hold.
- `regression.py` ? incommensurability guards, baseline store: correct.
- `calibration.py` ? kfold, select\_best, cross\_validate, LOVO: edge cases fine.
- `per\_user\_gate.py`, `per\_user\_regression.py`, `per\_user\_eval.py`, `promotion.py` ? floor/CI/sign-test logic: consistent and conservative.
- `experiment.py` ? variant prediction, LOVO, calibration/tuning: contracts hold.
- `score\_calibration.py`, `classifier.py`, `metrics.py`, `segmentation\_metrics.py`, `gt\_loader.py`, `annotations.py`, `harness.py`, `batch.py`, `training.py`, `splits.py`, `windowing.py` ? all verified.

This is a mature codebase with 848 passing tests and multiple hardening sweeps. After tracing concrete triggers for every edge case I suspected (empty inputs, n=0/1, NaN/div-by-zero, Hungarian/greedy boundaries, bootstrap/sign-test correctness, loader key access), I found no real, reachable defect that a maintainer would agree is a bug. Every numeric edge case I probed is explicitly guarded, and the statistics match their documented contracts and tests.

Per the instructions, returning an empty candidates list is a valid and respected answer when no genuine bug is found. I will not manufacture findings.

**59. user (2026-06-18T01:31:52.485Z)**

Structured output provided successfully

**60. assistant (2026-06-18T01:32:03.697Z)**

I completed a thorough adversarial review of the `backend/eval/` core scoring/stats area and found no real, reachable defects.

What I traced (with concrete triggers) and confirmed correct:

```
- **`eval_stats.py`** ? `bootstrap_ci` (n=0?(0,0), n=1?(v,v), identical-values?point interval),
`paired_sign_test` exact two-sided binomial (n_eff=0?p=1.0; capped at 1.0), `paired_delta_ci`
```



```
(min-length pairing), `grouped_folds`/`out_of_fold_predictions` (singleton groups yield no fold
/ stay None). All edge cases guarded; matches its test pins.
- **`tolerance_metrics.py`** ? `match_1to1` empty-input guard (returns zeros), Hungarian cost
matrix correctly drops ineligible/padding via `cost<big`, greedy fallback deterministic and
(verified) agrees with Hungarian on the near-diagonal graphs rally intervals produce;
`_percentile` single-element safe; `boundary_error_report` `max(..., default=None)` guarded;
`over_seg_guard` `_aligned` by-name path is filtered (`if b["name"] in cmap`).
- **`rally_seg_eval.py`** ? `aggregate` returns `None` per-key on empty rows and `_ci(None)`
renders "?"; all `_AGG_KEYS` are emitted by `score_intervals`.
- **`regression.py` / `per_user_.py` / `promotion.py` / `calibration.py` / `experiment.py`** ?
incommensurability guards, minimum-N floor, CI/sign-test gating, kfold/LOVO splits, and the
threshold/calibration LOVO path are all internally consistent and conservative.
- **Supporting modules** (`metrics.py`, `segmentation_metrics.py`, `score_calibration.py` incl.
PAVA length-alignment verified empirically, `classifier.py`, `gt_loader.py`, `annotations.py`,
`windowing.py`, `batch.py`, `harness.py`, `training.py`, `splits.py`) ? clean.
```

I ran the relevant test slices (206 tests across these modules) ? all green ? and verified  
numpy/scipy/math.comb` availability so the Hungarian and exact-binomial paths actually execute  
here.

Consistent with the brief (mature repo, 848 passing tests, prior hardening sweeps ? "expect FEW  
or ZERO real bugs"), I returned an empty `candidates` list rather than manufacture low-  
confidence findings.